

FLAT FUNCTORS AND CLASSIFYING TOPOSES

In this chapter we develop a general theory of extensions of flat functors along geometric morphisms of toposes. This theory will be applied in the next chapter to characterize the class of theories whose classifying topos is equivalent to a presheaf topos. In fact, the necessary and sufficient conditions for a theory $\mathbb{T}$ to be of presheaf type (i.e. classified by a presheaf topos) provided by the characterization theorem of Chapter 6 arise precisely from the requirement that for any Grothendieck topos $\mathcal{E}$ the operation of extension of flat functors with values in $\mathcal{E}$ from the opposite of the category of finitely presentable models of $\mathbb{T}$ to the geometric syntactic category of $\mathbb{T}$ should yield an equivalence of categories onto the category of $\mathbb{T}$ -models in $\mathcal{E}$, naturally in $\mathcal{E}$.

The plan of the chapter is as follows.

In the first section, we investigate $\mathcal{E}$ -indexed colimits of internal diagrams in a Grothendieck topos $\mathcal{E}$, analysing in particular their behaviour with respect to final $\mathcal{E}$ -indexed functors (cf. section 5.1.5) and establishing explicit characterizations for an $\mathcal{E}$ -indexed cocone to be colimiting (cf. section 5.1.6). Also, we exploit the abstract interpretation of colimits as generalized tensor products to derive commutativity results which allow us to interpret certain set-indexed colimits arising in the context of the above-mentioned characterization theorem as special kinds of filtered indexed colimits (cf. section 5.1.4).

In the second section, we investigate the properties of the operation on flat functors induced by a geometric morphism via Diaconescu's equivalence. We focus in particular on geometric morphisms between presheaf toposes induced by embeddings between small categories and on geometric morphisms to the classifying topos of a geometric theory induced by a small category of set-based models of the theory. We also establish, in section 5.2.4, a general 'hom-tensor' adjunction between categories of $\mathcal{E}$ -valued functors (for $\mathcal{E}$ a Grothendieck topos) $[\mathcal{C}, \mathcal{E}]$ and $[\mathcal{D}, \mathcal{E}]$ induced by a functor $P : \mathcal{C} \rightarrow [\mathcal{D}^{\text{op}}, \mathbf{Set}]$; this generalizes the well-known adjunction induced by left Kan extensions along a given functor.

In the third section, we describe a way of representing flat functors into arbitrary Grothendieck toposes by using a suitable internalized version of the Yoneda lemma.

5.1 Preliminary results on indexed colimits in toposes

5.1.1 Background on indexed categories

In this section we recall some standard notions and facts from the theory of indexed categories that will be useful in the sequel; we refer the reader to [55] (especially sections B1.2, B2.3 and B3 therein) and to [67] for the relevant prerequisites.

Given an internal category $\mathbb{C}$ in a topos $\mathcal{E}$, we denote by $\mathbb{C}_0$ (resp. $\mathbb{C}_1$) its object of objects (resp. of arrows), and by $d_0^{\mathbb{C}}, d_1^{\mathbb{C}} : \mathbb{C}_1 \rightarrow \mathbb{C}_0$ the domain and codomain arrows.

Given a small category $\mathcal{C}$ and a Grothendieck topos $\mathcal{E}$, we can always internalize $\mathcal{C}$ into $\mathcal{E}$ by means of the inverse image functor $\gamma_{\mathcal{E}}^*$ of the unique geometric morphism $\gamma_{\mathcal{E}} : \mathcal{E} \rightarrow \mathbf{Set}$; the resulting internal category in $\mathcal{E}$ will be called the *internalization* of $\mathcal{C}$ in $\mathcal{E}$ and denoted by the symbol $\mathbb{C}$.

Given a category $\mathcal{E}$, an $\mathcal{E}$ -indexed category is by definition a pseudofunctor $\mathcal{E}^{\text{op}} \rightarrow \mathbf{Cat}$, where $\mathbf{Cat}$ is the meta-2-category of categories. Indexed categories will be denoted by underlined letters, possibly with a subscript indicating the indexing category; the fibre at an object E of an $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}$ will be denoted by $\mathcal{A}_E$, and the functor $\mathcal{A}_E \rightarrow \mathcal{A}_{E'}$ corresponding to an arrow $e : E' \rightarrow E$ in $\mathcal{E}$ will be denoted by $\mathcal{A}_e$.

An $\mathcal{E}$ -indexed subcategory $\underline{\mathcal{B}}_{\mathcal{E}}$ of an $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ consists, for each object $E \in \mathcal{E}$, of a subcategory $\mathcal{B}_E$ of the category $\mathcal{A}_E$ such that for any arrow $\alpha : E' \rightarrow E$ in $\mathcal{E}$ the functor $\mathcal{A}_{\alpha} : \mathcal{A}_E \rightarrow \mathcal{A}_{E'}$ restricts to the subcategories $\mathcal{B}_E$ and $\mathcal{B}_{E'}$. This clearly defines an $\mathcal{E}$ -indexed category $\underline{\mathcal{B}}_{\mathcal{E}}$ with an $\mathcal{E}$ -indexed inclusion functor $\underline{\mathcal{B}}_{\mathcal{E}} \hookrightarrow \underline{\mathcal{A}}_{\mathcal{E}}$.

An $\mathcal{E}$ -indexed functor $F : \underline{\mathcal{B}}_{\mathcal{E}} \rightarrow \underline{\mathcal{A}}_{\mathcal{E}}$ is said to be *full* if for every object E of $\mathcal{E}$ the functor $F_E : \mathcal{B}_E \rightarrow \mathcal{A}_E$ is full. An $\mathcal{E}$ -indexed subcategory $\underline{\mathcal{B}}_{\mathcal{E}}$ of an $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ is said to be a *full $\mathcal{E}$ -indexed subcategory* of $\underline{\mathcal{A}}_{\mathcal{E}}$ if the associated $\mathcal{E}$ -indexed inclusion functor is full.

Every Grothendieck topos $\mathcal{E}$ gives rise to an $\mathcal{E}$ -indexed category $\underline{\mathcal{E}}_{\mathcal{E}}$ obtained by indexing $\mathcal{E}$ over itself, i.e. by setting $\mathcal{E}_E = \mathcal{E}/E$ for any $E \in \mathcal{E}$ and $\mathcal{E}_e = e^* : \mathcal{E}/E \rightarrow \mathcal{E}/E'$ for any arrow $e : E' \rightarrow E$ in $\mathcal{E}$.

Given an internal category $\mathbb{C} = (d_0^{\mathbb{C}}, d_1^{\mathbb{C}} : \mathbb{C}_1 \rightarrow \mathbb{C}_0)$ in a topos $\mathcal{E}$, a *diagram of shape $\mathbb{C}$ in $\mathcal{E}$* is a pair $(f : F \rightarrow \mathbb{C}_0, \phi : \mathbb{C}_1 \times_{\mathbb{C}_0} F \rightarrow F)$ of arrows in $\mathcal{E}$ satisfying appropriate conditions which generalize those defining the notion of set-based diagram (cf. Definition 2.14 [50]), where the pullback $\mathbb{C}_1 \times_{\mathbb{C}_0} F$ is taken relatively to the arrow $d_0^{\mathbb{C}} : \mathbb{C}_1 \rightarrow \mathbb{C}_0$ and the arrow $f : F \rightarrow \mathbb{C}_0$:

$$\begin{array}{ccc} \mathbb{C}_1 \times_{\mathbb{C}_0} F & \xrightarrow{\pi_F} & F \\ \downarrow \pi_1 & & \downarrow f \\ \mathbb{C}_1 & \xrightarrow{d_0^{\mathbb{C}}} & \mathbb{C}_0. \end{array}$$

For any Grothendieck topos $\mathcal{E}$ and any internal category $\mathbb{C}$ in $\mathcal{E}$, we have an $\mathcal{E}$ -indexed category $[\mathbb{C}, \mathcal{E}]$ whose underlying category is the category $[\mathbb{C}, \mathcal{E}]$ of diagrams of shape $\mathbb{C}$ in $\mathcal{E}$ and morphisms between them.

Any internal category $\mathbb{C} = (d_0^{\mathbb{C}}, d_1^{\mathbb{C}} : \mathbb{C}_1 \rightarrow \mathbb{C}_0)$ in $\mathcal{E}$ naturally gives rise to an $\mathcal{E}$ -indexed category $\underline{\mathbb{C}}_{\mathcal{E}}$, called the $\mathcal{E}$ -externalization of $\mathbb{C}$, whose fibre $\mathbb{C}_E$ at any object E of $\mathcal{E}$ is the category whose objects are the arrows $E \rightarrow \mathbb{C}_0$ in $\mathcal{E}$ and whose arrows $x \rightarrow x'$ are the arrows $a : E \rightarrow \mathbb{C}_1$ in $\mathcal{E}$ such that $d_0^{\mathbb{C}} \circ a = x$ and $d_1^{\mathbb{C}} \circ a = x'$. An $\mathcal{E}$ -indexed category is said to be *small* if it is of the form $\underline{\mathbb{C}}_{\mathcal{E}}$ for some internal category $\mathbb{C}$ in $\mathcal{E}$.

The category $[\mathbb{C}, \mathcal{E}]$ is equivalent (naturally in $\mathcal{E}$) to the category $[\underline{\mathbb{C}}_{\mathcal{E}}, \underline{\mathcal{E}}_{\mathcal{E}}]_{\mathcal{E}}$ of $\mathcal{E}$ -indexed functors $\underline{\mathbb{C}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ and indexed natural transformations between them (by Lemma B2.3.13 [55]). If $\mathcal{C}$ is a small category and $\mathbb{C}$ is its internalization in a Grothendieck topos $\mathcal{E}$ then the category $[\mathbb{C}, \mathcal{E}]$ is also equivalent to the category $[\mathcal{C}, \mathcal{E}]$ (by Corollary B2.3.14 [55]). The equivalence between $[\mathbb{C}, \mathcal{E}]$ and $[\mathcal{C}, \mathcal{E}]$ restricts to an equivalence between the full subcategories **Tors**($\mathbb{C}, \mathcal{E}$) of $\mathbb{C}$ -torsors in $\mathcal{E}$ (as in section B3.2 of [55]) and **Flat**($\mathcal{C}, \mathcal{E}$) of flat functors $\mathcal{C} \rightarrow \mathcal{E}$ (as in Chapter VII of [63]). For any internal functor between internal categories in $\mathcal{E}$ or internal diagram F in $\mathcal{E}$ we denote the corresponding $\mathcal{E}$ -indexed functor by the symbol $\underline{F}_{\mathcal{E}}$. For any functor $F : \mathcal{C} \rightarrow \mathcal{E}$ we denote the internal diagram in $[\mathbb{C}, \mathcal{E}]$ corresponding to it under the equivalence $[\mathbb{C}, \mathcal{E}] \simeq [\mathcal{C}, \mathcal{E}]$ by the symbol $\overline{F}$.

The *discrete opfibration* $p : \mathbb{F} \rightarrow \mathbb{C}$ over $\mathbb{C}$ corresponding to a diagram $(f : F \rightarrow \mathbb{C}_0, \phi : \mathbb{C}_1 \times_{\mathbb{C}_0} F \rightarrow F)$ of shape $\mathbb{C}$ in $\mathcal{E}$ (where the pullback $\mathbb{C}_1 \times_{\mathbb{C}_0} F$ is taken with respect to the domain arrow $\mathbb{C}_1 \rightarrow \mathbb{C}_0$ and the arrow f) is defined as follows: $\mathbb{F}_0 = F$, $\mathbb{F}_1 = \mathbb{C}_1 \times_{\mathbb{C}_0} F$, $d_0^{\mathbb{F}} = \pi_F : \mathbb{F}_1 = \mathbb{C}_1 \times_{\mathbb{C}_0} F \rightarrow F = \mathbb{F}_0$, $d_1^{\mathbb{F}} = \phi : \mathbb{F}_1 = \mathbb{C}_1 \times_{\mathbb{C}_0} F \rightarrow F = \mathbb{F}_0$, $p_0 = f : \mathbb{F}_0 = F \rightarrow \mathbb{C}_0$, $p_1 = \pi_1 : \mathbb{F}_1 = \mathbb{C}_1 \times_{\mathbb{C}_0} F \rightarrow \mathbb{C}_1$. The discrete opfibration corresponding to a diagram $F \in [\mathbb{C}, \mathcal{E}]$ will also be denoted by $\pi_F^{\text{opf}} : \int^{\text{opf}} F \rightarrow \mathbb{C}$.

It is also natural to consider the *discrete fibration* corresponding to an internal diagram $G \in [\mathbb{C}^{\text{op}}, \mathcal{E}]$, i.e. the opposite functor $(\pi_G^{\text{opf}})^{\text{op}} : (\int^{\text{opf}} G)^{\text{op}} \rightarrow (\mathbb{C}^{\text{op}})^{\text{op}} = \mathbb{C}$. We shall denote this functor by $\pi_G^{\text{f}} : \int^{\text{f}} G \rightarrow \mathbb{C}$, or simply by $\pi_G : \int G \rightarrow \mathbb{C}$.

For any internal category $\mathbb{C}$ in a topos $\mathcal{E}$, the category $[\mathbb{C}, \mathcal{E}]$ of diagrams of shape $\mathbb{C}$ in $\mathcal{E}$ is equivalent to the category **DOpf**/ $\mathbb{C}$ of *discrete opfibrations* over $\mathbb{C}$ (cf. Proposition B2.5.3 [55]); we shall denote this equivalence by

$$\tau_{\mathcal{E}}^{\mathbb{C}} : [\mathbb{C}, \mathcal{E}] \rightarrow \mathbf{DOpf}/\mathbb{C}.$$

A diagram of shape $\mathbb{C}$ in $\mathcal{E}$ lies in the subcategory **Tors**($\mathbb{C}, \mathcal{E}$) of $[\mathbb{C}, \mathcal{E}]$ if and only if the domain of the corresponding discrete opfibration is a filtered internal category in $\mathcal{E}$.

Any internal functor $H : \mathbb{C} \rightarrow \mathbb{D}$ between internal categories $\mathbb{C}$ and $\mathbb{D}$ in a topos $\mathcal{E}$ induces a functor $[\mathbb{D}, \mathcal{E}] \rightarrow [\mathbb{C}, \mathcal{E}]$ whose action on objects is indicated by $D \rightarrow D \circ H$. This functor corresponds, under the equivalences $[\mathbb{D}, \mathcal{E}] \simeq [\underline{\mathbb{D}}_{\mathcal{E}}, \underline{\mathcal{E}}_{\mathcal{E}}]_{\mathcal{E}}$ and $[\mathbb{C}, \mathcal{E}] \simeq [\underline{\mathbb{C}}_{\mathcal{E}}, \underline{\mathcal{E}}_{\mathcal{E}}]_{\mathcal{E}}$, to the composition functor with the indexed functor associated with H , and, under the equivalences $[\mathbb{D}, \mathcal{E}] \simeq \mathbf{DOpf}/\mathbb{D}$ and

$[\mathbb{C}, \mathcal{E}] \simeq \mathbf{D}\mathbf{Opf}/\mathbb{C}$, to the pullback along the functor H . The latter pullbacks in the category of internal categories in $\mathcal{E}$ are computed ‘pointwise’ as pullbacks in $\mathcal{E}$ and are preserved by the dualizing functor $\mathbb{C} \rightarrow \mathbb{C}^{\text{op}}$.

Let us recall from [67] the notion of (indexed) *colimit of an $\mathcal{E}$ -indexed functor*, where $\mathcal{E}$ is a cartesian category. We shall denote by $!_E$ the unique arrow from an object E of $\mathcal{E}$ to the terminal object 1 of $\mathcal{E}$. Given an $\mathcal{E}$ -indexed functor $\Gamma : \underline{\mathcal{X}} \rightarrow \underline{\mathcal{A}}$ and an object A of $\mathcal{A}_1$, we denote by $\Delta_{\underline{\mathcal{A}}}^{\underline{\mathcal{X}}} A$ the $\mathcal{E}$ -indexed functor $\underline{\mathcal{X}} \rightarrow \underline{\mathcal{A}}$ assigning to any $E \in \mathcal{E}$ the constant functor $\Delta(\mathcal{A}_{!_E}(A))$ on X_E with value $\mathcal{A}_{!_E}(A)$.

Let $\underline{\mathcal{X}}$ be an indexed category and $\Gamma : \underline{\mathcal{X}} \rightarrow \underline{\mathcal{A}}$ an indexed functor. An indexed cocone $\mu : \Gamma \rightarrow A$ consists of an object A in $\mathcal{A}_1$ together with an indexed natural transformation $\mu : \Gamma \rightarrow \Delta_{\underline{\mathcal{A}}}^{\underline{\mathcal{X}}} A$, i.e., for each E in $\mathcal{E}$ an ordinary cocone $\mu_E : \Gamma_E \rightarrow \Delta(\mathcal{A}_{!_E}(A))$ such that for each arrow $\alpha : E' \rightarrow E$ in $\mathcal{E}$, $\alpha^* \circ \mu_E = \mu_{E'} \circ \alpha^*$. If $\mu : \Gamma \rightarrow \Delta_{\underline{\mathcal{A}}}^{\underline{\mathcal{X}}} A$ is a universal such cocone we say that it is a *colimit* of the indexed functor Γ with vertex A . If, furthermore, for any object E of $\mathcal{E}$, the localization $\mu/E : \Gamma/E \rightarrow (\Delta_{\underline{\mathcal{A}}}^{\underline{\mathcal{X}}} A)/E = \Delta_{\underline{\mathcal{A}/E}}^{\underline{\mathcal{X}/E}}(\mathcal{A}_{!_E}(A))$ of μ at E is a colimit we say that μ is an ($\mathcal{E}$ -) *indexed colimit* of Γ .

We can describe this notion more explicitly in the particular case of $\mathcal{E}$ -indexed functors with values in the $\mathcal{E}$ -indexed category $\underline{\mathcal{E}}_{\mathcal{E}}$. Let us suppose that $\underline{\mathcal{A}}_{\mathcal{E}}$ is an $\mathcal{E}$ -indexed category and $D : \underline{\mathcal{A}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ is an $\mathcal{E}$ -indexed functor. A cocone μ over D consists of an object U of $\mathcal{E}$ and, for each object E of $\mathcal{E}$, of a cocone $\mu_E : D_E \rightarrow \Delta(!_E^*(U))$ over the diagram $D_E : \mathcal{A}_E \rightarrow \mathcal{E}/E$ such that for any arrow $\alpha : E' \rightarrow E$ in $\mathcal{E}$ we have $\alpha^*(\mu_E(c)) = \mu_{E'}(\mathcal{A}_{\alpha}(c))$ for all $c \in \mathcal{A}_E$, that is $\alpha^* \mu_E = \mu_{E'} \mathcal{A}_{\alpha}$ as arrows $D_{E'}(\mathcal{A}_{\alpha}(c)) = \alpha^*(D_E(c)) \rightarrow !_{E'}^*(U) = \alpha^*(!_E^*(U))$ in $\mathcal{E}/E'$, where $\alpha^* : \mathcal{E}/E \rightarrow \mathcal{E}/E'$ is the pullback functor (notice that $D_{E'} \circ \mathcal{A}_{\alpha} = \alpha^* \circ D_E$ since D is an $\mathcal{E}$ -indexed functor).

If $\underline{\mathcal{A}}_{\mathcal{E}}$ is the $\mathcal{E}$ -externalization of an internal category $\mathbb{C}$ in $\mathcal{E}$ and $D : \underline{\mathcal{A}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ is an $\mathcal{E}$ -indexed functor corresponding to an internal diagram $D \in [\mathbb{C}, \mathcal{E}]$, the discrete opfibration associated with the internal diagram in $[_E^*(\mathbb{C}), \mathcal{E}/E]$ corresponding to the localization $D/E : \underline{\mathcal{A}}_{\mathcal{E}}/E \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}/E \cong \underline{\mathcal{E}}/E_{\mathcal{E}/E}$ is the image of the discrete opfibration associated with D under the pullback functor $!_E^* : \mathcal{E} \rightarrow \mathcal{E}/E$ along the unique arrow $!_E : E \rightarrow 1_{\mathcal{E}}$.

The colimit $\text{colim}_{\mathcal{E}}(D)$ in $\mathcal{E}$ of an internal diagram $D \in [\mathbb{C}, \mathcal{E}]$, where $\mathbb{C}$ is an internal category in $\mathcal{E}$, is by definition the coequalizer of the two arrows $d_0^{\text{opf } D}, d_1^{\text{opf } D}$ (recall that $\int^{\text{opf } D}$ is the domain of the discrete opfibration over $\mathbb{C}$ corresponding to the diagram D).

For any internal diagram $G \in [\mathbb{F}, \mathcal{E}]$, $\text{colim}_{\mathcal{E}}(G)$ is isomorphic to $\underline{\text{colim}}_{\mathcal{E}}(\underline{G}_{\mathcal{E}})$, where $\underline{G}_{\mathcal{E}} : \underline{\mathbb{F}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ is the $\mathcal{E}$ -indexed functor corresponding to G under the equivalence $[\mathbb{F}, \mathcal{E}] \simeq [\underline{\mathbb{F}}_{\mathcal{E}}, \underline{\mathcal{E}}_{\mathcal{E}}]$.

If $D \in [\mathbb{C}, \mathcal{E}]$ is an internal diagram in $\mathcal{E}$, the colimiting $\mathcal{E}$ -indexed cocone $(\text{colim}_{\mathcal{E}}(D) = \text{coeq}(d_1^{\text{opf } D}, d_1^{\text{opf } D}), \mu)$ of the corresponding $\mathcal{E}$ -indexed functor $\underline{D}_{\mathcal{E}} : \underline{\mathbb{C}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ can be described as follows. Let us denote by c the canonical arrow

$(\int^{\text{opf}} D)_0 \rightarrow \text{coeq}(d_0^{\int^{\text{opf}} D}, d_1^{\int^{\text{opf}} D})$ in $\mathcal{E}$. For any objects E of $\mathcal{E}$ and $x : E \rightarrow \mathbb{C}_0$ of $\mathbb{C}_E$, the arrow $\mu_E(x) : \overline{D}_E = r_x \rightarrow !^*_E(\text{colim}_{\mathcal{E}}(D)) = \text{coeq}(d_0^{\int^{\text{opf}} D}, d_1^{\int^{\text{opf}} D}) \times E$ is equal to $< c \circ z_x, r_x >$ where the arrows r_x and z_x are defined by the following pullback square:

$$\begin{array}{ccc} R_x & \xrightarrow{z_x} & (\int^{\text{opf}} D)_0 \\ \downarrow r_x & & \downarrow (\pi_D^{\text{opf}})_0 \\ E & \xrightarrow{x} & \mathbb{C}_0. \end{array}$$

Given an $\mathcal{E}$ -indexed category $\underline{\mathbb{D}}$ and an object $E \in \mathcal{E}$, we have an $\mathcal{E}/E$ -indexed category $\underline{\mathbb{D}}/E$ (defined in the obvious way) called the *localization of $\underline{\mathbb{D}}$ at E* .

Given a topos $\mathcal{E}$, we denote by $!^*_E : \mathcal{E} \rightarrow \mathcal{E}/E$ the pullback functor along the unique arrow $!_E : E \rightarrow 1$, that is the inverse image functor of the local homeomorphism $\mathcal{E}/E \rightarrow \mathcal{E}$.

An $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ is said to have *small homs at 1* if for any objects F, G in $\mathcal{A}_1$ there exists an object $\text{Hom}_{\underline{\mathcal{A}}_{\mathcal{E}}}^{\mathcal{E}}(F, G)$ in $\mathcal{E}$ (also denoted by $\text{Hom}_{\underline{\mathcal{A}}}^{\mathcal{E}}(F, G)$) satisfying the universal property that, for every E in $\mathcal{E}$, there is a bijection between the arrows $E \rightarrow \text{Hom}_{\underline{\mathcal{A}}_{\mathcal{E}}}^{\mathcal{E}}(F, G)$ in $\mathcal{E}$ and the arrows $\mathcal{A}_{!_E}(F) \rightarrow \mathcal{A}_{!_E}(G)$ in $\mathcal{A}_E$ which natural in E . An $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ is said to be *locally small* if, for every object E of $\mathcal{E}$, the $\mathcal{E}/E$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}/E$ has small homs at 1.

For any Grothendieck topos $\mathcal{E}$ and any internal category $\mathbb{C}$ in $\mathcal{E}$, the $\mathcal{E}$ -indexed category $[\underline{\mathbb{C}}, \underline{\mathcal{E}}]$ is locally small (by Lemma B2.3.15 [55]); from this it follows that there exists an $\mathcal{E}$ -indexed hom functor

$$\text{Hom}_{[\underline{\mathbb{C}}, \underline{\mathcal{E}}]}^{\mathcal{E}} : [\underline{\mathbb{C}}, \underline{\mathcal{E}}]^{\text{op}} \times [\underline{\mathbb{C}}, \underline{\mathcal{E}}] \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$$

whose underlying functor assigns to a pair of diagrams F and G in $[\underline{\mathbb{C}}, \underline{\mathcal{E}}]$ an object $\text{Hom}_{[\underline{\mathbb{C}}, \underline{\mathcal{E}}]}^{\mathcal{E}}(F, G)$ of $\mathcal{E}$, which we call the *object of morphisms* from F to G in $[\underline{\mathbb{C}}, \underline{\mathcal{E}}]$; also, there is a Yoneda $\mathcal{E}$ -indexed functor $\mathbb{Y} : \underline{\mathbb{C}} \rightarrow [\underline{\mathbb{C}}^{\text{op}}, \underline{\mathcal{E}}]$ which plays in this context the same role as that of a Yoneda functor in ordinary category theory. If $\mathbb{C}$ is the internalization in a Grothendieck topos $\mathcal{E}$ of a small category $\mathcal{C}$ then, since $[\underline{\mathcal{C}}, \underline{\mathcal{E}}] \simeq [\underline{\mathbb{C}}, \underline{\mathcal{E}}]$, we have a hom functor

$$\text{Hom}_{[\underline{\mathcal{C}}, \underline{\mathcal{E}}]}^{\mathcal{E}} : [\underline{\mathcal{C}}, \underline{\mathcal{E}}]^{\text{op}} \times [\underline{\mathcal{C}}, \underline{\mathcal{E}}] \rightarrow \underline{\mathcal{E}}$$

which assigns to each pair of functors $F, G \in [\underline{\mathcal{C}}, \underline{\mathcal{E}}]$ an object $\text{Hom}_{[\underline{\mathcal{C}}, \underline{\mathcal{E}}]}^{\mathcal{E}}(F, G)$ of $\mathcal{E}$ such that for each $E \in \mathcal{E}$ the arrows $E \rightarrow \text{Hom}_{[\underline{\mathcal{C}}, \underline{\mathcal{E}}]}^{\mathcal{E}}(F, G)$ in $\mathcal{E}$ are in natural bijection with the arrows in $[\underline{\mathcal{C}}, \underline{\mathcal{E}}/E]$ from $!^*_E \circ F$ to $!^*_E \circ G$, that is with the natural transformations $!^*_E \circ F \Rightarrow !^*_E \circ G$. Note that, since $\mathbf{Flat}(\underline{\mathcal{C}}, \underline{\mathcal{E}})$ is a full subcategory of $[\underline{\mathcal{C}}, \underline{\mathcal{E}}]$, we may use the objects $\text{Hom}_{[\underline{\mathcal{C}}, \underline{\mathcal{E}}]}^{\mathcal{E}}(F, G)$ for $F, G \in \mathbf{Flat}(\underline{\mathcal{C}}, \underline{\mathcal{E}})$ as the objects of morphisms from F to G , so as to obtain a locally small $\mathcal{E}$ -indexed category $\mathbf{Flat}(\underline{\mathcal{C}}, \underline{\mathcal{E}})$.

5.1.2 $\mathcal{E}$ -filtered indexed categories

The following definition will be important in the sequel:

Definition 5.1.1 Let $\mathcal{E}$ be a Grothendieck topos and $\mathcal{A}$ an $\mathcal{E}$ -indexed category. We say that $\mathcal{A}$ is $\mathcal{E}$ -filtered if the following conditions are satisfied:

- (a) For any object E of $\mathcal{E}$, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object b_i of the category $\mathcal{A}_{E_i}$.
- (b) For any $E \in \mathcal{E}$ and any objects a and b of the category $\mathcal{A}_E$, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object c_i of the category $\mathcal{A}_{E_i}$ and arrows $f_i : \mathcal{A}_{e_i}(a) \rightarrow c_i$ and $g_i : \mathcal{A}_{e_i}(b) \rightarrow c_i$ in $\mathcal{A}_{E_i}$.
- (c) For any object E of $\mathcal{E}$ and any arrows $u, v : a \rightarrow b$ in the category $\mathcal{A}_E$, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object c_i of the category $\mathcal{A}_{E_i}$ and an arrow $w_i : \mathcal{A}_{e_i}(b) \rightarrow c_i$ in $\mathcal{A}_{E_i}$ such that $w_i \circ \mathcal{A}_{e_i}(u) = w_i \circ \mathcal{A}_{e_i}(v)$.

The $\mathcal{E}$ -externalization of any internal filtered category in $\mathcal{E}$ is $\mathcal{E}$ -filtered, but if the externalization of an internal category $\mathbb{C}$ in $\mathcal{E}$ is $\mathcal{E}$ -filtered then $\mathbb{C}$ is not necessarily filtered as an internal category in $\mathcal{E}$.

A standard source of indexed, $\mathcal{E}$ -filtered categories is provided by the indexed categories of elements of flat functors with values in $\mathcal{E}$ in the sense of the following definition:

Definition 5.1.2 Let $P : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ be a functor. The $\mathcal{E}$ -indexed category $\underline{\int} P_{\mathcal{E}}$ of elements of P assigns

- to any object E of $\mathcal{E}$ the category $\underline{\int} P_E$ whose objects are the pairs (c, x) where c is an object of $\mathcal{C}$ and $x : E \rightarrow F(c)$ is an arrow in $\mathcal{E}$ and whose arrows $(c, x) \rightarrow (d, y)$ are the arrows $f : c \rightarrow d$ in $\mathcal{C}$ such that $F(f) \circ y = x$;
- to any arrow $e : E' \rightarrow E$ in $\mathcal{E}$ the functor $\underline{\int} P_e : \underline{\int} P_E \rightarrow \underline{\int} P_{E'}$ sending any object (c, x) of $\underline{\int} P_E$ to the object $(c, x \circ e)$ of $\underline{\int} P_{E'}$ and acting accordingly on arrows.

Proposition 5.1.3 Let $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ be a flat functor. Then the indexed category $\underline{\int} F_{\mathcal{E}}$ is $\mathcal{E}$ -filtered.

Proof Straightforward from the characterization of flat functors as filtering functors given in Proposition 1.1.32. $\square$

In the sequel we shall work both with the $\mathcal{E}$ -indexed category $\underline{\int} F_{\mathcal{E}}$ (for a flat functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$) and with the $\mathcal{E}$ -externalization $\underline{\int}^{\text{f}} \overline{F}_{\mathcal{E}}$ of the domain $\int^{\text{f}} \overline{F}$ of the discrete fibration corresponding to the $\mathbb{C}$ -torsor $\overline{F}$, which is $\mathcal{E}$ -filtered since $\int^{\text{f}} \overline{F}$ is filtered as an internal category in $\mathcal{E}$.

5.1.3 Indexation of internal diagrams

Given an internal diagram $D \in [\mathbb{C}, \mathcal{E}]$, the corresponding $\mathcal{E}$ -indexed functor

$$\overline{D}_{\mathcal{E}} : \underline{\mathbb{C}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$$

can be described as follows. For any object E of $\mathcal{E}$, $\overline{D}_{\mathcal{E}} : \mathbb{C}_E \rightarrow \mathcal{E}/E$ sends any object $x : E \rightarrow \mathbb{C}_0$ of $\mathbb{C}_E$ to the object $r_x : R_x \rightarrow E$ of $\mathcal{E}/E$ obtained by pulling $(\pi_D^{\text{opf}})_0 : (\int^{\text{opf}} D)_0 \rightarrow \mathbb{C}_0$ back along x and any arrow $h : E \rightarrow \mathbb{C}_1$ of $\mathbb{C}_E$ from $x : E \rightarrow \mathbb{C}_0$ to $x' : E \rightarrow \mathbb{C}_0$ to the arrow $\overline{D}_{\mathcal{E}}(h) : r_x \rightarrow r_{x'}$ in $\mathcal{E}/E$ defined as follows. Consider the pullback squares

$$\begin{array}{ccc} S & \xrightarrow{u} & (\int^{\text{opf}} D)_1 \\ \downarrow s & & \downarrow (\pi_D^{\text{opf}})_1 \\ E & \xrightarrow{h} & \mathbb{C}_1 \end{array}$$

and

$$\begin{array}{ccc} R_{x'} & \xrightarrow{z_{x'}} & (\int^{\text{opf}} D)_0 \\ \downarrow r_{x'} & & \downarrow (\pi_D^{\text{opf}})_0 \\ E & \xrightarrow{x'} & \mathbb{C}_0. \end{array}$$

Since $d_1^{\mathbb{C}} \circ h = x'$ we have that $(\pi_D^{\text{opf}})_0 \circ d_1^{\int^{\text{opf}} D} \circ u = x' \circ s$; thus, by the universal property of the first pullback square, there exists a unique arrow $\beta : S \rightarrow R_{x'}$ such that $r_{x'} \circ \beta = s$ and $z_{x'} \circ \beta = d_1^{\int^{\text{opf}} D} \circ u$.

As $\pi_F^{\text{opf}} : \int^{\text{opf}} F \rightarrow \mathbb{C}$ is a discrete opfibration, the diagram

$$\begin{array}{ccc} (\int^{\text{opf}} D)_1 & \xrightarrow{d_0^{\int^{\text{opf}} D}} & (\int^{\text{opf}} D)_0 \\ \downarrow (\pi_D^{\text{opf}})_1 & & \downarrow (\pi_D^{\text{opf}})_0 \\ \mathbb{C}_1 & \xrightarrow{d_0^{\mathbb{C}}} & \mathbb{C}_0 \end{array}$$

is a pullback. ‘Composing’ it with the first of the above pullback squares thus yields a pullback square

$$\begin{array}{ccc} S & \xrightarrow{d_0^{\int^{\text{opf}} D} \circ u} & (\int^{\text{opf}} D)_0 \\ \downarrow s & & \downarrow (\pi_D^{\text{opf}})_0 \\ E & \xrightarrow{d_0^{\mathbb{C}} \circ h} & \mathbb{C}_0. \end{array}$$

Now, since $(\pi_D^{\text{opf}})_0 \circ z_x = x \circ r_x = d_0^{\mathbb{C}} \circ h \circ r_x$, the universal property of this pullback square provides a unique arrow $\gamma : R_x \rightarrow S$ such that $s \circ \gamma = r_x$ and

$d_0^{\int^{\text{opf}} D} \circ u \circ \gamma = z_x$. We define $\overline{D}_{\mathcal{E}}(h) : r_x \rightarrow r_{x'}$ in $\mathcal{E}/E$ to be equal to the composite arrow $\beta \circ \gamma : R_x \rightarrow R_{x'}$.

5.1.4 Colimits and tensor products

For any internal category $\mathbb{F}$ in $\mathcal{E}$, we denote by $\text{coeq}(\mathbb{F})$ the coequalizer of the two arrows $d_0^{\mathbb{F}}$ and $d_1^{\mathbb{F}}$.

From the above discussion it follows that for any $D, P \in [\mathbb{C}, \mathcal{E}]$, $\text{colim}_{\mathcal{E}}(P \circ \pi_D^{\text{opf}}) \cong \text{colim}_{\mathcal{E}}(D \circ \pi_P^{\text{opf}})$. Indeed, if we consider the pullback square

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{q^P} & \int^{\text{opf}} P \\ \downarrow q^D & & \downarrow \pi_P^{\text{opf}} \\ \int^{\text{opf}} D & \xrightarrow{\pi_D^{\text{opf}}} & \mathbb{C} \end{array}$$

in the category $\mathbf{cat}(\mathcal{E})$ of internal categories in $\mathcal{E}$ we have that $\text{colim}_{\mathcal{E}}(P \circ \pi_D^{\text{opf}}) \cong \text{coeq}(\text{dom}(q^D)) = \text{coeq}(\mathbb{R}) = \text{coeq}(\text{dom}(q^P)) \cong \text{colim}_{\mathcal{E}}(D \circ \pi_P^{\text{opf}})$.

Similarly, by exploiting the fact that the operation $\mathbb{F} \rightarrow \text{coeq}(\mathbb{F})$ on internal categories in $\mathcal{E}$ is invariant under the dualization functor $\mathbb{F} \rightarrow \mathbb{F}^{\text{op}}$, we obtain another commutativity result, which we shall use in the sequel: for any internal diagrams $F \in [\mathbb{C}^{\text{op}}, \mathcal{E}]$ and $P \in [\mathbb{C}, \mathcal{E}]$, we have a natural isomorphism

$$\text{colim}_{\mathcal{E}}(P \circ \pi_F^{\text{f}}) \cong \text{colim}_{\mathcal{E}}(F \circ \pi_P^{\text{f}}).$$

To prove this, we consider the following pullback squares:

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{q^P} & \int^{\text{opf}} P \\ \downarrow q^F & & \downarrow \pi_P^{\text{opf}} \\ \int^{\text{f}} F & \xrightarrow{\pi_F^{\text{f}}} & \mathbb{C}, \end{array} \quad \begin{array}{ccc} \mathbb{S} & \xrightarrow{r^F} & \int^{\text{opf}} F \\ \downarrow r^P & & \downarrow \pi_F^{\text{opf}} \\ \int^{\text{f}} P & \xrightarrow{\pi_P^{\text{f}}} & \mathbb{C}^{\text{op}}. \end{array}$$

We have that

$$\text{colim}_{\mathcal{E}}(P \circ \pi_F^{\text{f}}) \cong \text{coeq}(\text{dom}(q^F)) = \text{coeq}(\mathbb{R}),$$

while

$$\text{colim}_{\mathcal{E}}(F \circ \pi_P^{\text{f}}) = \text{coeq}(\text{dom}(r^P)) = \text{coeq}(\mathbb{S}).$$

But the fact that the dualization functor preserves pullbacks in $\mathbf{cat}(\mathcal{E})$ implies that $\mathbb{S} \simeq \mathbb{R}^{\text{op}}$, whence $\text{coeq}(\mathbb{R}) \cong \text{coeq}(\mathbb{S})$, as required.

Summarizing, we have the following result:

Proposition 5.1.4 *Let $\mathbb{C}$ be an internal category in a Grothendieck topos $\mathcal{E}$, P and D internal diagrams in $[\mathbb{C}, \mathcal{E}]$ and F an internal diagram in $[\mathbb{C}^{\text{op}}, \mathcal{E}]$. Then we have natural isomorphisms*

- (i) $\text{colim}_{\mathcal{E}}(P \circ \pi_D^{\text{opf}}) \cong \text{colim}_{\mathcal{E}}(D \circ \pi_P^{\text{opf}});$
- (ii) $\text{colim}_{\mathcal{E}}(P \circ \pi_F^{\text{f}}) \cong \text{colim}_{\mathcal{E}}(F \circ \pi_P^{\text{f}}).$

□

Let us now proceed to apply this proposition in the context of a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$, where $\mathcal{C}$ is a small category, $\mathcal{E}$ is a Grothendieck topos and P is a functor $\mathcal{C} \rightarrow \mathbf{Set}$. To this end, we explicitly describe the discrete opfibration corresponding to the diagram $(g : \overline{G} \rightarrow \mathbb{C}_0, \phi : \mathbb{C}_1 \times_{\mathbb{C}_0} \overline{G} \rightarrow \overline{G})$ of shape $\mathbb{C}$ in $\mathcal{E}$ associated with a functor $G : \mathcal{C} \rightarrow \mathcal{E}$, where $\mathbb{C}$ is the internalization of $\mathcal{C}$ in $\mathcal{E}$.

Let $\text{Ob}(\mathcal{C})$ denote the set of objects of $\mathcal{C}$ and $\text{Arr}(\mathcal{C})$ the set of arrows of $\mathcal{C}$. We have that $\overline{G} = \coprod_{c \in \text{Ob}(\mathcal{C})} G(c)$ and that $g : \overline{G} \rightarrow \mathbb{C}_0 = \coprod_{c \in \text{Ob}(\mathcal{C})} 1_{\mathcal{E}}$ is equal to $\coprod_{c \in \text{Ob}(\mathcal{C})} !_{G(c)}$, where $!_{G(c)}$ is the unique arrow $G(c) \rightarrow 1_{\mathcal{E}}$ in $\mathcal{E}$ (for any $c \in \mathcal{C}$).

Let $J_f : G(\text{dom}(f)) \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} G(\text{dom}(f))$, $\mu_c : G(c) \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} G(c)$, $\kappa_f : 1_{\mathcal{E}} \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} 1_{\mathcal{E}}$, and $\lambda_c : 1_{\mathcal{E}} \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} 1_{\mathcal{E}}$ be the canonical coproduct arrows.

First, let us show that the diagram

$$\begin{array}{ccc} \coprod_{f \in \text{Arr}(\mathcal{C})} G(\text{dom}(f)) & \xrightarrow{d_0^G} & \coprod_{c \in \text{Ob}(\mathcal{C})} G(c) \\ \downarrow \scriptstyle f \in \text{Arr}(\mathcal{C}) \quad !_{G(\text{dom}(f))} & & \downarrow \scriptstyle c \in \text{Ob}(\mathcal{C}) \quad !_{G(c)} \\ \coprod_{f \in \text{Arr}(\mathcal{C})} 1_{\mathcal{E}} & \xrightarrow{d_0^{\mathcal{C}}} & \coprod_{c \in \text{Ob}(\mathcal{C})} 1_{\mathcal{E}} \end{array}$$

is a pullback, where the arrow d_0^G is defined by setting $d_0^G \circ J_f = \mu_{\text{dom}(f)}$ for any $f \in \text{Arr}(\mathcal{C})$. We have to prove that, for any object E of $\mathcal{E}$ and arrows $\alpha : E \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} 1_{\mathcal{E}}$ and $\beta : E \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} G(c)$ such that $\coprod_{c \in \text{Ob}(\mathcal{C})} !_{G(c)} \circ \beta = d_0^{\mathcal{C}} \circ \alpha$, there exists a unique arrow $\gamma : E \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} !_{G(\text{dom}(f))}$ such that $\alpha = \coprod_{f \in \text{Arr}(\mathcal{C})} !_{G(\text{dom}(f))} \circ \gamma$ and $\beta = d_0^G \circ \gamma$. To this end, consider, for any $c \in \text{Ob}(\mathcal{C})$ and $f \in \text{Arr}(\mathcal{C})$, the commutative diagram

$$\begin{array}{ccccc} E_{(f,c)} & \xrightarrow{q_f} & E_f & \xrightarrow{\alpha_f} & 1_{\mathcal{E}} \\ \downarrow p_c & & \downarrow y_f & & \downarrow \kappa_f \quad \searrow id \\ E_c & \xrightarrow{z_c} & E & \xrightarrow{\alpha} & \coprod_{f \in \text{Arr}(\mathcal{C})} 1_{\mathcal{E}} \\ \downarrow \beta_c & & \downarrow \beta & & \downarrow d_0^{\mathcal{C}} \quad \swarrow \lambda_{\text{dom}(f)} \\ G(c) & \xrightarrow{\kappa_c} & \coprod_{c \in \text{Ob}(\mathcal{C})} G(c) & \xrightarrow{\coprod_{c \in \text{Ob}(\mathcal{C})} !_{G(c)}} & \coprod_{c \in \text{Ob}(\mathcal{C})} 1_{\mathcal{E}} \\ & \searrow !_{G(c)} & & \nearrow \lambda_c & \\ & & 1_{\mathcal{E}} & & \end{array}$$

where all the squares except for the lower right one are pullbacks. The commutativity of the diagram, combined with the fact that distinct coproduct arrows are disjoint from each other, immediately implies that for any pair (f, c) such that $c \neq \text{dom}(f)$ we have $E_{(f,c)} \cong 0_{\mathcal{E}}$. On the other hand, the stability of coproducts under pullbacks implies that $E \cong \coprod_{c \in \text{Ob}(\mathcal{C})} E_c$ and $E \cong \coprod_{f \in \text{Arr}(\mathcal{C})} E_f$, whence $E \cong \coprod_{(f,c) \in \text{Arr}(\mathcal{C}) \times \text{Ob}(\mathcal{C})} E_{(f,c)} \cong \coprod_{(f,c) \mid \text{dom}(f)=c} E_{(f,c)}$ with canonical coproduct arrows $\xi_{(f,c)} = y_f \circ q_f = z_c \circ p_c : E_{(f,c)} \rightarrow E$. We define, for each pair (f, c) such that $c = \text{dom}(f)$, the arrow $\gamma_{(f,c)} : E_{(f,c)} \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} G(\text{dom}(f))$ as the composite $J_f \circ \beta_c \circ p_c$, and set γ equal to the arrow $\coprod_{(f,c) \mid \text{dom}(f)=c} \gamma_{(f,c)} : E \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} G(\text{dom}(f))$. We have to verify that $\alpha = \coprod_{f \in \text{Arr}(\mathcal{C})} !_{G(\text{dom}(f))} \circ \gamma$ and $\beta = d_0^G \circ \gamma$ or, equivalently, that for any pair (f, c) such that $\text{dom}(f) = c$ we have:

- (1) $\alpha \circ \xi_{(f,c)} = \coprod_{f \in \text{Arr}(\mathcal{C})} !_{G(\text{dom}(f))} \circ \gamma_{(f,c)}$ and
- (2) $\beta \circ \xi_{(f,c)} = d_0^G \circ \gamma_{(f,c)}$.

To prove (1) it suffices to observe that for any pair (f, c) such that $c = \text{dom}(f)$, we have $\alpha_f \circ q_f = !_{G(c)} \circ \beta_c \circ p_c$, whence $\alpha \circ \xi_{(f,c)} = \alpha \circ y_f \circ q_f = \kappa_f \circ \alpha_f \circ q_f = \kappa_f \circ !_{G(c)} \circ \beta_c \circ p_c = \coprod_{f \in \text{Arr}(\mathcal{C})} !_{G(\text{dom}(f))} \circ \gamma_{(f,c)}$. Further, $\beta \circ \xi_{(f,c)} = \beta \circ z_c \circ p_c = \mu_c \circ \beta_c \circ p_c = \mu_{\text{dom}(f)} \circ \beta_c \circ p_c = d_0^G \circ J_f \circ \beta_c \circ p_c = d_0^G \circ \gamma_{(f,c)}$, which proves (2).

Let us define an arrow

$$d_1^G : \coprod_{f \in \text{Arr}(\mathcal{C})} G(\text{dom}(f)) \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} G(c)$$

by setting, for each $f \in \text{Arr}(\mathcal{C})$, $d_1^G \circ J_f = \mu_{\text{cod}(f)} \circ G(f)$.

The discrete opfibration $p : \mathbb{F} \rightarrow \mathbb{C}$ corresponding to G can be described as follows: $\mathbb{F}_0 = \coprod_{c \in \text{Ob}(\mathcal{C})} G(c)$, $\mathbb{F}_1 = \coprod_{f \in \text{Arr}(\mathcal{C})} G(\text{dom}(f))$, the domain and codomain arrows $d_0^{\mathbb{F}}, d_1^{\mathbb{F}} : \mathbb{F}_1 \rightarrow \mathbb{F}_0$ are respectively equal to d_0^G and to d_1^G , $p_0 : \mathbb{F}_0 \rightarrow \mathbb{C}_0$ is equal to $\coprod_{c \in \text{Ob}(\mathcal{C})} !_{G(c)}$, $p_1 : \mathbb{F}_1 \rightarrow \mathbb{C}_1$ is equal to $\coprod_{f \in \text{Arr}(\mathcal{C})} !_{G(\text{dom}(f))}$ and the composition law in the internal category $\mathbb{F}$ is defined in the obvious way.

We leave to the reader the straightforward task of verifying that this is indeed the discrete opfibration corresponding to the functor G via the composite of the equivalence $[\mathcal{C}, \mathcal{E}] \simeq [\mathbb{C}, \mathcal{E}]$ with the equivalence $\tau_{\mathcal{E}}^{\mathbb{C}} : [\mathbb{C}, \mathcal{E}] \rightarrow \mathbf{DOPf}/\mathbb{C}$. Recalling that for any internal diagram $F : \mathbb{C}^{\text{op}} \rightarrow \mathcal{E}$ the discrete fibration $\pi_F^f : \int^f F \rightarrow \mathbb{C}$ associated with it is equal to $(\pi_F^{\text{opf}})^{\text{op}} : (\int^{\text{opf}} F)^{\text{op}} \rightarrow (\mathbb{C}^{\text{op}})^{\text{op}} = \mathbb{C}$, we deduce the following explicit description of the discrete fibration $\pi_F^f : \int^f F \rightarrow \mathbb{C}$ associated with a functor $F : \mathbb{C}^{\text{op}} \rightarrow \mathcal{E}$: $(\int^f F)_0 = \coprod_{c \in \text{Ob}(\mathcal{C})} F(c)$, $(\int^f F)_1 = \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f))$,

the domain and codomain arrows $d_0^F, d_1^F : (\int^f F)_1 \rightarrow (\int^f F)_0$ are defined by the conditions $d_0^F \circ J_f = \mu_{\text{cod}(f)}$ and $d_1^F \circ J_f = \mu_{\text{dom}(f)} \circ F(f)$ for all $f \in \text{Arr}(\mathcal{C})$ (where $\mu_c : F(c) \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} F(c)$ and $J_f : F(\text{cod}(f)) \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f))$ are

the canonical coproduct arrows), $(\pi_F^f)_0 : (\int^f F)_0 \rightarrow \mathbb{C}_0$ is equal to $\coprod_{c \in \text{Ob}(\mathcal{C})} !_{F(c)}$

and $(\pi_F^f)_1 : (\int^f F)_1 \rightarrow \mathbb{C}_1$ is equal to $\coprod_{f \in \text{Arr}(\mathcal{C})} !_{F(\text{cod}(f))}$. The composition law in

the internal category $\int^f F$ is defined in the obvious way.

Theorem 5.1.5 *Let $\mathcal{C}$ be a small category, $\mathcal{E}$ a Grothendieck topos and $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}, P : \mathcal{C} \rightarrow \mathbf{Set}$ two functors. Then the following three objects are naturally isomorphic:*

- (i) $\text{colim}(F \circ \pi_P^f)$
- (ii) $\text{colim}_{\mathcal{E}}(\overline{F} \circ \pi_{P_{\mathcal{E}}}^f) \cong \text{colim}_{\mathcal{E}}(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f)$ (cf. Proposition 5.1.4)
- (iii) $\underline{\text{colim}}_{\mathcal{E}}(\underline{P}_{\mathcal{E}_{\mathcal{E}}} \circ \pi_{\overline{F}_{\mathcal{E}}}^f)$

where $\mathbb{C}$ is the internalization of $\mathcal{C}$ in $\mathcal{E}$ and $P_{\mathcal{E}}$ is the internal diagram in $[\mathbb{C}, \mathcal{E}]$ given by $\gamma_{\mathcal{E}}^* \circ \overline{P}$.

Proof The isomorphism between $\text{colim}_{\mathcal{E}}(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f)$ and $\underline{\text{colim}}_{\mathcal{E}}(\underline{P}_{\mathcal{E}_{\mathcal{E}}} \circ \pi_{\overline{F}_{\mathcal{E}}}^f)$ follows from the general fact that for any internal diagram $G \in [\mathbb{D}, \mathcal{E}]$, its colimit $\text{colim}_{\mathcal{E}}(G)$ is isomorphic to the $\mathcal{E}$ -indexed colimit $\underline{\text{colim}}_{\mathcal{E}}(\underline{G}_{\mathcal{E}})$. It thus remains to prove the isomorphism between $\text{colim}(F \circ \pi_P^f)$ and $\text{colim}_{\mathcal{E}}(\overline{F} \circ \pi_{P_{\mathcal{E}}}^f)$. To this end, we recall the following three general facts:

- (1) For any functor $G : \mathcal{D} \rightarrow \mathcal{E}$, its colimit $\text{colim}(G)$ is naturally isomorphic to the colimit $\text{colim}_{\mathcal{E}}(\overline{G})$.
- (2) For any functor $H : \mathcal{D} \rightarrow \mathcal{C}$ between small categories $\mathcal{C}$ and $\mathcal{D}$ and any functor $F : \mathcal{C} \rightarrow \mathcal{E}$, we have a natural isomorphism $\overline{F} \circ \overline{H} \cong \overline{F} \circ \gamma_{\mathcal{E}}^*(H)$.
- (3) For any geometric morphism $f : \mathcal{F} \rightarrow \mathcal{E}$, the diagram

$$\begin{array}{ccc} [\mathbb{C}, \mathcal{E}] & \xrightarrow{\tau_{\mathcal{E}}^{\mathbb{C}}} & \mathbf{DOpf}/\mathcal{C} \\ \downarrow f^*(-) & & \downarrow f^*(-) \\ [f^*(\mathbb{C}), \mathcal{F}] & \xrightarrow{\tau_{\mathcal{F}}^{f^*(\mathbb{C})}} & \mathbf{DOpf}/f^*(\mathbb{C}) \end{array}$$

commutes.

We therefore have that

$$\text{colim}(F \circ \pi_P^f) \cong \text{colim}_{\mathcal{E}}(\overline{F} \circ \pi_{P_{\mathcal{E}}}^f) \cong \text{colim}_{\mathcal{E}}(\overline{F} \circ \gamma_{\mathcal{E}}^*(\pi_P^f)) \cong \text{colim}_{\mathcal{E}}(\overline{F} \circ \pi_{P_{\mathcal{E}}}^f),$$

where the first isomorphism follows from (1), the second from (2) and the third from (3). $\square$

We shall denote the three isomorphic objects of Theorem 5.1.5 by the symbol $F \otimes_{\mathcal{C}} P$.

The following lemma is essentially contained in the proof of Giraud's theorem (cf. [42] or, for instance, the Appendix of [63]); we provide a proof of it for the reader's convenience.

Lemma 5.1.6 *Let $\mathcal{E}$ be a Grothendieck topos and $\{e_i : E_i \rightarrow E \mid i \in I\}$ an epimorphic family in $\mathcal{E}$. Then the arrow $\coprod_{i \in I} e_i : \coprod_{i \in I} E_i \rightarrow E$ yields an isomorphism $(\coprod_{i \in I} E_i)/R \cong E$, where R is the equivalence relation in $\mathcal{E}$ on $\coprod_{i \in I} E_i$ given by the subobject $\coprod_{(i,j) \in I \times I} E_i \times_E E_j \rightarrow \coprod_{(i,j) \in I \times I} E_i \times E_j \cong \coprod_{i \in I} E_i \times \coprod_{j \in I} E_j$; in particular, the arrow $\coprod_{i \in I} e_i : \coprod_{i \in I} E_i \rightarrow E$ is the coequalizer in $\mathcal{E}$ of the two canonical arrows $\coprod_{(i,j) \in I \times I} E_i \times_E E_j \rightarrow \coprod_{i \in I} E_i$.*

Proof Let R be the kernel pair of the epimorphism $\coprod_{i \in I} e_i$, that is the pullback of this arrow along itself; then, by the well-known exactness properties of Grothendieck toposes, R is an equivalence relation on $\coprod_{i \in I} E_i$ such that the coequalizer in $\mathcal{E}$ of the two associated projections is isomorphic to $\coprod_{i \in I} e_i$. Now, the fact that pullbacks preserve coproducts in a Grothendieck topos implies that R is isomorphic to the subobject $\coprod_{(i,j) \in I \times I} E_i \times_E E_j \rightarrow \coprod_{(i,j) \in I \times I} E_i \times E_j \cong \coprod_{i \in I} E_i \times \coprod_{j \in I} E_j$, as required. $\square$

Corollary 5.1.7 *Let $a : A \rightarrow E$ and $l : L \rightarrow E$ be objects of the slice topos $\mathcal{E}/E$, and $\{e_i : E_i \rightarrow E \mid i \in I\}$ an epimorphic family in $\mathcal{E}$. Then a family of arrows $\{f_i : e_i^*(a) \rightarrow e_i^*(l) \mid i \in I\}$ in the toposes $\mathcal{E}/E_i$ defines a (unique) arrow $f : a \rightarrow l$ in $\mathcal{E}/E$ such that $e_i^*(f) = f_i$ for all $i \in I$ if and only if for every $i, j \in I$, $q_i^*(f_i) = q_j^*(f_j)$, where the arrows q_i and q_j are defined by the following pullback square:*

$$\begin{array}{ccc} E_{i,j} & \xrightarrow{q_i} & E_i \\ \downarrow q_j & & \downarrow e_i \\ E_j & \xrightarrow{e_j} & E. \end{array}$$

$\square$

5.1.5 $\mathcal{E}$ -final functors

The notions introduced in the following definition will be important in the sequel.

Definition 5.1.8 Let $\underline{\mathcal{A}}_{\mathcal{E}}$ be an $\mathcal{E}$ -indexed category and $i : \underline{\mathcal{B}}_{\mathcal{E}} \rightarrow \underline{\mathcal{A}}_{\mathcal{E}}$ an $\mathcal{E}$ -indexed functor.

- (a) We say that i is $\mathcal{E}$ -final if, for every $E \in \mathcal{E}$ and every $x \in \mathcal{A}_E$, there exists a non-empty set $\mathcal{E}_E^x$ of triples $\{(e_i, b_i, f_i) \mid i \in I\}$, where the family $\{e_i : E_i \rightarrow E \mid i \in I\}$ is epimorphic, b_i is an object of $\mathcal{B}_{E_i}$ and $f_i : \mathcal{A}_{e_i}(x) \rightarrow i_{E_i}(b_i)$ is an arrow in $\mathcal{A}_{E_i}$ (for each $i \in I$), with the property that for any triples $\{(e_i, b_i, f_i) \mid i \in I\}$ and $\{(e'_j, c_j, f'_j) \mid j \in J\}$ in $\mathcal{E}_E^x$ there exist an epimorphic family $\{g_k^{i,j} : E_k^{i,j} \rightarrow E_{i,j} \mid k \in K_{i,j}\}$ in $\mathcal{E}$ and for each $k \in K_{i,j}$ an object $d_k^{i,j}$ of $\mathcal{B}_{E_k^{i,j}}$ and arrows $r_k^{i,j} : \mathcal{B}_{q_i \circ g_k^{i,j}}(b_i) \rightarrow d_k^{i,j}$ and $s_k^{i,j} : \mathcal{B}_{q_j \circ g_k^{i,j}}(c_j) \rightarrow d_k^{i,j}$ in $\mathcal{B}_{E_k^{i,j}}$ such that $i_{E_k^{i,j}}(r_k^{i,j}) \circ \mathcal{A}_{q_i}(f_i) = i_{E_k^{i,j}}(s_k^{i,j}) \circ \mathcal{A}_{q_j}(f'_j)$.
- (b) We say that an $\mathcal{E}$ -indexed subcategory $\underline{\mathcal{B}}_{\mathcal{E}}$ of an $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ is an $\mathcal{E}$ -final subcategory of $\underline{\mathcal{A}}_{\mathcal{E}}$ if the canonical $\mathcal{E}$ -indexed inclusion functor $i : \underline{\mathcal{B}}_{\mathcal{E}} \hookrightarrow \underline{\mathcal{A}}_{\mathcal{E}}$ is $\mathcal{E}$ -final.
- (c) We say that an $\mathcal{E}$ -indexed subcategory $\underline{\mathcal{B}}_{\mathcal{E}}$ of an $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ is an $\mathcal{E}$ -strictly final subcategory of $\underline{\mathcal{A}}_{\mathcal{E}}$ if for every arrow $f : a \rightarrow b$ in $\mathcal{A}_E$ there exists an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ such that the arrow $\mathcal{A}_{e_i}(f) : \mathcal{A}_{e_i}(a) \rightarrow \mathcal{A}_{e_i}(b)$ lies in $\mathcal{B}_{E_i}$ for every $i \in I$.
- (d) We say that an $\mathcal{E}$ -indexed subcategory $\underline{\mathcal{B}}_{\mathcal{E}}$ of an $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ is $\mathcal{E}$ -full if for every arrow $f : b \rightarrow b'$ in $\mathcal{A}_E$, where b and b' are objects of $\mathcal{B}_E$, there exists an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ such that the arrow $\mathcal{A}_{e_i}(f)$ lies in $\mathcal{B}_{E_i}$ for every $i \in I$.

Remarks 5.1.9 (a) Let $\underline{\mathcal{B}}_{\mathcal{E}}$ be an $\mathcal{E}$ -indexed subcategory of $\underline{\mathcal{A}}_{\mathcal{E}}$ with the property that for every $E \in \mathcal{E}$ and $a \in \mathcal{A}_E$ there exists an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ with the property that for every $i \in I$ the object $\mathcal{A}_{e_i}(a)$ lies in $\mathcal{B}_{E_i}$ and for every arrow $f : a \rightarrow b$ in $\mathcal{A}_E$, where $a, b \in \mathcal{B}_E$, there exists an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ such that for every $i \in I$ the arrow $\mathcal{A}_{e_i}(f) : \mathcal{A}_{e_i}(a) \rightarrow \mathcal{A}_{e_i}(b)$ lies in $\mathcal{B}_{E_i}$. Then $\underline{\mathcal{B}}_{\mathcal{E}}$ is an $\mathcal{E}$ -strictly final subcategory of $\underline{\mathcal{A}}_{\mathcal{E}}$.

(b) Every $\mathcal{E}$ -strictly final subcategory is an $\mathcal{E}$ -final subcategory.

(c) If i is an $\mathcal{E}$ -full embedding of an $\mathcal{E}$ -indexed category $\underline{\mathcal{B}}_{\mathcal{E}}$ into an $\mathcal{E}$ -filtered $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ then i is $\mathcal{E}$ -final if and only if for every $E \in \mathcal{E}$ and $x \in \mathcal{A}_E$ there exist a non-empty epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object b_i of $\mathcal{B}_{E_i}$ and an arrow $f_i : \mathcal{A}_{e_i}(x) \rightarrow b_i$ of $\mathcal{A}_{E_i}$.

Proposition 5.1.10 *Let i be the embedding of an $\mathcal{E}$ -full and $\mathcal{E}$ -final subcategory $\underline{\mathcal{B}}_{\mathcal{E}}$ of an $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ into $\underline{\mathcal{A}}_{\mathcal{E}}$. Then $\underline{\mathcal{B}}_{\mathcal{E}}$ is $\mathcal{E}$ -filtered if and only if $\underline{\mathcal{A}}_{\mathcal{E}}$ is $\mathcal{E}$ -filtered.*

Proof The proof is entirely analogous to that of the classical result in ordinary category theory and is left to the reader. $\square$

The following result represents an $\mathcal{E}$ -indexed version of the classical theorem formalizing the behaviour of colimits with respect to final functors:

Theorem 5.1.11 *Let $i : \mathcal{B}_{\mathcal{E}} \rightarrow \mathcal{A}_{\mathcal{E}}$ be an $\mathcal{E}$ -final functor and $D : \mathcal{A}_{\mathcal{E}} \rightarrow \mathcal{E}_{\mathcal{E}}$ an $\mathcal{E}$ -indexed functor. Then D admits a colimit (resp. an $\mathcal{E}$ -indexed colimit) if and only if $D \circ i$ admits a colimit (resp. an $\mathcal{E}$ -indexed colimit) and the two colimits are equal.*

Proof First, let us show that any cocone λ over $D \circ i$ with vertex V can be (uniquely) extended to a cocone $\tilde{\lambda}$ over D . For any $E \in \mathcal{E}$ and $a \in \mathcal{A}_E$, we have to define an arrow $\tilde{\lambda}_E(a) : D_E(a) \rightarrow !^*_E(V)$ in $\mathcal{E}/E$. By our hypotheses there exists a non-empty set $\mathcal{E}_E^a$ of triples $\{(e_i, b_i, f_i) \mid i \in I\}$, where the family $\{e_i : E_i \rightarrow E \mid i \in I\}$ is epimorphic, b_i is an object of $\mathcal{B}_{E_i}$ and $f_i : \mathcal{A}_{e_i}(a) \rightarrow i_{E_i}(b_i)$ is an arrow in $\mathcal{A}_{E_i}$ (for each $i \in I$), with the property that for any triples $\{(e_i, b_i, f_i) \mid i \in I\}$ and $\{(e'_j, c_j, f'_j) \mid j \in J\}$ in $\mathcal{E}_E^a$, there exist an epimorphic family $\{g_k^{i,j} : E_k^{i,j} \rightarrow E_{i,j} \mid k \in K_{i,j}\}$ and for each $k \in K_{i,j}$ an object $d_k^{i,j}$ of $\mathcal{B}_{E_k^{i,j}}$ and arrows $r_k^{i,j} : \mathcal{B}_{q_i \circ g_k^{i,j}}(b_i) \rightarrow d_k^{i,j}$ and $s_k^{i,j} : \mathcal{B}_{q_j \circ g_k^{i,j}}(c_j) \rightarrow d_k^{i,j}$ in $\mathcal{B}_{E_k^{i,j}}$ such that $i_{E_k^{i,j}}(r_k^{i,j}) \circ \mathcal{A}_{q_i}(f_i) = i_{E_k^{i,j}}(s_k^{i,j}) \circ \mathcal{A}_{q_j}(f'_j)$.

Consider, for each $i \in I$, the arrow $\lambda_{E_i}(b_i) \circ D_{E_i}(f_i) : D_{E_i}(\mathcal{A}_{e_i}(a)) = e_i^*(D_E(a)) \rightarrow !^*_{E_i}(V) = e_i^*(!^*_E(V))$. By applying the above property to the pair $((e_i, f_i, b_i), (e_i, f_i, b_i))$ and exploiting the fact that λ is a cocone over $D \circ i$, we obtain that $q_i^*(\lambda_{E_i}(b_i) \circ D_{E_i}(f_i)) = q_j^*(\lambda_{E_j}(b_j) \circ D_{E_j}(f'_j))$ for any $i, j \in I$ (where the notation is that of Corollary 5.1.7). By Corollary 5.1.7, the family of arrows $\lambda_{E_i}(\mathcal{A}_{e_i}(a))$ (for $i \in I$) thus induces a unique arrow $u_{\mathcal{E}} = D_E(a) \rightarrow !^*_E(V)$. To be able to set $\tilde{\lambda}_E(a)$ equal to this arrow, we have to show that such a definition does not depend on the choice of the family $\{(e_i, b_i, f_i) \mid i \in I\}$. But this follows similarly to above, by applying the condition in the definition of an $\mathcal{E}$ -final functor and recalling that λ is a cocone over $D \circ i$.

Notice that if $b \in \mathcal{B}_E$ then $\tilde{\lambda}_E(i_E(b)) = \lambda_E(b)$, and that for any cocone ξ over D , $\xi = \tilde{\xi} \circ i$.

Now that we have shown that for any $a \in \mathcal{A}_E$ the definition of the arrow $\tilde{\lambda}_E(a) : D_E(a) \rightarrow !^*_E(V)$ in the topos $\mathcal{E}/E$ is well-posed, it remains to prove that the assignment $a \rightarrow \tilde{\lambda}_E(a)$ defines a cocone over the diagram D_E with vertex $!^*_E(V)$ i.e. that for any arrow $f : a \rightarrow b$ in $\mathcal{A}_E$, $\tilde{\lambda}_E(b) \circ D_E(f) = \tilde{\lambda}_E(a)$ as arrows in $\mathcal{E}/E$. Further, we have to show that the assignment $E \rightarrow \tilde{\lambda}_E$ defines an $\mathcal{E}$ -indexed cocone on the $\mathcal{E}$ -indexed functor D , i.e. that for any arrow $\alpha : E' \rightarrow E$ in $\mathcal{E}$ and any $a \in \mathcal{A}_E$ we have $\alpha^*(\tilde{\lambda}_E(a)) = \tilde{\lambda}_{E'}(\mathcal{A}_{\alpha}(a))$. But this can be easily deduced from the fact that λ is an $\mathcal{E}$ -indexed cocone over $D \circ i$.

The second part of the theorem, for indexed colimits, follows from the first part by noticing that any localization of an indexed $\mathcal{E}$ -final functor with respect to a localization functor $\mathcal{E} \rightarrow \mathcal{E}/E$ is an $(\mathcal{E}/E)$ -final functor. So it will be sufficient to show the first part.

To complete the proof of the theorem, it thus remains to show that for any cocone (U, μ) over D , (U, μ) is colimiting over D if and only if $(U, \mu \circ i)$ is colimiting over D .

Suppose that D has a colimiting cocone μ with vertex U . We want to prove that $\mu \circ i$ is a colimiting cocone over $D \circ i$. Let λ be a cocone over $D \circ i$ with vertex V ; then, as we have just proved, $\tilde{\lambda}$ is a cocone over D with vertex V and hence $(V, \tilde{\lambda})$ factors through (U, μ) , say by an arrow $z : U \rightarrow V$ in $\mathcal{E}$. Clearly, z yields a factorization of (V, λ) across $(U, \mu \circ i)$. The uniqueness of the factorization of (V, λ) across $(U, \mu \circ i)$ follows from the fact that for any two cocones ξ and χ over $D \circ i$, if ξ factors through χ by an arrow w then $\tilde{\xi}$ factors through $\tilde{\chi}$ by the same arrow. This proves that if (U, μ) is a colimiting cocone over D then $(U, \mu \circ i)$ is a colimiting cocone over $D \circ i$.

Conversely, suppose that $(U, \mu \circ i)$ is a colimiting cocone over $D \circ i$. For any cocone (Z, χ) over D , we have that $(Z, \chi \circ i)$ is a cocone over $D \circ i$; therefore $(Z, \chi \circ i)$ factors uniquely through $(U, \mu \circ i)$ or equivalently (by arguments similar to the above ones), (Z, χ) factors uniquely through (U, μ) . Therefore (U, μ) is a colimiting cocone over D , as required. $\square$

Let $\mathbf{Set}_{\mathcal{E}}$ be the $\mathcal{E}$ -indexed category given by $\mathbf{Set}_E = \mathbf{Set}$ for all $E \in \mathcal{E}$ and $\mathbf{Set}_{\alpha} = 1_{\mathbf{Set}}$ (where $1_{\mathbf{Set}}$ is the identical functor on $\mathbf{Set}$) for all arrows α in $\mathcal{E}$. There is an $\mathcal{E}$ -indexed functor

$$\gamma_{\mathcal{E}} : \mathbf{Set}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$$

defined by setting $\gamma_{\mathcal{E}E} = \gamma_{\mathcal{E}/E}^* : \mathbf{Set} \rightarrow \mathcal{E}/E$ (for any object E of $\mathcal{E}$).

We shall consider $\mathcal{E}$ -indexed functors obtained by composing the $\mathcal{E}$ -indexed functor $\gamma_{\mathcal{E}} : \mathbf{Set}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ with an $\mathcal{E}$ -indexed functor $D : \mathcal{A}_{\mathcal{E}} \rightarrow \mathbf{Set}_{\mathcal{E}}$. Notice that an $\mathcal{E}$ -indexed functor $D : \mathcal{A}_{\mathcal{E}} \rightarrow \mathbf{Set}_{\mathcal{E}}$ consists of a functor $D_E : \mathcal{A}_E \rightarrow \mathbf{Set}$ for each object E of $\mathcal{E}$ such that, for any arrow $\alpha : E' \rightarrow E$ in $\mathcal{E}$, $D_{E'} \circ \mathcal{A}_{\alpha} = D_E$.

Let us give an explicit description of cocones over the $\mathcal{E}$ -indexed functor $\gamma_{\mathcal{E}} \circ D$. Specializing the general definition, we obtain that a cocone (U, μ) over $\gamma_{\mathcal{E}} \circ D$ consists of an object U of $\mathcal{E}$ and of an arrow $\mu_E(a) : \gamma_{\mathcal{E}/E}^*(D_E(a)) \rightarrow !_{E}^*(U)$ in $\mathcal{E}/E$ (for any objects $E \in \mathcal{E}$ and $a \in \mathcal{A}_E$) such that, for any arrow $f : a \rightarrow b$ in $\mathcal{A}_E$, $\mu_E(b) \circ \gamma_{\mathcal{E}/E}^*(D_E(f)) = \mu_E(a)$ and, for any arrow $\alpha : E' \rightarrow E$ in $\mathcal{E}$, $\alpha^*(\mu_E(a)) = \mu_{E'}(\mathcal{A}_{\alpha}(a))$.

By using the well-known adjunction between $\gamma_{\mathcal{E}}^*$ and the global sections functor $\mathcal{E} \rightarrow \mathbf{Set}$, we can alternatively present the above set of data as follows: a cocone (U, μ) over $\gamma_{\mathcal{E}} \circ D$ consists of an object U of $\mathcal{E}$ and of a function $\mu_E(a) : D_E(a) \rightarrow \mathrm{Hom}_{\mathcal{E}}(E, U)$ (for any objects E of $\mathcal{E}$ and a of $\mathcal{A}_E$) such that, for any arrow $f : a \rightarrow b$ in $\mathcal{A}_E$, $\mu_E(b) \circ D_E(f) = \mu_E(a)$ and, for any arrow $\alpha : E' \rightarrow E$ in $\mathcal{E}$, $\mathrm{Hom}_{\mathcal{E}}(\alpha, U) \circ \mu_E(a) = \mu_{E'}(\mathcal{A}_{\alpha}(a))$ (notice that the domains of these two arrows are the same since $D_E(a) = D_{E'}(\mathcal{A}_{\alpha}(a))$).

Theorem 5.1.12 *Let $\mathcal{C}$ be a small category, $\mathcal{E}$ a Grothendieck topos and $F : \mathcal{C}^{\mathrm{op}} \rightarrow \mathcal{E}$ a functor. Then the $\mathcal{E}$ -indexed subcategory $\int F_{\mathcal{E}}$ of $\int \overline{F}_{\mathcal{E}}^f$ is $\mathcal{E}$ -strictly final.*

Proof By definition of $\int \overline{F}^f_{\mathcal{E}}$, for any object E of $\mathcal{E}$ the category $(\int \overline{F}^f)_E$ has as objects the arrows $x : E \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} F(c)$ and as arrows $x \rightarrow x'$ the arrows $z : E \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f))$ such that $d_0^F \circ z = x$ and $d_1^F \circ z = x'$. For any $E \in \mathcal{E}$,

the category $(\int F)_E$ thus identifies with a subcategory of the category $(\int \overline{F}^f)_E$, through the assignment sending any pair (c, x) , where $x : E \rightarrow F(c)$, to the arrow $\mu_c \circ x : E \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} F(c)$ and any arrow $(c, x) \rightarrow (c', x')$ to the arrow

$$J_f \circ x' : E \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f)) \quad (\text{where the notation is that of section 5.1.4}).$$

These assignments clearly make $\int F_{\mathcal{E}}$ into an $\mathcal{E}$ -indexed subcategory of $\int \overline{F}^f_{\mathcal{E}}$.

To prove our thesis, we shall apply the criterion of Remark 5.1.9(a). To show that the $\mathcal{E}$ -indexed subcategory $\int F_{\mathcal{E}}$ satisfies the first condition in the Remark, we observe that for any object $x : E \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} F(c)$ of the category $(\int \overline{F}^f)_E$, if we consider the pullbacks

$$\begin{array}{ccc} E_c & \xrightarrow{x_c} & F(c) \\ \downarrow e_c & & \downarrow \mu_c \\ E & \xrightarrow{x} & \coprod_{c \in \text{Ob}(\mathcal{C})} F(c) \end{array}$$

of x along the coproduct arrows $\mu_c : F(c) \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} F(c)$, we obtain an epimorphic family $\{e_c : E_c \rightarrow E \mid c \in \text{Ob}(\mathcal{C})\}$ such that $(\int \overline{F}^f)_{e_c}(x)$ lies in the subcategory $(\int F)_{E_c}$ for each $c \in \text{Ob}(\mathcal{C})$.

It remains to prove that the second condition of Remark 5.1.9(a) is satisfied. Suppose that (c, x) and (c', x') are objects of the subcategory $(\int F)_E$ and that $\alpha : E \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f))$ is an arrow such that $d_0^F \circ \alpha = \mu_c \circ x$ and $d_1^F \circ \alpha = \mu_{c'} \circ x'$:

$$\begin{array}{ccc} E & \xrightarrow{\alpha} & \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f)) \\ \downarrow x & & \downarrow d_0^F \\ F(c) & \xrightarrow{\mu_c} & \coprod_{c \in \text{Ob}(\mathcal{C})} F(c), \end{array} \quad \begin{array}{ccc} E & \xrightarrow{\alpha} & \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f)) \\ \downarrow x' & & \downarrow d_1^F \\ F(c') & \xrightarrow{\mu_{c'}} & \coprod_{c \in \text{Ob}(\mathcal{C})} F(c). \end{array}$$

Consider, for each $f \in \text{Arr}(\mathcal{C})$, the following pullback square:

$$\begin{array}{ccc}
E_f & \xrightarrow{\alpha_f} & F(\text{cod}(f)) \\
\downarrow e_f & & \downarrow J_f \\
E & \xrightarrow{\alpha} & \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f)).
\end{array}$$

The commutativity of the three diagrams above, combined with the fact that distinct coproduct arrows are disjoint, implies that for any arrow f of $\mathcal{C}$ such that $\text{dom}(f) \neq c$ or $\text{cod}(f) \neq c'$ we have $E_f \cong 0_{\mathcal{E}}$. We can therefore restrict our attention to the arrows f such that $\text{dom}(f) = c$ and $\text{cod}(f) = c'$. For any such arrow f , we have $\alpha_f = x' \circ e_f$ (equivalently, since $\mu_{c'} = \mu_{\text{cod}(f)}$ is monic, $\mu_{c'} \circ \alpha_f = \mu_{c'} \circ x' \circ e_f$). Indeed, $\mu_{c'} \circ \alpha_f = d_1^F \circ J_f \circ \alpha_f = d_1^F \circ \alpha \circ e_f = \mu_{c'} \circ x' \circ e_f$. Therefore $\alpha \circ e_f = J_f \circ x' \circ e_f$, since $\alpha \circ e_f = J_f \circ \alpha_f = J_f \circ x' \circ e_f$. Lastly, we observe that f defines an arrow $(c, x \circ e_f) \rightarrow (c', x' \circ e_f)$ in the subcategory $(\int F)_{E_f}$, i.e. $F(f) \circ x' \circ e_f = x \circ e_f$. Indeed, the arrow $\mu_{\text{dom}(f)} = \mu_c$ is monic and we have $\mu_{\text{dom}(f)} \circ F(f) \circ x' \circ e_f = d_0^F \circ J_f \circ x' \circ e_f = J_f \circ \alpha \circ e_f = \mu_c \circ x \circ e_f$.

The epimorphic family $\{e_f : E_f \rightarrow E \mid \text{dom}(f) = c \text{ and } \text{cod}(f) = c'\}$ thus satisfies the property that the arrow $(\int \overline{F})_{e_f}(\alpha)$ lies in the subcategory $(\int F)_{E_f}$ for each $f : c \rightarrow c'$. This completes our proof. $\square$

Theorem 5.1.13 *Let $\mathcal{C}$ be a small category, $\mathcal{E}$ a Grothendieck topos, $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ and $P : \mathcal{C} \rightarrow \mathbf{Set}$ functors and $P_{\mathcal{E}}$ the internal diagram in $[\mathcal{C}, \mathcal{E}]$ given by $\gamma_{\mathcal{E}}^* \circ \overline{P}$. Then the restriction of the $\mathcal{E}$ -indexed functor $(\underline{P}_{\mathcal{E}} \circ \pi_{\overline{F}}^f) \cong \underline{P}_{\mathcal{E}} \circ \pi_{\overline{F}}^f$ to the $\mathcal{E}$ -indexed subcategory $\int \underline{F}_{\mathcal{E}}$ of $\int \overline{F}_{\mathcal{E}}^f$ is naturally isomorphic to $\gamma_{\mathcal{E}} \circ z_{\mathcal{E}}$, where $z_{\mathcal{E}} : \int \underline{F}_{\mathcal{E}} \rightarrow \mathbf{Set}_{\mathcal{E}}$ is the $\mathcal{E}$ -indexed functor defined by setting $(z_{\mathcal{E}})_E((c, x)) = P(c)$ and $(z_{\mathcal{E}})_E(f) = P(f)$ (for any $E \in \mathcal{E}$, object (c, x) and arrow f of the category $(\int F)_E$).*

Proof We shall exhibit an isomorphism

$$(\underline{P}_{\mathcal{E}} \circ \pi_{\overline{F}}^f)_E((c, x)) \cong (\gamma_{\mathcal{E}} \circ z_{\mathcal{E}})_E((c, x))$$

natural in $E \in \mathcal{E}$ and $(c, x) \in (\int F)_E$.

Consider the following pullback square in $\mathbf{cat}(\mathcal{E})$:

$$\begin{array}{ccc}
\overline{F} \otimes_{\mathcal{C}} P & \xrightarrow{t_P} & \int^{\text{opf}} P_{\mathcal{E}} \\
\downarrow t_{\overline{F}} & & \downarrow \pi_{P_{\mathcal{E}}}^{\text{opf}} \\
\int^f \overline{F} & \xrightarrow{\pi_{\overline{F}}} & \mathcal{C}.
\end{array}$$

Since coproducts are stable under pullback in any Grothendieck topos, we have that

$$(\overline{F} \otimes_{\mathbb{C}} P)_0 = \coprod_{c \in \text{Ob}(\mathcal{C})} F(c) \times \gamma_{\mathcal{E}}^*(P(c)),$$

$$(\overline{F} \otimes_{\mathbb{C}} P)_1 = \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f)) \times \gamma_{\mathcal{E}}^*(P(\text{dom}(f))),$$

$$(t_{\overline{F}})_0 : (\overline{F} \otimes_{\mathbb{C}} P)_0 \rightarrow \left(\int^f \overline{F} \right)_0 = \coprod_{c \in \text{Ob}(\mathcal{C})} F(c)$$

is equal to the arrow $\coprod_{c \in \text{Ob}(\mathcal{C})} \pi_{F(c)}$, where $\pi_{F(c)}$ is the canonical projection $F(c) \times \gamma_{\mathcal{E}}^*(P(c)) \rightarrow F(c)$, $(t_{\overline{F}})_1 : (\overline{F} \otimes_{\mathbb{C}} P)_1 \rightarrow \left(\int^f \overline{F} \right)_1 = \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f))$ is equal to the arrow $\coprod_{f \in \text{Arr}(\mathcal{C})} \pi'_{F(\text{cod}(f))}$, where $\pi'_{F(\text{cod}(f))}$ is the canonical projection $F(\text{cod}(f)) \times \gamma_{\mathcal{E}}^*(P(\text{dom}(f))) \rightarrow F(\text{cod}(f))$, and the domain and codomain arrows

$$d_0^{\overline{F} \otimes_{\mathbb{C}} P}, d_1^{\overline{F} \otimes_{\mathbb{C}} P} : (\overline{F} \otimes_{\mathbb{C}} P)_1 \rightarrow (\overline{F} \otimes_{\mathbb{C}} P)_0$$

are defined by the following conditions:

$$d_0^{\overline{F} \otimes_{\mathbb{C}} P} \circ W_f = Z_{\text{cod}(f)} \circ \pi'_{F(\text{cod}(f))} \circ \gamma_{\mathcal{E}}^*(P(f)) \circ \mu_{\gamma_{\mathcal{E}}^*(P(\text{dom}(f)))} >$$

and

$$d_1^{\overline{F} \otimes_{\mathbb{C}} P} \circ W_f = Z_{\text{dom}(f)} \circ F(f) \circ \pi'_{F(\text{cod}(f))} \circ \mu_{\gamma_{\mathcal{E}}^*(P(\text{dom}(f)))} >,$$

where $\mu_{\gamma_{\mathcal{E}}^*(P(\text{dom}(f)))}$ is the canonical projection

$$F(\text{cod}(f)) \times \gamma_{\mathcal{E}}^*(P(\text{dom}(f))) \rightarrow \gamma_{\mathcal{E}}^*(P(\text{dom}(f)))$$

and

$$W_f : F(\text{cod}(f)) \times \gamma_{\mathcal{E}}^*(P(\text{dom}(f))) \rightarrow \coprod_{f \in \text{Arr}(\mathcal{C})} F(\text{cod}(f)) \times \gamma_{\mathcal{E}}^*(P(\text{dom}(f))),$$

$$Z_c : F(c) \times \gamma_{\mathcal{E}}^*(P(c)) \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} F(c) \times \gamma_{\mathcal{E}}^*(P(c))$$

are the canonical coproduct arrows (resp. for $f \in \text{Arr}(\mathcal{C})$ and $c \in \text{Ob}(\mathcal{C})$).

Now, the internal diagram $P_{\mathcal{E}} \circ \pi_{\overline{F}}^f$ corresponds precisely to the discrete op-fibration $t_{\overline{F}}$ and hence the $\mathcal{E}$ -indexed functor $P_{\mathcal{E}} \circ \pi_{\overline{F}}^f$ sends any generalized element $x : E \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} F(c)$ to the object of $\mathcal{E}/\overline{E}$ given by the pullback of $(t_{\overline{F}})_0$ along it. In particular, since in a Grothendieck topos any diagram of the form

$$\begin{array}{ccc} A_i & \longrightarrow & \coprod_{i \in I} A_i \\ \downarrow f_i & & \downarrow \coprod_{i \in I} f_i \\ B_i & \longrightarrow & \coprod_{i \in I} B_i, \end{array}$$

where the horizontal arrows are the canonical coproduct arrows, is a pullback, and products commute with pullbacks, the functor $(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f)_E$ sends any object (c, x) of the category $(\int F)_E$ to the object given by the canonical projection $E \times \gamma_{\mathcal{E}}^*(P(c)) \rightarrow E$. But this object is canonically isomorphic to $\gamma_{\mathcal{E}/E}^*(P(c))$ in $\mathcal{E}/E$, that is to the value at (c, x) of the functor $(\gamma_{\mathcal{E}} \circ z_{\mathcal{E}})_E$. $\square$

Let us now explicitly describe the $\mathcal{E}$ -indexed colimiting cocone on the restriction to the $\mathcal{E}$ -indexed subcategory $\int \overline{F}_{\mathcal{E}}$ of the $\mathcal{E}$ -indexed diagram $P_{\mathcal{E}} \circ \pi_{\overline{F}}^f$. The vertex of this colimiting cocone is the codomain of the coequalizer $w : (\overline{F} \otimes_{\mathbb{C}} P)_0 \rightarrow \text{colim}_{\mathcal{E}}(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f)$ of the pair of arrows $d_0^{\overline{F} \otimes_{\mathbb{C}} P}, d_1^{\overline{F} \otimes_{\mathbb{C}} P} : (\overline{F} \otimes_{\mathbb{C}} P)_1 \rightarrow (\overline{F} \otimes_{\mathbb{C}} P)_0$. For any object (c, x) of the category $(\int \overline{F})_E$, the colimit arrow $(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f)_E((c, x)) \cong \gamma_{\mathcal{E}/E}^*(P(c)) \rightarrow \text{colim}_{\mathcal{E}}(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f)$ is given by the composite $!_E^*(w) \circ h_{(c, x)}$, where $h_{(c, x)} = !_E^*(Z_c) \circ < x \times 1_{\gamma_{\mathcal{E}}^*(P(c))}, \pi_E > : \gamma_{\mathcal{E}/E}^*(P(c)) \cong E \times \gamma_{\mathcal{E}}^*(P(c)) \rightarrow \coprod_{c \in \text{Ob}(\mathcal{C})} F(c) \times \gamma_{\mathcal{E}}^*(P(c)) \times E \cong !_E^*(\coprod_{c \in \text{Ob}(\mathcal{C})} F(c) \times \gamma_{\mathcal{E}}^*(P(c)))$; in other words, the function $\xi_{(c, x)} : P(c) \rightarrow \text{Hom}_{\mathcal{E}}(E, \text{colim}_{\mathcal{E}}(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f))$ corresponding to it assigns to every element $a \in P(c)$ the arrow $w \circ Z_c \circ < x, r_a \circ !_E >$, where $r_a : 1_{\mathcal{E}} \rightarrow \gamma_{\mathcal{E}}^*(P(c)) \cong \coprod_{a \in P(c)} 1_{\mathcal{E}}$ is the coproduct arrow corresponding to the element a .

It is immediate to verify that, under the isomorphism

$$\text{colim}_{\mathcal{E}}(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f) \cong \text{colim}(F \circ \pi_P^f)$$

of Theorem 5.1.5, the functions

$$\xi_{(c, x)} : P(c) \rightarrow \text{Hom}_{\mathcal{E}}(E, \text{colim}_{\mathcal{E}}(P_{\mathcal{E}} \circ \pi_{\overline{F}}^f)) \cong \text{Hom}_{\mathcal{E}}(E, \text{colim}(F \circ \pi_P^f))$$

admit the following description in terms of the colimit arrows $\chi_{(c, a)} : F(c) \rightarrow \text{colim}(F \circ \pi_P^f)$: for any $a \in P(c)$, $\xi_{(c, x)}(a) = \chi_{(c, a)} \circ x$.

5.1.6 A characterization of $\mathcal{E}$ -indexed colimits

In this section we establish, for a Grothendieck topos $\mathcal{E}$, necessary and sufficient conditions for an $\mathcal{E}$ -indexed cocone on a small diagram in $\mathcal{E}$ to be an ($\mathcal{E}$ -indexed) colimit.

Before stating and proving the main theorem of this section we need to introduce some relevant definitions and a few technical lemmas.

Given an $\mathcal{E}$ -indexed functor $D : \mathcal{A}_{\mathcal{E}} \rightarrow \mathcal{E}_{\mathcal{E}}$ and an object R of $\mathcal{E}$, we denote by $\mathcal{I}_R^D$ the collection of pairs of the form (x, y) , where x is an object of the category $\mathcal{A}_R$ and y is an arrow $1_{\mathcal{E}/R} \rightarrow D_R(x)$ in the topos $\mathcal{E}/R$. On such a set we consider the equivalence relation $\mathcal{R}_R^D$ generated by the pairs $((x, y), (x', y'))$ with the property that there exists an arrow $f : x \rightarrow x'$ in the category $\mathcal{A}_R$ such that $y' = D_R(f) \circ y$.

The following remarks are useful in connection with the application of the localization technique.

- Remarks 5.1.14** (a) For any $E \in \mathcal{E}$ we have a natural bijection $\mathcal{I}_E^D \cong \mathcal{I}_{1_{\mathcal{E}/E}}^{D/E}$ and the relation $\mathcal{R}_E^D$ corresponds to the relation $\mathcal{R}_{1_{\mathcal{E}/E}}^{D/E}$ under this bijection.
- (b) For any arrow $f : F \rightarrow E$ in $\mathcal{E}$ and any object $x \in \mathcal{A}_E$, $D_F(\mathcal{A}_f(x)) = f^*(D_E(x))$ (since D is an $\mathcal{E}$ -indexed functor). We shall denote the canonical arrow $\text{dom}(D_F(\mathcal{A}_f(x))) \rightarrow \text{dom}(D_E(x))$ in $\mathcal{E}$ by the symbol r_f .
- (c) For any object x of $\mathcal{A}_E$ and any arrow $y : (f : F \rightarrow E) \rightarrow D_E(x)$ in the topos $\mathcal{E}/E$, there exists a unique arrow $y_f : (1_F : F \rightarrow F) \rightarrow D_F(\mathcal{A}_f(x))$ in the topos $\mathcal{E}/F$ such that $r_f \circ y_f = y$ (where r_f is regarded here as an arrow in $\mathcal{E}/E$ in the obvious way).
- (d) For any $\mathcal{E}$ -indexed cocone μ over D with vertex U , any object x of $\mathcal{A}_E$ and any arrow $y : (f : F \rightarrow E) \rightarrow D_E(x)$ in the topos $\mathcal{E}/E$, we have $\mu_E(x) \circ y = (1_U \times f) \circ \mu_F(\mathcal{A}_f(x)) \circ y_f$. Moreover, we have pullback squares

$$\begin{array}{ccc}
 D_F(\mathcal{A}_f(x)) & \xrightarrow{r_f} & D_E(x) \\
 \downarrow \mu_F(\mathcal{A}_f(x)) & & \downarrow \mu_E(x) \\
 U \times F & \xrightarrow{1_U \times f} & U \times E \\
 \downarrow \pi_F & & \downarrow \pi_E \\
 F & \xrightarrow{f} & E
 \end{array}$$

in $\mathcal{E}$, where π_E and π_F are the obvious canonical projections. In particular, for any other such pair (x', y') , $\mu_E(x) \circ y = \mu_E(x') \circ y'$ if and only if $\mu_F(\mathcal{A}_f(x)) \circ y_f = \mu_F(\mathcal{A}_f(x')) \circ y'_f$ (where y_f is the arrow defined in Remark 5.1.14(c)).

- (e) If $\mathcal{A}_{\mathcal{E}}$ is the $\mathcal{E}$ -externalization of an internal category $\mathbb{C}$ in $\mathcal{E}$, x is a generalized element $E \rightarrow \mathbb{C}_0$ and y is an arrow $(f : F \rightarrow E) \rightarrow D_E(x)$ in $\mathcal{E}/E$, there exists a unique arrow $y_f : (1_F : F \rightarrow F) \rightarrow D_F(x \circ f)$ in $\mathcal{E}/F$ (provided by the universal property of the pullback square defining $D_F(x \circ f)$) such that $z_{x \circ f} \circ y_f = z_x \circ y$ (cf. section 5.1.3 for the notation), which coincides with the arrow y_f defined in Remark 5.1.14(c).
- (f) If a pair $((x, y), (x', y'))$ belongs to $\mathcal{R}_R^D$ then, for any $\mathcal{E}$ -indexed cocone λ over D with vertex V , $\lambda_R(x) \circ y = \lambda_R(x') \circ y'$. Indeed, for any pair $((x, y), (x', y'))$ with the property that there exists an arrow $f : x \rightarrow x'$ in the category $\mathcal{A}_R$ such that $y' = D_E(f) \circ y$, we have that $\lambda_R(x') \circ y' = \lambda_R(x') \circ D_E(f) \circ y = \lambda_R(x) \circ y$.

Lemma 5.1.15 *Let $\mathcal{E}$ be a Grothendieck topos, $\mathbb{C}$ an internal category in $\mathcal{E}$, $D \in [\mathbb{C}, \mathcal{E}]$ an internal diagram, $x : 1_{\mathcal{E}} \rightarrow \mathbb{C}_0$ and $f : 1_{\mathcal{E}} \rightarrow \mathbb{C}_1$ generalized elements such that $d_0^{\mathbb{C}} \circ f = x$ and $m : F \rightarrow D_{1_{\mathcal{E}}}(x)$ an arrow in $\mathcal{E}$. Then the unique arrow $\chi_{m,f} : F \rightarrow (\int^{\text{opf}} D)_1$ in $\mathcal{E}$ such that $d_0^{\int^{\text{opf}} D} \circ \chi_{m,f} = z_x \circ m$ and $(\pi_D^{\text{opf}})_1 \circ \chi_{m,f} = f \circ !_F$ (provided by the universal property of the pullback square arising in the definition of the discrete opfibration associated with D – cf.*

section 5.1.3 for the notation) satisfies the property that $z_x \circ m = d_0^{\int^{\text{opf}} D} \circ \chi_{m,f}$ (by definition) and $z_{x'} \circ D_1(f) \circ m = d_1^{\int^{\text{opf}} D} \circ \chi_{m,f}$, where x' is equal to $d_1^{\mathbb{C}} \circ f$ and $d_0^{\int^{\text{opf}} D}, d_1^{\int^{\text{opf}} D}$ are respectively the domain and codomain arrows $(\int^{\text{opf}} D)_1 \rightarrow (\int^{\text{opf}} D)_0$.

Proof By definition of $D_1(f)$ we have that $z_{x'} \circ D_1(f) = d_1^{\int^{\text{opf}} D} \circ r \circ \xi$, where r is defined by the pullback square

$$\begin{array}{ccc} H & \xrightarrow{r} & (\int^{\text{opf}} D)_1 \\ \downarrow !_H & & \downarrow (\pi_D^{\text{opf}})_1 \\ 1_{\mathcal{E}} & \xrightarrow{f} & \mathbb{C}_1 \end{array}$$

and ξ is the unique arrow $D_1(x) \rightarrow H$ such that $z_x = d_0^{\int^{\text{opf}} D} \circ r \circ \xi$, provided by the universal property of the pullback square

$$\begin{array}{ccc} H & \xrightarrow{d_0^{\int^{\text{opf}} D} \circ r} & (\int^{\text{opf}} D)_0 \\ \downarrow s & & \downarrow (\pi_D^{\text{opf}})_0 \\ 1_{\mathcal{E}} & \xrightarrow{d_0^{\mathbb{C}} \circ f} & \mathbb{C}_0. \end{array}$$

It thus remains to show that $\chi_{m,f} = r \circ \xi \circ m$. But this immediately follows from the universal property of the pullback square

$$\begin{array}{ccc} (\int^{\text{opf}} D)_1 & \xrightarrow{d_0^{\int^{\text{opf}} D}} & (\int^{\text{opf}} D)_0 \\ \downarrow (\pi_D^{\text{opf}})_1 & & \downarrow (\pi_D^{\text{opf}})_0 \\ \mathbb{C}_1 & \xrightarrow{d_0^{\mathbb{C}}} & \mathbb{C}_0, \end{array}$$

since $d_0^{\int^{\text{opf}} D} \circ \chi_{m,f} = d_0^{\int^{\text{opf}} D} \circ (r \circ \xi \circ m)$ and $(\pi_D^{\text{opf}})_1 \circ \chi_{m,f} = (\pi_D^{\text{opf}})_1 \circ (r \circ \xi \circ m)$. $\square$

Lemma 5.1.16 *Let $\mathcal{E}$ be a Grothendieck topos, $\mathbb{C}$ an internal category in $\mathcal{E}$, D an internal diagram in $[\mathbb{C}, \mathcal{E}]$ and χ an arrow $1_{\mathcal{E}} \rightarrow (\int^{\text{opf}} D)_1$. Let $x = (\pi_D^{\text{opf}})_0 \circ d_0^{\int^{\text{opf}} D} \circ \chi$, $x' = (\pi_D^{\text{opf}})_0 \circ d_1^{\int^{\text{opf}} D} \circ \chi$ and $f = (\pi_D^{\text{opf}})_1 \circ \chi$. Then $d_0^{\mathbb{C}} \circ f = x$, $d_1^{\mathbb{C}} \circ f = x'$, and $D_1(f) \circ \chi_0 = \chi_1$, where χ_0 is the unique arrow $1_{\mathcal{E}} \rightarrow D_1(x)$ such that $z_x \circ \chi_0 = d_0^{\int^{\text{opf}} D} \circ \chi$ (which exists by the universal property of the pullback square defining $D_1(x)$) and χ_1 is the unique arrow $1_{\mathcal{E}} \rightarrow D_1(x')$ such that $z_{x'} \circ \chi_1 = d_1^{\int^{\text{opf}} D} \circ \chi$ (which exists by the universal property of the pullback square defining $D_1(x')$).*

Proof The first two identities follow immediately from the definition of π_D^{opf} . It remains to prove that $D_1(f) \circ \chi_0 = \chi_1$. Let us employ the notation of the proof of Lemma 5.1.15. By the universal property of the pullback square defining $D_1(x')$, it is equivalent to verify that $z_{x'} \circ D_1(f) \circ \chi_0 = z_{x'} \circ \chi_1$. But $z_{x'} \circ D_1(f) \circ \chi_0 = d_1^{\int^{\text{opf}} D} \circ r \circ \xi \circ \chi_0$, while $z_{x'} \circ \chi_1 = d_1^{\int^{\text{opf}} D} \circ \chi$; so to prove the desired equality it suffices to show that $r \circ \xi \circ \chi_0 = \chi$. This follows from the fact that $d_0^{\int^{\text{opf}} D} \circ r \circ \xi \circ \chi_0 = z_x \circ \chi_0 = d_0^{\int^{\text{opf}} D} \circ \chi$ by virtue of the universal property of the pullback square given by the domain of the discrete opfibration associated with D . $\square$

Proposition 5.1.17 *Let $\mathcal{E}$ be a Grothendieck topos, $\mathbb{C}$ an internal category in $\mathcal{E}$, $D \in [\mathbb{C}, \mathcal{E}]$ an internal diagram, $x, x' : 1_{\mathcal{E}} \rightarrow \mathbb{C}_0$ generalized elements and $y : 1_{\mathcal{E}} \rightarrow D_1(x)$, $y' : 1_{\mathcal{E}} \rightarrow D_1(x')$ arrows in $\mathcal{E}$. Then $(z_x \circ y, z_{x'} \circ y')$ belongs to the equivalence relation on $\text{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, (\int^{\text{opf}} D)_0)$ generated by the pairs of the form $(d_0^{\int^{\text{opf}} D} \circ a, d_1^{\int^{\text{opf}} D} \circ a)$ for some arrow $a : 1_{\mathcal{E}} \rightarrow (\int^{\text{opf}} D)_1$ if and only if $((x, y), (x', y'))$ belongs to $\mathcal{R}_{1_{\mathcal{E}}}^D$.*

Proof Let us first prove the ‘if’ direction. We have that $((x, y), (x', y'))$ belongs to $\mathcal{R}_{1_{\mathcal{E}}}^D$ if and only if there exists a finite sequence of pairs

$$(x_0, y_0) = (x, y), \dots, (x_n, y_n) = (x', y')$$

such that for any $i \in \{0, \dots, n-1\}$

- (1) either there exists $f_i : 1_{\mathcal{E}} \rightarrow \mathbb{C}_1$ such that $d_0^{\mathbb{C}} \circ f_i = x_i$, $d_1^{\mathbb{C}} \circ f_i = x_{i+1}$ and $D_1(f_i) \circ y_i = y_{i+1}$ or
- (2) there exists $g_i : 1_{\mathcal{E}} \rightarrow \mathbb{C}_1$ such that $d_0^{\mathbb{C}} \circ g_i = x_{i+1}$, $d_1^{\mathbb{C}} \circ g_i = x_i$ and $D_1(g_i) \circ y_{i+1} = y_i$.

In case (1), Lemma 5.1.15 implies (by taking m to be y_i and f to be f_i) that $z_{x_i} \circ y_i = d_0^{\int^{\text{opf}} D} \circ \chi_{y_i, f_i}$ and $z_{x_{i+1}} \circ y_{i+1} = d_1^{\int^{\text{opf}} D} \circ \chi_{y_i, f_i}$ (here the notation is that of the Lemma). In case (2), Lemma 5.1.15 implies (by taking m to be y_{i+1} and f to be g_i) that $z_{x_{i+1}} \circ y_{i+1} = d_0^{\int^{\text{opf}} D} \circ \chi_{y_{i+1}, g_i}$ and $z_{x_i} \circ y_i = d_1^{\int^{\text{opf}} D} \circ \chi_{y_{i+1}, g_i}$.

From this it clearly follows that $(z_x \circ y, z_{x'} \circ y')$ belongs to the equivalence relation T on $\text{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, (\int^{\text{opf}} D)_0)$ generated by the pairs of the form $(d_0^{\int^{\text{opf}} D} \circ a, d_1^{\int^{\text{opf}} D} \circ a)$ for some arrow $a : 1_{\mathcal{E}} \rightarrow (\int^{\text{opf}} D)_1$, as required.

Let us now prove the ‘only if’ direction. If $(z_x \circ y, z_{x'} \circ y')$ belongs to T , there exists a finite sequence $\xi_0, \dots, \xi_n$ of arrows $1_{\mathcal{E}} \rightarrow (\int^{\text{opf}} D)_1$ and of arrows $e_0, \dots, e_n$ (which are either $d_0^{\int^{\text{opf}} D}$ or $d_1^{\int^{\text{opf}} D}$ – we denote by e_i^{op} the arrow $d_1^{\int^{\text{opf}} D}$ if e_i is $d_0^{\int^{\text{opf}} D}$ and the arrow $d_0^{\int^{\text{opf}} D}$ if e_i is $d_1^{\int^{\text{opf}} D}$) such that $e_0 \circ \xi_0 = z_x \circ y$, $e_{i+1} \circ \xi_{i+1} = e_i^{\text{op}} \circ \xi_i$ for all $i \in \{0, \dots, n-1\}$ and $e_n \circ \xi_n = z_{x'} \circ y'$.

To deduce our thesis, it suffices to apply Lemma 5.1.16 noticing that if $e_i \circ \chi = z_{x''} \circ y''$ then $\chi_0 = y''$ if e_i is $d_0^{\int^{\text{opf}} D}$ and $\chi_1 = y''$ if e_i is $d_1^{\int^{\text{opf}} D}$ (where χ_0 and

χ_1 are the arrows defined in the statement of Lemma 5.1.16) since $z_{x''} \circ \chi_0 = d_0^{\int^{\text{opf}} D} \circ \chi = z_{x''} \circ y''$ (if e_i is $d_0^{\int^{\text{opf}} D}$) and $z_{x''} \circ \chi_1 = d_1^{\int^{\text{opf}} D} \circ \chi = z_{x''} \circ y''$ (if e_i is $d_1^{\int^{\text{opf}} D}$). $\square$

Theorem 5.1.18 *Let $D : \underline{\mathcal{A}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ be an $\mathcal{E}$ -indexed functor, where $\underline{\mathcal{A}}_{\mathcal{E}}$ is equivalent to the $\mathcal{E}$ -externalization of an internal category in $\mathcal{E}$. Then an $\mathcal{E}$ -indexed cocone μ over D with vertex U is an indexed colimiting cocone for D if and only if the following conditions are satisfied:*

- (i) *For any arrow $h : F \rightarrow U$ in $\mathcal{E}$, there exist an epimorphic family $\{f_i : F_i \rightarrow F \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object $x_i \in \mathcal{A}_{F_i}$ and an arrow $\alpha_i : 1_{\mathcal{E}/F_i} \rightarrow D_{F_i}(x_i)$ in the topos $\mathcal{E}/F_i$ such that $\langle h \circ f_i, 1_{F_i} \rangle = \mu_{F_i}(x_i) \circ \alpha_i$ as arrows $1_{\mathcal{E}/F_i} \rightarrow F_i^*(U)$ in $\mathcal{E}/F_i$.*
- (ii) *For any pairs (x, y) and (x', y') , where x and x' are objects of $\mathcal{A}_E$, y is an arrow $(f : F \rightarrow E) \rightarrow D_E(x)$ in $\mathcal{E}/E$ and y' is an arrow $(f : F \rightarrow E) \rightarrow D_E(x')$ in $\mathcal{E}/E$, we have $\mu_E(x) \circ y = \mu_E(x') \circ y'$ if and only if there exists an epimorphic family $\{f_i : F_i \rightarrow F \mid i \in I\}$ in $\mathcal{E}$ such that the pair $((\mathcal{A}_{f_i \circ f_i}(x), f_i^*(y)), (\mathcal{A}_{f_i \circ f_i}(x'), f_i^*(y')))$ belongs to the relation $\mathcal{R}_{F_i}^D$.*

Proof First, let us prove that, under the assumption that $\underline{\mathcal{A}}_{\mathcal{E}}$ is the $\mathcal{E}$ -externalization of an internal category $\mathbb{C}$ in $\mathcal{E}$, any colimiting cocone μ for D satisfies the two conditions of the theorem.

Under this assumption, condition (i) for μ rewrites as follows: for any object F of $\mathcal{E}$ and arrow $h : F \rightarrow \text{colim}(D)$ in $\mathcal{E}$, there exist an epimorphic family $\{f_i : F_i \rightarrow F \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a generalized element $x_i : F_i \rightarrow \mathbb{C}_0$ and an arrow $\alpha_i : 1_{\mathcal{E}/F_i} \rightarrow D_{F_i}(x_i)$ in the topos $\mathcal{E}/F_i$ such that $\langle h \circ f_i, 1_{F_i} \rangle = c \circ z_{x_i} \circ \alpha_i$ as arrows $1_{\mathcal{E}/F_i} \rightarrow !_{F_i}^*(\text{colim}(D))$ in $\mathcal{E}/F_i$ (where the notation is that of section 5.1.1).

Consider the following pullback square:

$$\begin{array}{ccc} F' & \xrightarrow{f'} & F \\ \downarrow h' & & \downarrow h \\ (\int^{\text{opf}} D)_0 & \xrightarrow{c} & \text{colim}(D). \end{array}$$

Since c is an epimorphism, $f' : F' \rightarrow F$ is also an epimorphism. This arrow will form, by itself, the unique element of an epimorphic family satisfying condition (i). We set x' equal to the composite $(\pi_D^{\text{opf}})_0 \circ h'$ (recall that $(\pi_D^{\text{opf}})_0 : (\int^{\text{opf}} D)_0 \rightarrow \mathbb{C}_0$ is the object component of the discrete opfibration associated with D). Consider the following pullback square defining $D_{F'}(x')$:

$$\begin{array}{ccc} D_{F'}(x') & \xrightarrow{u} & (\int^{\text{opf}} D)_0 \\ \downarrow v & & \downarrow (\pi_D^{\text{opf}})_0 \\ F' & \xrightarrow{x'} & \mathbb{C}_0. \end{array}$$

By the universal property of this pullback diagram, there exists a unique arrow $\alpha_{x'} : F' \rightarrow D_{F'}(x')$ in $\mathcal{E}$ such that $u \circ \alpha_{x'} = h'$ and $v \circ \alpha_{x'} = 1_{F'}$. So $\alpha_{x'}$ is an arrow $1_{\mathcal{E}/F'} \rightarrow D_{F'}(x')$ in the topos $\mathcal{E}/F'$ and $\langle h \circ f', 1_{F'} \rangle = \langle c \circ u \circ \alpha_{x'}, v \circ \alpha_{x'} \rangle$, as required.

Under the assumption that $\mathcal{A}_{\mathcal{E}}$ is the $\mathcal{E}$ -externalization of an internal category $\mathbb{C}$ in $\mathcal{E}$, condition (ii) for μ rewrites as follows: for any pairs (x, y) and (x', y') , where x and x' are generalized elements $E \rightarrow \mathbb{C}_0$, y is an arrow $(f : F \rightarrow E) \rightarrow D_E(x)$ in $\mathcal{E}/E$ and y' is an arrow $(f : F \rightarrow E) \rightarrow D_E(x')$ in $\mathcal{E}/E$, we have $c \circ z_x \circ y = c \circ z_{x'} \circ y'$ if and only if there exists an epimorphic family $\{f_i : F_i \rightarrow F \mid i \in I\}$ in $\mathcal{E}$ such that the pair $((x \circ f \circ f_i, y \circ f_i), (x' \circ f \circ f_i, y' \circ f_i))$ belongs to the relation $\mathcal{R}_{F_i}^D$ for each $i \in I$.

Thanks to the localization technique, we can suppose $E = 1_{\mathcal{E}}$ without loss of generality. Indeed, condition (ii) for the diagram D , the cocone μ and the object E is equivalent to condition (ii) for the diagram D/E , the cocone μ/E and the object $1_{\mathcal{E}/E}$, and μ is, by our assumption, an indexed colimiting cocone and hence stable under localization.

We have that $c \circ z_x \circ y = c \circ z_{x'} \circ y'$ if and only if there exists an epimorphic family $\{f_i : F_i \rightarrow F \mid i \in I\}$ in $\mathcal{E}$ such that, for each $i \in I$, $(z_x \circ y \circ f_i, z_{x'} \circ y' \circ f_i)$ belongs to the equivalence relation on the set $\text{Hom}_{\mathcal{E}}(F_i, (\int^{\text{opf}} D)_0)$ generated by the set of pairs of the form $(d_0^{\int^{\text{opf}} D} \circ a, d_1^{\int^{\text{opf}} D} \circ a)$ for a generalized element $a : F_i \rightarrow (\int^{\text{opf}} D)_0$ in $\mathcal{E}$ (cf. Lemma 5.1.28 below).

By Remark 5.1.14(e), we have that $(z_x \circ y \circ f_i, z_{x'} \circ y' \circ f_i) = (z_{x \circ f \circ f_i} \circ (y \circ f_i), z_{x' \circ f \circ f_i} \circ (y' \circ f_i))$. Our thesis thus follows from Proposition 5.1.17, applied to the toposes $\mathcal{E}/F_i$ and the pairs $((x \circ f \circ f_i, y \circ f_i), (x' \circ f \circ f_i, y' \circ f_i))$.

Let us now prove that the conditions of the theorem are sufficient for an indexed cocone μ over D to be an indexed colimiting cocone. For this part of the theorem, we shall not need to assume the $\mathcal{E}$ -indexed category $\mathcal{A}_{\mathcal{E}}$ to be small.

Since conditions (i) and (ii) are both stable under localization, it suffices to prove that μ is a colimiting cocone over the diagram D . Suppose that λ is an indexed cocone over D with vertex V . We have to prove that there exists a unique arrow $l : U \rightarrow V$ in $\mathcal{E}$ such that for any $E \in \mathcal{E}$ and $x \in \mathcal{A}_E$, $(l \times 1_E) \circ \mu_E(x) = \lambda_E(x)$. To define the arrow l we shall define a function $L : \text{Hom}_{\mathcal{E}}(E, U) \rightarrow \text{Hom}_{\mathcal{E}}(E, V)$ natural in $E \in \mathcal{E}$. Given $h \in \text{Hom}_{\mathcal{E}}(E, U)$, by condition (i) in the statement of the theorem, there exist an epimorphic family $\mathcal{I} = \{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object x_i of $\mathcal{A}_{E_i}$ and an arrow $\alpha_i : 1_{\mathcal{E}/E_i} \rightarrow D_{E_i}(x_i)$ in $\mathcal{E}/E_i$ such that $\langle h \circ e_i, 1_{E_i} \rangle = \mu_{E_i}(x_i) \circ \alpha_i$. The family of arrows $\{\lambda_{E_i}(x_i) \circ \alpha_i \mid i \in I\}$ satisfies the hypotheses of Corollary 5.1.7. Indeed, using the notation of the corollary, we have that $q_i^*(\lambda_{E_i}(x_i) \circ \alpha_i) = q_j^*(\lambda_{E_j}(x_j) \circ \alpha_j)$ for all $i, j \in I$; this can be easily proved by using condition (ii) of the theorem, the fact that the same identity holds for the cocone μ (in place of λ) and Remark 5.1.14(d). Therefore there exists a unique arrow $L_h^T : E \rightarrow V$ such that $\langle L_h^T \circ e_i, 1_{E_i} \rangle = \lambda_{E_i}(x_i) \circ \alpha_i$ for all $i \in I$. In order to be able to define $L(h)$ as equal to L_h^T we need to check that for any other epimorphic family

$\mathcal{J} = \{e'_j : E'_j \rightarrow E \mid j \in J\}$ in $\mathcal{E}$, we have $L_h^{\mathcal{I}} = L_h^{\mathcal{J}}$. To this end, consider the fibred product of the epimorphic families $\mathcal{I}$ and $\mathcal{J}$, that is the family of arrows $p_{i,j} = e_i \circ f_i = e'_j \circ f'_j : F_{i,j} \rightarrow E$ (for $(i,j) \in I \times J$), where the arrows f_i and f'_j are defined by the following pullback square:

$$\begin{array}{ccc} F_{i,j} & \xrightarrow{f_i} & E_i \\ \downarrow f'_j & & \downarrow e_i \\ E'_j & \xrightarrow{e'_j} & E. \end{array}$$

It clearly suffices to verify that for any $i \in I, j \in J$, we have $L_h^{\mathcal{I}} \circ p_{i,j} = L_h^{\mathcal{J}} \circ p_{i,j}$. Now, we have that $\langle L_h^{\mathcal{I}} \circ e_i, 1_{E_i} \rangle = \lambda_{E_i}(x_i) \circ \alpha_i$, while $\langle L_h^{\mathcal{J}} \circ e'_j, 1_{E'_j} \rangle = \lambda_{E'_j}(x'_j) \circ \beta_j$. Applying respectively f_i^* and $f'_j{}^*$ to these two identities, we obtain that $\langle L_h^{\mathcal{I}} \circ p_{i,j}, 1_{F_{i,j}} \rangle = \lambda_{F_{i,j}}(x_{i,j}) \circ f_i^*(\alpha_i)$ and $\langle L_h^{\mathcal{J}} \circ p_{i,j}, 1_{F_{i,j}} \rangle = \lambda_{F_{i,j}}(x'_{i,j}) \circ f'_j{}^*(\beta_j)$, where $x_{i,j} = \mathcal{A}_{f_i}(x_i)$ and $x'_{i,j} = \mathcal{A}_{f'_j}(x'_j)$, whence the condition $L_h^{\mathcal{I}} \circ p_{i,j} = L_h^{\mathcal{J}} \circ p_{i,j}$ is equivalent to the condition $\lambda_{F_{i,j}}(x_{i,j}) \circ f_i^*(\alpha_i) = \lambda_{F_{i,j}}(x'_{i,j}) \circ f'_j{}^*(\beta_j)$. This equality can be proved by using condition (ii) of the theorem, by using the fact that the same identity holds with the cocone μ in place of λ and Remark 5.1.14(d). More specifically, the identities $\langle h \circ e_i, 1_{E_i} \rangle = \mu_{E_i}(x_i) \circ \alpha_i$ and $\langle h \circ e'_j, 1_{E'_j} \rangle = \mu_{E'_j}(x'_j) \circ \beta_j$ imply, applying respectively f_i^* and $f'_j{}^*$ to them, the identity $\mu_{F_{i,j}}(x_{i,j}) \circ f_i^*(\alpha_i) = \mu_{F_{i,j}}(x'_{i,j}) \circ f'_j{}^*(\beta_j)$. From condition (ii) of the theorem, it thus follows that for any $(i,j) \in I \times J$, there exists an epimorphic family $\{g_k^{i,j} : G_k^{i,j} \rightarrow F_{i,j} \mid k \in K_{i,j}\}$ in $\mathcal{E}$ such that the pair $((\mathcal{A}_{g_k^{i,j}}(x_{i,j}), f_i^*(\alpha_i) \circ g_k^{i,j}), (\mathcal{A}_{g_k^{i,j}}(x'_{i,j}), f'_j{}^*(\beta_j) \circ g_k^{i,j}))$ belongs to the relation $\mathcal{R}_{G_k^{i,j}}^D$. This in turn implies (by Remark 5.1.14(d)) that $\lambda_{G_k^{i,j}}(\mathcal{A}_{g_k^{i,j}}(x_{i,j})) \circ (f_i^*(\alpha_i) \circ g_k^{i,j}) = \lambda_{G_k^{i,j}}(\mathcal{A}_{g_k^{i,j}}(x'_{i,j})) \circ (f'_j{}^*(\beta_j) \circ g_k^{i,j})$. But $\lambda_{G_k^{i,j}}(\mathcal{A}_{g_k^{i,j}}(x_{i,j})) \circ (f_i^*(\alpha_i) \circ g_k^{i,j}) = (1_V \times g_k^{i,j})^*(\lambda_{F_{i,j}}(x_{i,j}) \circ f_i^*(\alpha_i))$ and $\lambda_{G_k^{i,j}}(\mathcal{A}_{g_k^{i,j}}(x'_{i,j})) \circ (f'_j{}^*(\beta_j) \circ g_k^{i,j}) = (1_V \times g_k^{i,j})^*(\lambda_{F_{i,j}}(x'_{i,j}) \circ f'_j{}^*(\beta_j))$, whence $(1_V \times g_k^{i,j})^*(\lambda_{F_{i,j}}(x_{i,j}) \circ f_i^*(\alpha_i)) = (1_V \times g_k^{i,j})^*(\lambda_{F_{i,j}}(x'_{i,j}) \circ f'_j{}^*(\beta_j))$, equivalently (since the $g_k^{i,j}$ are jointly epimorphic) $\lambda_{F_{i,j}}(x_{i,j}) \circ f_i^*(\alpha_i) = \lambda_{F_{i,j}}(x'_{i,j}) \circ f'_j{}^*(\beta_j)$, as required.

It is clear that the assignment $h \rightarrow L(h)$ defined above is natural in $E \in \mathcal{E}$; it therefore remains to show that the resulting arrow $l : U \rightarrow V$ satisfies the required property that for any $E \in \mathcal{E}$ and any $x \in \mathcal{A}_E$, $(l \times 1_E) \circ \mu_E(x) = \lambda_E(x)$. Take s to be the canonical arrow $s : D_E(x) \rightarrow E$ in $\mathcal{E}$ and set $x' = \mathcal{A}_s(x)$, $E' = D_E(x)$; then $x' \in \mathcal{A}_{E'}$ and, considering the pullback square

$$\begin{array}{ccc} D_{E'}(x') & \xrightarrow{r} & D_E(x) \\ \downarrow t & & \downarrow s \\ E' & \xrightarrow{s} & E, \end{array}$$

the unique arrow $\alpha : E' \rightarrow D_{E'}(x')$ such that $r \circ \alpha = 1_{E'}$ and $t \circ \alpha = 1_{E'}$ satisfies the property that $\langle h, 1_{E'} \rangle = \mu_{E'}(x') \circ \alpha$. So the epimorphic family $\{1_{E'} : E' \rightarrow E'\}$ satisfies condition (i) of the theorem with respect to the arrow $h = \pi_U \circ \mu_E(x)$, where $\pi_U : U \times E \rightarrow U$ is the canonical projection, and hence $l \circ h = L(h) = \lambda_{E'}(x') \circ \alpha$; therefore $(l \times 1_E) \circ \mu_E(x) = \lambda_E(x)$, as required. $\square$

Remarks 5.1.19 (a) The proof of the theorem shows that the sufficiency of its conditions holds more generally for any (i.e. not necessarily small) $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$.

(b) The conditions in the statement of the theorem are both stable under localization; that is, if the cocone μ over the diagram D satisfies them then, for any $E \in \mathcal{E}$, the cocone μ/E over the diagram D/E does so too.

Proposition 5.1.20 *Let $\underline{\mathcal{B}}_{\mathcal{E}}$ be an $\mathcal{E}$ -strictly final subcategory of an indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$, $D : \underline{\mathcal{A}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ an $\mathcal{E}$ -indexed functor and μ an $\mathcal{E}$ -indexed cocone over D . Then μ satisfies the conditions of Theorem 5.1.18 with respect to the diagram D if and only if $\mu \circ i$ satisfies them with respect to the diagram $D \circ i$, where i is the canonical embedding $\underline{\mathcal{B}}_{\mathcal{E}} \hookrightarrow \underline{\mathcal{A}}_{\mathcal{E}}$.*

Proof Suppose that μ satisfies the conditions of Theorem 5.1.18 with respect to the diagram D . The fact that $\mu \circ i$ satisfies condition (i) of the theorem with respect to the diagram $D \circ i$ immediately follows from the fact that μ does so with respect to the diagram D , by using the fact that for any object $x_i \in \mathcal{A}_{F_i}$ there exists an epimorphic family $\{g_k^i : G_k^i \rightarrow F_i \mid k \in K_i\}$ such that $\mathcal{A}_{g_k^i}(x_i)$ lies in $\mathcal{B}_{G_k^i}$ for each $k \in K_i$. Let us now show that $\mu \circ i$ satisfies condition (ii) with respect to the diagram $D \circ i$.

Let us first establish the following fact (*): for any pairs (x, y) and (x', y') in $\mathcal{I}_R^D$, if $((x, y), (x', y')) \in \mathcal{R}_R^D$, there exists an epimorphic family $\{r_i : R_i \rightarrow R \mid i \in I\}$ in $\mathcal{E}$ such that $((\mathcal{A}_{r_i}(x), r_i^*(y)), (\mathcal{A}_{r_i}(x'), r_i^*(y')))) \in \mathcal{R}_{R_i}^{D \circ i}$ for each $i \in I$.

If $((x, y), (x', y')) \in \mathcal{R}_R^D$, there exists a finite sequence

$$(x, y) = (x_0, y_0), \dots, (x_n, y_n) = (x', y')$$

of pairs in $\mathcal{I}_R^D$ such that for any $j \in \{0, \dots, n-1\}$, either there exists an arrow $f_j : x_j \rightarrow x_{j+1}$ in $\mathcal{A}_R$ such that $y_{j+1} = D_R(f_j) \circ y_j$ or there exists an arrow $f_j : x_{j+1} \rightarrow x_j$ in $\mathcal{A}_R$ such that $y_j = D_R(f_j) \circ y_{j+1}$. By applying the definition of an $\mathcal{E}$ -strictly final subcategory a finite number of times (using the fact that the fibred product of epimorphic families is again an epimorphic family), we can find an epimorphic family $\{r_i : R_i \rightarrow R \mid i \in I\}$ in $\mathcal{E}$ such that for every $i \in I$ and every $j \in \{0, \dots, n-1\}$ the arrow $\mathcal{A}_{r_i}(f_j)$ lies in $\mathcal{B}_{R_i}$. From this it immediately follows that $((\mathcal{A}_{r_i}(x), r_i^*(y)), (\mathcal{A}_{r_i}(x'), r_i^*(y')))) \in \mathcal{R}_{R_i}^{D \circ i}$, as required.

Using (*), the proof of the fact that $\mu \circ i$ satisfies condition (ii) of the theorem with respect to the diagram $D \circ i$ follows readily from the fact that μ satisfies it with respect to the diagram D .

Conversely, let us suppose that $\mu \circ i$ satisfies the conditions of the theorem with respect to $D \circ i$ and deduce that μ satisfies them with respect to D . The

fact that the validity of condition (i) for $\mu \circ i$ with respect to $D \circ i$ implies the validity of condition (i) for μ with respect to D is obvious. Concerning condition (ii) for (μ, D) , this can be deduced from condition (ii) for $(\mu \circ i, D \circ i)$ by using the fact that, given (x, y) and (x', y') as in the statement of the condition for (μ, D) , there exists an epimorphic family $\{e_j : E_j \rightarrow E \mid j \in J\}$ in $\mathcal{E}$ such that $\mathcal{A}_{e_j}(x)$ and $\mathcal{A}_{e_j}(x')$ lie in $\mathcal{B}_{E_j}$ for all $j \in J$. $\square$

Remark 5.1.21 The proposition shows, in view of Theorem 5.1.18, that for any $\mathcal{E}$ -strictly final subcategory $\underline{\mathcal{B}}_{\mathcal{E}}$ of a small indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$ (in particular, any $\mathcal{E}$ -indexed subcategory of the form $\int \underline{F}_{\mathcal{E}} \hookrightarrow \int \overline{F}_{\mathcal{E}}^f$ for a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ as in Theorem 5.1.12), any $\mathcal{E}$ -indexed functor $D : \underline{\mathcal{A}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ and any $\mathcal{E}$ -indexed cocone μ over D , μ is an indexed colimiting cocone for D if and only if $\mu \circ i$ satisfies the conditions of Theorem 5.1.18 with respect to the diagram $D \circ i$, where i is the canonical embedding $\underline{\mathcal{B}}_{\mathcal{E}} \hookrightarrow \underline{\mathcal{A}}_{\mathcal{E}}$.

Proposition 5.1.22 *Let $D : \underline{\mathcal{A}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ be an indexed functor, where $\underline{\mathcal{A}}_{\mathcal{E}}$ is an $\mathcal{E}$ -filtered category, R an object of $\mathcal{E}$ and $(x, y), (x', y')$ pairs in $\mathcal{I}_R^D$. Then there exists an epimorphic family $\{r_i : R_i \rightarrow R \mid i \in I\}$ in $\mathcal{E}$ such that, for every $i \in I$, $((\mathcal{A}_{r_i}(x), r_i^*(y)), (\mathcal{A}_{r_i}(x'), r_i^*(y')))) \in \mathcal{R}_{R_i}^D$ if and only if there exists an epimorphic family $\{e_j : E_j \rightarrow R \mid j \in J\}$ in $\mathcal{E}$ with the property that for any $j \in J$ there exist arrows $f_j : \mathcal{A}_{e_j}(x) \rightarrow z$ and $g_j : \mathcal{A}_{e_j}(x') \rightarrow z$ in the category $\mathcal{A}_{E_j}$ such that $D_{E_j}(f_j) \circ e_j^*(y) = D_{E_j}(g_j) \circ e_j^*(y')$.*

Proof The ‘if’ direction is obvious, so it remains to prove the ‘only if’ one. It clearly suffices to show, by induction on the length of a sequence $(x_1, y_1), \dots, (x_n, y_n)$ of pairs in $\mathcal{I}_R^D$ with the property that for any $i \in \{1, \dots, n-1\}$ either there exists an arrow $f_j : x_j \rightarrow x_{j+1}$ in $\mathcal{A}_R$ such that $y_{j+1} = D_R(f_j) \circ y_j$ or there exists an arrow $f_j : x_{j+1} \rightarrow x_j$ in $\mathcal{A}_R$ such that $y_j = D_R(f_j) \circ y_{j+1}$, that there exists an epimorphic family $\{r_i : R_i \rightarrow R \mid i \in I\}$ in $\mathcal{E}$ such that for any $i \in I$ there are arrows $f_i : \mathcal{A}_{r_i}(x_1) \rightarrow z$ and $g_i : \mathcal{A}_{r_i}(x_n) \rightarrow z$ in the category $\mathcal{A}_{R_i}$ satisfying $D_{R_i}(f_i) \circ r_i^*(y_1) = D_{R_i}(g_i) \circ r_i^*(y_n)$. The case $n = 1$ is obvious. Suppose now that the claim is valid for n and show that it holds for $n+1$. There exists an epimorphic family $\{e_j : E_j \rightarrow R \mid j \in J\}$ in $\mathcal{E}$ such that for any $j \in J$ there are arrows $f_j : \mathcal{A}_{e_j}(x_1) \rightarrow z_j$ and $g_j : \mathcal{A}_{e_j}(x_n) \rightarrow z_j$ in $\mathcal{A}_{E_j}$ such that $D_{E_j}(f_j) \circ e_j^*(y_1) = D_{E_j}(g_j) \circ e_j^*(y_n)$. On the other hand, there exists either an arrow $g : x_n \rightarrow x_{n+1}$ in $\mathcal{A}_R$ such that $D_R(g) \circ e_j^*(y_n) = e_j^*(y_{n+1})$ or an arrow $h : x_{n+1} \rightarrow x_n$ in $\mathcal{A}_R$ such that $D_R(h) \circ e_j^*(y_{n+1}) = e_j^*(y_n)$. In the latter case, our thesis follows at once; so we can concentrate on the case where there exists an arrow $g : x_n \rightarrow x_{n+1}$ in $\mathcal{A}_R$ such that $D_R(g) \circ e_j^*(y_n) = e_j^*(y_{n+1})$.

By definition of an $\mathcal{E}$ -filtered category, for each $j \in J$ there exist an epimorphic family $\{f_k^j : F_k^j \rightarrow E_j \mid k \in K_j\}$ in $\mathcal{E}$ and for each $k \in K_j$ arrows $m_k^j : \mathcal{A}_{f_k^j}(z) \rightarrow w_k^j$ and $n_k^j : \mathcal{A}_{f_k^j}(\mathcal{A}_{e_j}(x_{n+1})) \rightarrow w_k^j$ such that $D_{F_k^j}(m_k^j) \circ D_{F_k^j}(\mathcal{A}_{f_k^j}(g_j)) \circ (f_k^j \circ e_j)^*(y_n) = D_{F_k^j}(n_k^j) \circ D_{F_k^j}(\mathcal{A}_{f_k^j}(g)) \circ (f_k^j \circ e_j)^*(y_n)$. Again, by definition of an $\mathcal{E}$ -filtered category, for each $j \in J$ and $k \in K_j$ there exist an

epimorphic family $\{g_l^{k,j} : G_l^{k,j} \rightarrow F_k^j \mid l \in L_{k,j}\}$ in $\mathcal{E}$ and for each $l \in L_{k,j}$ an arrow $p_l^{k,j} : \mathcal{A}_{g_l^{k,j}}(w_k^j) \rightarrow v_l^{k,j}$ such that $p_l^{k,j} \circ \mathcal{A}_{g_l^{k,j}}(m_k^j \circ \mathcal{A}_{f_k^j}(g_j)) = p_l^{k,j} \circ \mathcal{A}_{g_l^{k,j}}(n_k^j \circ \mathcal{A}_{f_k^j}(\mathcal{A}_{e_j}(g)))$. For any $j \in J, k \in K_j$ and $l \in L_{k,j}$, set $a_{j,k,l} : \mathcal{A}_{e_j \circ f_k^j \circ g_l^{k,j}}(x_1) \rightarrow v_l^{k,j}$ equal to $p_l^{k,j} \circ \mathcal{A}_{g_l^{k,j}}(m_k^j) \circ \mathcal{A}_{f_k^j \circ g_l^{k,j}}(f_j)$ and $b_{j,k,l} : \mathcal{A}_{e_j \circ f_k^j \circ g_l^{k,j}}(x_{n+1}) \rightarrow v_l^{k,j}$ equal to $p_l^{k,j} \circ \mathcal{A}_{g_l^{k,j}}(n_k^j)$. It is readily seen that $D_{G_l^{k,j}}(a_{j,k,l}) \circ (e_j \circ f_k^j \circ g_l^{k,j})^*(y_1) = D_{G_l^{k,j}}(b_{j,k,l}) \circ (e_j \circ f_k^j \circ g_l^{k,j})^*(y_{n+1})$, whence the required condition is satisfied (by taking as epimorphic family $\{e_j \circ f_k^j \circ g_l^{k,j} \mid j \in J, k \in K_j, l \in L_{k,j}\}$). $\square$

Combining Theorem 5.1.18 with Proposition 5.1.22 in view of Remark 5.1.21, we immediately obtain the following result:

Corollary 5.1.23 *Let $D : \underline{\mathcal{A}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ be an indexed functor, where $\underline{\mathcal{A}}_{\mathcal{E}}$ is an $\mathcal{E}$ -filtered, $\mathcal{E}$ -strictly final subcategory of a small $\mathcal{E}$ -indexed category. Then an $\mathcal{E}$ -indexed cocone μ over D with vertex U is an indexed colimiting cocone for D if and only if the following conditions are satisfied:*

- (i) *For any arrow $h : F \rightarrow U$ in $\mathcal{E}$, there exist an epimorphic family $\{f_i : F_i \rightarrow F \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object $x_i \in \mathcal{A}_{F_i}$ and an arrow $\alpha_i : 1_{\mathcal{E}/F_i} \rightarrow D_{F_i}(x_i)$ in $\mathcal{E}/F_i$ such that $\langle h \circ f_i, 1_{F_i} \rangle = \mu_{F_i}(x_i) \circ \alpha_i$ as arrows $1_{\mathcal{E}/F_i} \rightarrow F_i^*(U)$ in $\mathcal{E}/F_i$.*
- (ii) *For any object E of $\mathcal{E}$ and any pairs (x, y) and (x', y') , where x and x' are objects of $\mathcal{A}_E$, y is an arrow $(f : F \rightarrow E) \rightarrow D_E(x)$ in $\mathcal{E}/E$ and y' is an arrow $(f : F \rightarrow E) \rightarrow D_E(x')$ in $\mathcal{E}/E$, we have that $\mu_E(x) \circ y = \mu_E(x') \circ y'$ if and only if there exist an epimorphic family $\{f_i : F_i \rightarrow F \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ arrows $g_i : \mathcal{A}_{f \circ f_i}(x) \rightarrow z_i$ and $h_i : \mathcal{A}_{f \circ f_i}(x') \rightarrow z_i$ in $\mathcal{A}_{F_i}$ such that $D_{F_i}(g_i) \circ f_i^*(y) = D_{F_i}(h_i) \circ f_i^*(y')$.*

Moreover, the $\mathcal{E}$ -filteredness of $\underline{\mathcal{A}}_{\mathcal{E}}$ implies the following ‘joint embedding property’: for any objects E of $\mathcal{E}$ and x, x' of $\mathcal{A}_E$, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ arrows $g_i : \mathcal{A}_{e_i}(x) \rightarrow z_i$ and $h_i : \mathcal{A}_{e_i}(x') \rightarrow z_i$ in $\mathcal{A}_{E_i}$ such that the diagram

$$\begin{array}{ccccc}
 D_{E_i}(\mathcal{A}_{e_i}(x)) & & & & \\
 \searrow \mu_{E_i}(\mathcal{A}_{e_i}(x)) & & & & \\
 D_{E_i}(g_i) & \searrow & D_{E_i}(z_i) & \xrightarrow{\mu_{E_i}(z_i)} & U \times E_i \\
 D_{E_i}(h_i) & \nearrow & & & \\
 D_{E_i}(\mathcal{A}_{e_i}(x')) & \nearrow \mu_{E_i}(\mathcal{A}_{e_i}(x')) & & &
 \end{array}$$

commutes. $\square$

Remarks 5.1.24 (a) In the statement of Corollary 5.1.23, we can suppose without loss of generality the object $f : F \rightarrow E$ of the topos $\mathcal{E}/E$ to be equal

to the terminal object $1_E : E \rightarrow E$. Indeed, by Remark 5.1.14(d), for any objects x and x' of $\mathcal{A}_E$ and any arrows $y : (f : F \rightarrow E) \rightarrow D_E(x)$ and $y' : (f : F \rightarrow E) \rightarrow D_E(x')$ in the topos $\mathcal{E}/E$, $\mu_E(x) \circ y = \mu_E(x') \circ y'$ if and only if $\mu_F(\mathcal{A}_f(x)) \circ y_f = \mu_F(\mathcal{A}_f(x')) \circ y_f$.

- (b) For any pairs (x, y) and (x', y') , where x and x' are objects of $\mathcal{A}_E$, y is an arrow $(f : F \rightarrow E) \rightarrow D_E(x)$ in $\mathcal{E}/E$ and y' is an arrow $(f : F \rightarrow E) \rightarrow D_E(x')$ in $\mathcal{E}/E$, and any epimorphic family $\{g_j : G_j \rightarrow F \mid j \in J\}$ in $\mathcal{E}$, condition (ii) of Corollary 5.1.23 is satisfied by the pair $((x, y), (x', y'))$ if and only if it is satisfied by the pair $((x, y \circ g_j), (x', y' \circ g_j))$ for all $j \in J$.

Let us now apply Corollary 5.1.23 in the context of a flat functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ and a functor $P : \mathcal{C} \rightarrow \mathbf{Set}$, where $\mathcal{C}$ is a small category and $\mathcal{E}$ is a Grothendieck topos. By Theorem 5.1.13, the restriction of the functor $(\underline{P}_{\mathcal{E}} \circ \underline{\pi}_{\overline{F}}^f) \cong \underline{P}_{\mathcal{E}} \circ \underline{\pi}_{\overline{F}}^f$

to the $\mathcal{E}$ -indexed subcategory $\int \underline{F}_{\mathcal{E}}$ of $\int \underline{F}_{\mathcal{E}}^f$ is naturally isomorphic to the composite $\mathcal{E}$ -indexed functor $\gamma_{\mathcal{E}} \circ z_{\mathcal{E}}$, where $z_{\mathcal{E}} : \int \underline{F}_{\mathcal{E}} \rightarrow \mathbf{Set}_{\mathcal{E}}$ is the $\mathcal{E}$ -indexed functor defined by setting $(z_{\mathcal{E}})_E((c, x)) = P(c)$ and $(z_{\mathcal{E}})_E(f) = P(f)$ (for any $E \in \mathcal{E}$, object (c, x) and arrow f of the category $(\int \underline{F})_E$). By the remarks preceding the statement of Theorem 5.1.12, an $\mathcal{E}$ -indexed cocone μ over $\gamma_{\mathcal{E}} \circ z_{\mathcal{E}}$ with vertex U can be identified with a family of functions $\mu_{(c,x)} : P(c) \rightarrow \text{Hom}_{\mathcal{E}}(E, U)$ indexed by the pairs (c, x) consisting of an object c of $\mathcal{C}$ and a generalized element $x : E \rightarrow F(c)$, which satisfies the following conditions:

- (i) For any generalized element $x : E \rightarrow F(c)$ and any arrow $f : d \rightarrow c$ in $\mathcal{C}$, $\mu_{(c,x)} \circ P(f) = \mu_{(c, F(f) \circ x)}$.
- (ii) For any generalized element $x : E \rightarrow F(c)$ and any arrow $e : E' \rightarrow E$ in $\mathcal{E}$, $\mu_{(c, x \circ e)} = \text{Hom}_{\mathcal{E}}(e, U) \circ \mu_{(c,x)}$.

The following theorem characterizes the cocones μ which are colimiting in terms of the functions $\mu_{(c,x)}$.

Theorem 5.1.25 *Let $\mathcal{C}$ be a small category, $\mathcal{E}$ a Grothendieck topos, $P : \mathcal{C} \rightarrow \mathbf{Set}$ a functor and $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ a flat functor. Then an $\mathcal{E}$ -indexed cocone $\mu = \{\mu_{(c,x)} : P(c) \rightarrow \text{Hom}_{\mathcal{E}}(E, U) \mid c \in \mathcal{C}, x : E \rightarrow F(c) \text{ in } \mathcal{E}\}$ with vertex U over the diagram given by the restriction of the $\mathcal{E}$ -indexed functor $(\underline{P}_{\mathcal{E}} \circ \underline{\pi}_{\overline{F}}^f) \cong \underline{P}_{\mathcal{E}} \circ \underline{\pi}_{\overline{F}}^f$ (where $\underline{P}_{\mathcal{E}}$ is the internal diagram $[\mathcal{C}, \mathcal{E}]$ given by $\overline{\gamma_{\mathcal{E}}^* \circ P}$) to the $\mathcal{E}$ -indexed subcategory $\int \underline{F}_{\mathcal{E}}$ is colimiting if and only if the following conditions are satisfied:*

- (i) *For any generalized element $x : E \rightarrow U$ in $\mathcal{E}$, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object c_i in $\mathcal{C}$, a generalized element $x_i : E_i \rightarrow F(c_i)$ and an element $y_i \in P(c_i)$ such that $\mu_{(c_i, x_i)}(y_i) = x \circ e_i$.*
- (ii) *For any pairs (a, x) and (b, x') , where a and b are objects of $\mathcal{C}$ and $x : E \rightarrow F(a)$, $x' : E \rightarrow F(b)$ are generalized elements, and any elements $y \in P(a)$ and $y' \in P(b)$, $\mu_{(a,x)}(y) = \mu_{(b,x')}(y')$ if and only if there exist an epimorphic*

family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object c_i of $\mathcal{C}$, arrows $f_i : a \rightarrow c_i$, $g_i : b \rightarrow c_i$ in $\mathcal{C}$ and a generalized element $x_i : E_i \rightarrow F(c_i)$ such that $\langle x, x' \rangle \circ e_i = \langle F(f_i), F(g_i) \rangle \circ x_i$ and $P(f_i)(y) = P(g_i)(y')$.

Moreover, the following ‘joint embedding property’ holds: for any pairs (a, x) and (b, x') , where a and b are objects of $\mathcal{C}$ and $x : E \rightarrow F(a)$, $x' : E \rightarrow F(b)$ are generalized elements in $\mathcal{E}$, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object c_i of $\mathcal{C}$, arrows $f_i : a \rightarrow c_i$, $g_i : b \rightarrow c_i$ in $\mathcal{C}$ and a generalized element $x_i : E_i \rightarrow F(c_i)$ such that $\langle x, x' \rangle \circ e_i = \langle F(f_i), F(g_i) \rangle \circ x_i$ and the following diagram commutes:

$$\begin{array}{ccccc}
 P(a) & \xrightarrow{\mu_{(a,x)}} & \text{Hom}_{\mathcal{E}}(E, U) & & \\
 & \searrow P(f_i) & & \searrow \text{Hom}_{\mathcal{E}}(e_i, M) & \\
 & & P(c_i) & \xrightarrow{\mu_{(c_i, x_i)}} & \text{Hom}_{\mathcal{E}}(E_i, U) \\
 & \nearrow P(g_i) & & \nearrow \text{Hom}_{\mathcal{E}}(e_i, U) & \\
 P(b) & \xrightarrow{\mu_{(b, x')}} & \text{Hom}_{\mathcal{E}}(E, U) & &
 \end{array}$$

In fact, the epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ can be taken to be the pullback of the family of arrows $\langle F(f), F(g) \rangle : F(c) \rightarrow F(a) \times F(b)$ (indexed by the spans $(f : a \rightarrow c, g : b \rightarrow c)$ in the category $\mathcal{C}$) along the arrow $\langle x, x' \rangle : E \rightarrow F(a) \times F(b)$.

Proof The theorem can be deduced from Corollary 5.1.23 by reasoning as follows.

Concerning condition (i), its equivalence with condition (i) of Corollary 5.1.23 follows from Remark 5.1.24(a) and the fact that for any object (c, z) of the category $(\int F)_E$ and any arrow $\alpha : 1_{\mathcal{E}/E} \rightarrow D_E((c, z))$ in the topos $\mathcal{E}/E$, denoting by $!_{f_r} : f_r : F_r \rightarrow E \rightarrow 1_{\mathcal{E}/E}$ the pullback along α of the coproduct arrow $u_r : 1_{\mathcal{E}/E} \rightarrow \gamma_{\mathcal{E}/E}^*(P(c)) \cong D_E((c, z))$ in $\mathcal{E}/E$ (for each $r \in P(c)$), the arrows $!_{f_r}$ define, for $r \in P(c)$, an epimorphic family of arrows in $\mathcal{E}/E$ with codomain $1_{\mathcal{E}/E}$. Indeed, this observation shows that we can suppose without loss of generality α to factor through one of the coproduct arrows u_r and hence to correspond to an element $y \in P(c)$.

Concerning condition (ii), we note that, by Remark 5.1.24, we can suppose without loss of generality that the pairs in condition (ii) of Corollary 5.1.23 have respectively the form $((x, a), y)$, where $y = u_r \circ !_{f_r}$, and $((x', b), y')$, where $y' = u_{r'} \circ !_{f_{r'}}$ (with the above notations), for some $r, r' \in P(c)$. Thus we have that $\mu_E(x) \circ y = \mu_E(x') \circ y'$ if and only if $(\text{Hom}_{\mathcal{E}}(f_r, U) \circ \mu_{(x,a)})(r) = (\text{Hom}_{\mathcal{E}}(f_{r'}, U) \circ \mu_{(x',b)})(r')$; but $(\text{Hom}_{\mathcal{E}}(f_r, U) \circ \mu_{(x,a)})(r) = \mu_{(a, x \circ f_r)}(r)$ and $(\text{Hom}_{\mathcal{E}}(f_{r'}, U) \circ \mu_{(x',b)})(r') = \mu_{(b, x' \circ f_{r'})}(r')$, whence $(\text{Hom}_{\mathcal{E}}(f_r, U) \circ \mu_{(x,a)})(r) = (\text{Hom}_{\mathcal{E}}(f_{r'}, U) \circ \mu_{(x',b)})(r')$ if and only if $\mu_{(a, x \circ f_r)}(r) = \mu_{(b, x' \circ f_{r'})}(r')$.

Finally, the ‘joint embedding property’ of the theorem is just an instance of that of Corollary 5.1.23. $\square$

5.1.7 Explicit calculation of set-indexed colimits

It is well known (cf. for instance p. 355 of [62]) that the colimit of a functor $H : \mathcal{I} \rightarrow \mathcal{E}$ defined over a small category $\mathcal{I}$ with values in a Grothendieck topos $\mathcal{E}$ can always be realized as the coequalizer $q : \coprod_{i \in I} H(i) \rightarrow \text{colim}(H)$ of the pair of arrows

$$a, b : \coprod_{u: i \rightarrow j \text{ in } \mathcal{I}} H(i) \rightarrow \coprod_{i \in I} H(i)$$

defined by the conditions

$$a \circ \lambda_u = \kappa_i$$

and

$$b \circ \lambda_u = \kappa_j \circ H(u)$$

for every arrow $u : i \rightarrow j$ in $\mathcal{I}$, where $\lambda_u : H(\text{dom}(u)) \rightarrow \coprod_{u: i \rightarrow j \text{ in } \mathcal{I}} H(i)$ and $\kappa_i : H(i) \rightarrow \coprod_{i \in I} H(i)$ are the canonical coproduct arrows.

Given two arrows $s, t : E \rightarrow \coprod_{i \in I} H(i)$ we can consider, for any objects $i, j \in \mathcal{I}$, the following pullback diagrams:

$$\begin{array}{ccc} E_i^s & \xrightarrow{a_i^s} & H(i) \\ \downarrow p_i^s & & \downarrow \kappa_i \\ E & \xrightarrow{s} & \coprod_{i \in I} H(i) \end{array} \quad \begin{array}{ccc} E_j^t & \xrightarrow{a_j^t} & H(j) \\ \downarrow p_j^t & & \downarrow \kappa_j \\ E & \xrightarrow{t} & \coprod_{i \in I} H(i) \end{array}$$

Let us also consider, for each $i, j \in \mathcal{I}$, the pullback square

$$\begin{array}{ccc} E_{i,j}^{s,t} & \xrightarrow{q_{i,j}^s} & E_i^s \\ \downarrow q_{i,j}^t & & \downarrow p_i^s \\ E_j^t & \xrightarrow{p_j^t} & E \end{array}$$

and denote by $r_{i,j}^{s,t} : E_{i,j}^{s,t} \rightarrow E$ the arrow $p_i^s \circ q_{i,j}^s = p_j^t \circ q_{i,j}^t$. Notice that the family of arrows $\{r_{i,j}^{s,t} : E_{i,j}^{s,t} \rightarrow E \mid i, j \in \mathcal{I}\}$ is epimorphic.

Lemma 5.1.26 *Let B be an object of a topos $\mathcal{E}$. Then an equivalence relation R in $\mathcal{E}$ on B can be identified with a function $E \rightarrow R_E$ sending each object E of $\mathcal{E}$ to an equivalence relation R_E on the set $\text{Hom}_{\mathcal{E}}(E, B)$ satisfying the following properties:*

- (i) *For any arrow $f : E \rightarrow E'$ in $\mathcal{E}$, if $(h, k) \in R_{E'}$ then $(h \circ f, k \circ f) \in R_E$.*
- (ii) *For any epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and any arrows $f, g \in \text{Hom}_{\mathcal{E}}(E, B)$, $(f \circ e_i, g \circ e_i) \in R_{E_i}$ for all $i \in I$ implies $(f, g) \in R_E$.*

Proof Clearly, for any object A of $\mathcal{C}$, the subobjects of A can be identified with the $J_{\mathcal{E}}$ -closed sieves on A , or equivalently with the $c_{J_{\mathcal{E}}}$ -closed subobjects of the representable $\text{Hom}_{\mathcal{E}}(-, A)$, where $c_{J_{\mathcal{E}}}$ is the closure operation on subobjects corresponding to the canonical Grothendieck topology $J_{\mathcal{E}}$ on $\mathcal{E}$. Applying this remark to $A = B \times B$ and noticing that the concept of equivalence relation is cartesian and hence preserved and reflected by cartesian fully faithful functors, in particular by the Yoneda embedding $\mathcal{E} \rightarrow [\mathcal{E}^{\text{op}}, \mathbf{Set}]$, we immediately deduce our thesis. $\square$

Lemma 5.1.26 implies that for any generalized elements $s, t : E \rightarrow \coprod_{i \in I} H(i)$, $(s, t) \in R_E$ if and only if, for every $i, j \in I$, $(\kappa_i \circ a_i^s \circ q_{i,j}^s, \kappa_j \circ a_j^t \circ q_{i,j}^t) \in R_{E_{i,j}^{s,t}}$; indeed, $s \circ r_{i,j}^{s,t} = \kappa_i \circ a_i^s \circ q_{i,j}^s$ and $t \circ r_{i,j}^{s,t} = \kappa_j \circ a_j^t \circ q_{i,j}^t$. Thus, in order to completely describe the relation R , it suffices to consider generalized elements of the form $\kappa_i \circ z$ and $\kappa_j \circ w$, where $z : E \rightarrow H(i)$ and $w : E \rightarrow H(j)$, and characterize when they belong to R_E .

The following lemma provides a characterization of the coequalizer

$$q : \coprod_{i \in I} H(i) \rightarrow \text{colim}(H),$$

which will be useful later on:

Lemma 5.1.27 *Under the above assumptions, let $p : \coprod_{i \in I} H(i) \rightarrow A$ be an epimorphism. Then p is isomorphic to the canonical map $q : \coprod_{i \in I} H(i) \rightarrow \text{colim}(H)$ if and only if, denoting by R the kernel pair of q , for any objects E of $\mathcal{E}$ and i, j of I and any generalized elements $z : E \rightarrow H(i)$ and $w : E \rightarrow H(j)$ in $\mathcal{E}$, $(\kappa_i \circ z, \kappa_j \circ w) \in R$ if and only if $p \circ \kappa_i \circ z = p \circ \kappa_j \circ w$.*

Proof Since in a topos every epimorphism is the coequalizer of its kernel pair, any two epimorphisms $\alpha : B \rightarrow C$ and $\beta : B \rightarrow D$ in $\mathcal{E}$ are isomorphic in $B/\mathcal{E}$ if and only if their kernel pairs are isomorphic as subobjects of $B \times B$. In particular, p is isomorphic to q if and only if the kernel pair of p is isomorphic to R .

The kernel pair S of p is defined by the property that for any generalized elements $s, t : E \rightarrow \coprod_{i \in I} H(i)$, $(s, t) \in S$ if and only if $p \circ s = p \circ t$. Our thesis thus follows from the remarks following the proof of Lemma 5.1.26. $\square$

Lemma 5.1.28 *Let $\mathcal{E}$ be a topos, $f, g : A \rightarrow B$ arrows in $\mathcal{E}$ with coequalizer $q : B \rightarrow C$ and $R \rightrightarrows B \times B$ the kernel pair of q . Then, for any object E of $\mathcal{E}$ and any elements $h, k \in \text{Hom}_{\mathcal{E}}(E, B)$, $\langle h, k \rangle : E \rightarrow B \times B$ factors through R if and only if there exists an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ such that, for each $i \in I$, $(h \circ e_i, k \circ e_i)$ belongs to the equivalence relation on the set $\text{Hom}_{\mathcal{E}}(E_i, B)$ generated by the set of pairs of the form $(f \circ a, g \circ a)$ for a generalized element $a : E_i \rightarrow A$ in $\mathcal{E}$.*

Proof For any object E of $\mathcal{E}$, consider the equivalence relation $R_E^{f,g}$ on the set $\text{Hom}_{\mathcal{E}}(E, B)$ consisting of the pairs (h, k) with the property that there exists an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ such that, for each $i \in I$, $(h \circ e_i, k \circ e_i)$ belongs to the equivalence relation on the set $\text{Hom}_{\mathcal{E}}(E_i, B)$ generated by the set of pairs of the form $(f \circ a, g \circ a)$ for a generalized element $a : E_i \rightarrow A$ in $\mathcal{E}$. The assignment $E \rightarrow R_E^{f,g}$ clearly satisfies the conditions of Lemma 5.1.26 and hence defines an equivalence relation $R^{f,g}$ on B in $\mathcal{E}$. To prove that $R = R^{f,g}$, we argue as follows. Denoting by $\pi_1, \pi_2 : R^{f,g} \rightarrow B$ the canonical projections, we have that for any $E \in \mathcal{E}$, $\text{Hom}_{\mathcal{E}}(E, q) \circ \text{Hom}_{\mathcal{E}}(E, \pi_1) = \text{Hom}_{\mathcal{E}}(E, q) \circ \text{Hom}_{\mathcal{E}}(E, \pi_2)$; therefore $q \circ \pi_1 = q \circ \pi_2$ in $\mathcal{E}$. To prove that q is actually the coequalizer of π_1 and π_2 in $\mathcal{E}$, we observe that any arrow $p : B \rightarrow D$ such that $p \circ \pi_1 = p \circ \pi_2$ satisfies $p \circ f = p \circ g$ and hence, by definition of q , there exists a unique factorization of p through q . As q is the coequalizer of $R^{f,g}$ and in a topos any equivalence relation is the kernel pair of its own coequalizer, we conclude that $R^{f,g} = R$, as required. $\square$

Remark 5.1.29 Under the hypotheses of Lemma 5.1.28, R can be characterized as the equivalence relation on B generated by the arrows f and g , i.e. as the smallest equivalence relation on B containing the image of the arrow $\langle f, g \rangle : A \rightarrow B \times B$. Indeed, clearly R contains the image of this arrow, and if T is an equivalence relation on B containing the image of the arrow $\langle f, g \rangle$ then the coequalizer z of T factors through q and hence the kernel pair of q , namely R , is contained in the kernel pair of z , namely T .

By using the internal language $\Sigma_{\mathcal{E}}$ of the topos $\mathcal{E}$ (cf. section 1.3.1.1), we can reformulate Lemma 5.1.28 as follows:

Lemma 5.1.30 *Let $\mathcal{E}$ be a topos, $f, g : A \rightarrow B$ arrows in $\mathcal{E}$ with coequalizer $q : B \rightarrow C$ and R the kernel pair of q . Let $(z, z') \in G_{f,g}$ (for variables z, z' of type B) be an abbreviation for the formula*

$$z = z' \vee (\exists x)(\ulcorner f \urcorner(x) = z \wedge \ulcorner g \urcorner(x) = z') \vee (\exists x')(\ulcorner f \urcorner(x') = z' \wedge \ulcorner g \urcorner(x') = z)$$

over $\Sigma_{\mathcal{E}}$. Then the geometric bisequent

$$\left(\ulcorner R \urcorner(y, y') \dashv\vdash_{y^B, y'^B} \bigvee_{n \in \mathbb{N}} \left(\exists z_1 \dots \exists z_n \left(z_1 = x \wedge z_n = x' \wedge \bigwedge_{1 < i < n} (z_{i-1}, z_i) \in G_{f,g} \right) \right) \right)$$

is valid in the $\Sigma_{\mathcal{E}}$ -structure $\mathcal{S}_{\mathcal{E}}$.

Proof The interpretation of the formula $(z^B, z'^B) \in G_{f,g}$ is the reflexive and symmetric closure of the relation $S_{f,g}$ on B given by the image of the arrow $\langle f, g \rangle : A \rightarrow B \times B$, that is the union of the diagonal subobject $\Delta : B \rightarrow B \times B$ of $B \times B$, the subobject $S_{f,g} \rightarrow B \times B$ and the subobject $\tau \circ S_{f,g}$ of $B \times B$ given

by the composite of $S_{f,g}$ with the exchange isomorphism $\tau : B \times B \rightarrow B \times B$. The interpretation of the formula

$$\bigvee_{n \in \mathbb{N}} \left(\exists z_1 \dots \exists z_n \left(z_1 = x \wedge z_n = x' \wedge \bigwedge_{1 < i < n} (z_{i-1}, z_i) \in G_{f,g} \right) \right)$$

thus coincides with the equivalence relation on B generated by $S_{f,g}$, that is with R (cf. Remark 5.1.29). $\square$

Given a functor $H : \mathcal{I} \rightarrow \mathcal{E}$ from a small category $\mathcal{I}$ to a Grothendieck topos $\mathcal{E}$ and an object E of $\mathcal{E}$, we define $\mathcal{I}_H^E$ as the set of pairs (i, x) , where i is an object of $\mathcal{I}$ and x is a generalized element $E \rightarrow H(i)$ in $\mathcal{E}$. Below, abusing notation, we shall occasionally identify a generalized element $x : E \rightarrow H(i)$ with its composite $\kappa_i \circ x$ with the coproduct arrow $\kappa_i : H(i) \rightarrow \coprod_{i \in \mathcal{I}} H(i)$.

The following proposition can be established by similar means to those employed in the proof of Lemma 5.1.28.

Proposition 5.1.31 *Let $H : \mathcal{I} \rightarrow \mathcal{E}$ be a functor from a small category $\mathcal{I}$ to a Grothendieck topos $\mathcal{E}$. Then the equivalence relation R on the object $\coprod_{i \in \mathcal{I}} H(i)$ given by the kernel pair of the canonical arrow $\coprod_{i \in \mathcal{I}} H(i) \rightarrow \text{colim}(H)$ is characterized by the following property: for any object E of $\mathcal{E}$ and any pairs (i, x) and (i', x') in $\mathcal{I}_H^E$, we have that $(\kappa_i \circ x, \kappa_{i'} \circ x') \in R_E$ if and only if there exists an epimorphic family $\{e_k : E_k \rightarrow E \mid k \in K\}$ in $\mathcal{E}$ such that for every $k \in K$ the pair $((i, x \circ e_k), (i', x' \circ e_k))$ belongs to the equivalence relation on the set $\mathcal{I}_H^{E_k}$ generated by the pairs of the form $((j, y), (j', H(f) \circ y))$ where $f : j \rightarrow j'$ is an arrow in $\mathcal{I}$ and y is a generalized element $E_k \rightarrow H(j)$.*

Proof Let us consider the pair of arrows $a, b : \coprod_{u: i \rightarrow j \text{ in } \mathcal{I}} H(i) \rightarrow \coprod_{i \in \mathcal{I}} H(i)$ defined at the beginning of this section. The canonical arrow $\coprod_{i \in \mathcal{I}} H(i) \rightarrow \text{colim}(H)$ is the coequalizer of a and b .

For any object E of $\mathcal{E}$, let R_E be the relation on the set $\text{Hom}_{\mathcal{E}}(E, \coprod_{i \in \mathcal{I}} H(i))$ consisting of the pairs (x, x') with the property that for any $(i, j) \in \mathcal{I} \times \mathcal{I}$ there exists an epimorphic family $\{e_k^{(i,j)} : E_k^{(i,j)} \rightarrow E_{i,j}^{x,x'} \mid k \in K_{(i,j)}\}$ in $\mathcal{E}$ such that, for each $k \in K_{(i,j)}$, $(x \circ r_{i,j}^{x,x'} \circ e_k^{(i,j)}, x' \circ r_{i,j}^{x,x'} \circ e_k^{(i,j)})$ yields a pair of elements of $\mathcal{I}_H^{E_k^{(i,j)}}$ belonging to the equivalence relation $T_k^{(i,j)}$ on $\mathcal{I}_H^{E_k^{(i,j)}}$ generated by the set of pairs of the form $((n, y), (m, H(z) \circ y))$ (for $n \in \mathcal{I}$, $y : E_k^{(i,j)} \rightarrow H(n)$ and $z : n \rightarrow m$ in $\mathcal{I}$).

It is readily seen that the assignment $E \rightarrow R_E$ satisfies the hypotheses of Lemma 5.1.26, whence it defines an equivalence relation R in $\mathcal{E}$ on the object $\coprod_{i \in \mathcal{I}} H(i)$ such that for any generalized elements $x, x' : E \rightarrow \coprod_{i \in \mathcal{I}} H(i)$, $x < x'$ factors through R if and only if $(x, x') \in R_E$.

The arrow $\langle a, b \rangle$ factors through R since, for every $i, j \in \mathcal{I}$, $(a \circ r_{i,j}^{s,t}, b \circ r_{i,j}^{s,t})$ belongs to the equivalence relation $T_k^{(i,j)}$. Therefore, by Remark 5.1.29, R contains the kernel pair of q . The converse inclusion follows from the fact that, for any $(x, x') \in R_E$, $q \circ x = q \circ x'$; indeed, $\{e_k^{(i,j)} : E_k^{(i,j)} \rightarrow E_{i,j}^{x,x'} \mid k \in K_{(i,j)}\}$ is an epimorphic family in $\mathcal{E}$ such that, for each $k \in K_{(i,j)}$, $(x \circ r_{i,j}^{x,x'} \circ e_k^{(i,j)}, x' \circ r_{i,j}^{x,x'} \circ e_k^{(i,j)})$ belongs to the equivalence relation $T_k^{(i,j)}$ and hence we have $q \circ x \circ r_{i,j}^{x,x'} \circ e_k^{(i,j)} = q \circ x' \circ r_{i,j}^{x,x'} \circ e_k^{(i,j)}$ for all $k \in K_{(i,j)}$, that is $q \circ x = q \circ x'$. $\square$

Remark 5.1.32 Proposition 5.1.31 could alternatively be deduced from Lemma 5.1.28, but the proof of such a derivation would be more involved than the direct argument given above.

Let us now proceed to apply the above results in the case of a functor H of the form $F \circ \pi_P$, where F is a flat functor with values in a Grothendieck topos and π_P is the fibration associated with a set-valued functor P .

Recall from Proposition 1.1.32 that a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ is flat if and only if it is filtering. Notice that one can suppose $E = 1_{\mathcal{E}}$ in condition (i)(a) of the proposition without loss of generality, and that if all the arrows in the category $\mathcal{C}$ are monic then condition (i)(c) of the proposition rewrites as follows: for any two parallel arrows $u, v : d \rightarrow c$ in $\mathcal{C}$, either $u = v$ or the equalizer of $F(u)$ and $F(v)$ is zero.

Proposition 5.1.33 *Let $\mathcal{C}$ be a small category, $\mathcal{E}$ a Grothendieck topos, $P : \mathcal{C} \rightarrow \mathbf{Set}$ a functor and $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ a flat functor. Let $\pi_P : \int P \rightarrow \mathcal{C}^{\text{op}}$ be the canonical projection from the category of elements of P to $\mathcal{C}^{\text{op}}$. Then*

$$\text{colim}(F \circ \pi_P) \cong \left(\coprod_{(c,x) \in \int P} F(c) \right) / R,$$

where R is the equivalence relation in $\mathcal{E}$ defined by saying that, for any objects (c, x) and (d, y) of the category $\int P$, the geometric bisequent

$$\left(\ulcorner R^{\ulcorner \ulcorner \xi_{(c,x)} \urcorner (u), \ulcorner \xi_{(d,y)} \urcorner (v) \urcorner} \dashv \vdash_{u^{F(c)}, v^{F(d)}} \right. \\ \left. \bigvee_{\substack{c \xrightarrow{f} a \xrightarrow{g} d \\ P(f)(x) = P(g)(y)}} (\exists \xi^{F(a)}) (\ulcorner F(f) \urcorner (\xi) = u \wedge \ulcorner F(g) \urcorner (\xi) = v) \right)$$

is valid in the $\Sigma_{\mathcal{E}}$ -structure $\mathcal{S}_{\mathcal{E}}$, where $\xi_{(c,x)} : F(c) \rightarrow \coprod_{(c,x) \in \int P} F(c)$ are the canonical coproduct arrows. In particular, for any objects $(c, x), (c, y)$ of the category $\int P$, the geometric bisequent

$$\left(\ulcorner R \urcorner (\ulcorner \xi_{(c,x)} \urcorner (u), \ulcorner \xi_{(c,y)} \urcorner (u)) \dashv\vdash_{u^{F(c)}} \bigvee_{\substack{c \xrightarrow{f} a \\ P(f)(x)=P(f)(y)}} (\exists \xi^{F(a)}) (\ulcorner F(f) \urcorner (\xi) = u) \right)$$

is valid in $\mathcal{S}_{\mathcal{E}}$.

Proof Let T be the equivalence relation on $\coprod_{(c,x) \in \int P} F(c)$ corresponding to the quotient $\coprod_{(c,x) \in \int P} F(c) \rightarrow \text{colim}(F \circ \pi_P)$. Then $\text{colim}(F \circ \pi_P) \cong \left(\coprod_{(c,x) \in \int P} F(c) \right) / T$, and by Lemma 5.1.28 we have that T is the equivalence relation on $\coprod_{(c,x) \in \int P} F(c)$ generated by the image of the arrow $\langle a, b \rangle$, where $a, b : \coprod_{z: (c,x) \rightarrow (d,y) \text{ in } \int P} F(c) \rightarrow$

$\coprod_{(c,x) \in \int P} F(c)$ are the arrows defined at the beginning of this section for the functor $H = F \circ \pi_P$.

Let $\lambda_z : F(c) \rightarrow \coprod_{z: (c,x) \rightarrow (d,y) \text{ in } \int P} F(c)$ be the canonical coproduct arrows (for each arrow $z : (c, x) \rightarrow (d, y)$ of the category $\int P$). For any objects $(c, x), (d, y)$ of $\int P$, let the expression $(u^{F(c)}, v^{F(d)}) \in E_{(c,x), (d,y)}$ be an abbreviation of the formula $\bigvee_{\substack{c \xrightarrow{f} a \xrightarrow{g} d \\ P(f)(x)=P(g)(y)}} (\exists \xi^{F(a)}) (\ulcorner F(f) \urcorner (\xi) = u \wedge \ulcorner F(g) \urcorner (\xi) = v)$. Notice that, for any arrow $z : (c, x) \rightarrow (d, y)$ in $\int P$, the sequent

$$(\top \vdash_{u^{F(c)}} (\ulcorner a \urcorner (\ulcorner \lambda_z \urcorner (u') \urcorner), \ulcorner b \urcorner (\ulcorner \lambda_z \urcorner (u') \urcorner))) \in E_{(c,x), (d,y)}$$

is valid in $\mathcal{S}_{\mathcal{E}}$.

The fact that the right-to-left implication of the bisequent in the statement of the proposition holds in $\mathcal{S}_{\mathcal{E}}$ is clear.

The validity of the left-to-right implication will follow immediately from Lemma 5.1.30 once we have shown that for any objects $(c, x), (d, y), (e, z)$ in $\int P$ the sequent

$$((u, v) \in E_{(c,x), (d,y)} \wedge (v, w) \in E_{(d,y), (e,z)} \vdash_{u^{F(c)}, v^{F(d)}, w^{F(e)}} (u, w) \in E_{(c,x), (e,z)})$$

is valid in the $\Sigma_{\mathcal{E}}$ -structure $\mathcal{S}_{\mathcal{E}}$. For simplicity we shall give a proof only in the case $\mathcal{E} = \mathbf{Set}$, but all of our arguments are formalizable in the internal logic of the topos $\mathcal{E}$ and hence are valid in general; in fact, the lift of the proof from the set-theoretic to the topos-theoretic setting is made possible by the following logical characterization of flatness (cf. Proposition 1.1.32): a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ is flat if and only if the following sequents are satisfied in the $\Sigma_{\mathcal{E}}$ -structure $\mathcal{S}_{\mathcal{E}}$:

$$\left(\top \vdash_{\square} \bigvee_{c \in \text{Ob}(\mathcal{C})} (\exists x^{F(c)}) (x = x) \right);$$

$$\left(\top \vdash_{x^{F(a)}, y^{F(b)}} \bigvee_{a \xrightarrow{f} c \xrightarrow{g} b} (\exists z^{F(c)}) (\ulcorner F(f) \urcorner (z) = x \wedge \ulcorner F(g) \urcorner (z) = y) \right)$$

for any objects a, b of $\mathcal{C}$;

$$\left(\ulcorner F(f) \urcorner(x) = \ulcorner F(g) \urcorner(x) \vdash_{x^{F(a)}} \bigvee_{h:a \rightarrow c \mid h \circ f = h \circ g} (\exists z^{F(c)})(\ulcorner F(h) \urcorner(z) = x) \right)$$

for any pair of arrows $f, g : b \rightarrow a$ in $\mathcal{C}$ with common domain and codomain.

Now, given objects $(c, x), (d, y), (e, z) \in \int P$ and elements $u \in F(c), v \in F(d), w \in F(e)$, suppose that $(u, v) \in E_{(c,x),(d,y)}$ and $(v, w) \in E_{(d,y),(e,z)}$; we want to prove that $(u, w) \in E_{(c,x),(e,z)}$.

As $(u, v) \in E_{(c,x),(d,y)}$, there exist arrows $f : c \rightarrow a$ and $g : d \rightarrow a$ in $\mathcal{C}$ and an element $\xi \in F(a)$ such that $F(f)(\xi) = u$, $F(g)(\xi) = v$ and $P(f)(x) = P(g)(y)$. Similarly, as $(v, w) \in E_{(d,y),(e,z)}$, there exist arrows $h : d \rightarrow b$ and $k : e \rightarrow b$ in $\mathcal{C}$ and an element $\chi \in F(b)$ such that $F(h)(\chi) = v$, $F(k)(\chi) = w$ and $P(h)(y) = P(k)(z)$.

Using the second condition in the definition of a filtering functor, we obtain the existence of an object m of $\mathcal{C}$, arrows $r : a \rightarrow m$ and $t : b \rightarrow m$ in $\mathcal{C}$ and an element $\epsilon \in F(m)$ such that $F(r)(\epsilon) = \xi$ and $F(t)(\epsilon) = \chi$. Consider now the arrows $r \circ g, t \circ h : d \rightarrow m$; we have that $F(r \circ g)(\epsilon) = F(t \circ h)(\epsilon)$ and hence by the third condition in the definition of a flat functor there exist an arrow $s : m \rightarrow n$ in $\mathcal{C}$ such that $s \circ r \circ g = s \circ t \circ h$ and an element $\alpha \in F(n)$ such that $F(s)(\alpha) = \epsilon$.

The arrows $s \circ r \circ f$ and $s \circ t \circ k$ satisfy the appropriate conditions ensuring that $(u, w) \in E_{(c,x),(e,z)}$, i.e. $F(s \circ r \circ f)(\alpha) = u$, $F(s \circ t \circ k)(\alpha) = w$ and $P(s \circ r \circ f)(x) = P(s \circ t \circ k)(z)$. Indeed, $F(s \circ r \circ f)(\alpha) = F(f)(F(r)(F(s)(\alpha))) = F(f)(F(r)(\epsilon)) = F(f)(\xi) = u$, $F(s \circ t \circ k)(\alpha) = F(k)(F(t)(F(s)(\alpha))) = F(k)(F(t)(\epsilon)) = F(k)(\chi) = w$ and $P(s \circ r \circ f)(x) = P(s \circ r)(P(f)(x)) = P(s \circ r)(P(g)(y)) = P(s \circ r \circ g)(y) = P(s \circ t \circ h)(y) = P(s \circ t)(P(h)(y)) = P(s \circ t)(P(k)(z)) = P(s \circ t \circ k)(z)$, as required.

This completes the proof of the first statement of the proposition; the second statement follows from the first by an easy application of the third condition in the definition of a filtering functor. $\square$

The following proposition represents the translation of Proposition 5.1.33 into the categorical language of generalized elements:

Proposition 5.1.34 *Let $\mathcal{C}$ be a small category, $\mathcal{E}$ a Grothendieck topos with a separating family of objects $\mathcal{S}$, $P : \mathcal{C} \rightarrow \mathbf{Set}$ a functor and $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ a flat functor. Let $\pi_P : \int P \rightarrow \mathcal{C}^{\text{op}}$ be the canonical projection from the category of elements of P to $\mathcal{C}^{\text{op}}$. Then the equivalence relation R defined by the formula*

$$\text{colim}(F \circ \pi_P) \cong \left(\coprod_{(c,x) \in \int P} F(c) \right) / R$$

admits the following characterization: for any objects $(c, x), (d, y)$ of $\int P$ and generalized elements $u : E \rightarrow F(c), v : E \rightarrow F(d)$ (where $E \in \mathcal{S}$), $(\xi_{(c,x)} \circ u, \xi_{(d,y)} \circ v) \in R_E$ if and only if there exist an epimorphic family $\{e_i : E_i \rightarrow$

$E \mid i \in I, E_i \in \mathcal{S}\}$ and for each $i \in I$ an object $a_i \in \mathcal{C}$, a generalized element $h_i : E_i \rightarrow F(a_i)$ and two arrows $f_i : c \rightarrow a_i$ and $f'_i : d \rightarrow a_i$ in $\mathcal{C}$ such that $P(f_i)(x) = P(f'_i)(y)$ and $\langle F(f_i), F(f'_i) \rangle \circ h_i = \langle u, v \rangle \circ e_i$.

In particular, for any objects (c, x) and (c, y) of $\int P$ and any generalized element $u : E \rightarrow F(c)$ (where $E \in \mathcal{S}$), $(\xi_{(c,x)} \circ u, \xi_{(c,y)} \circ v) \in R_E$ if and only if there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I, E_i \in \mathcal{S}\}$ and for each $i \in I$ an object $a_i \in \mathcal{C}$, a generalized element $h_i : E_i \rightarrow F(a_i)$ and an arrow $f_i : c \rightarrow a_i$ in $\mathcal{C}$ such that $P(f_i)(x) = P(f'_i)(y)$ and $F(f_i) \circ h_i = u \circ e_i$. $\square$

5.2 Extensions of flat functors

In this section, we investigate the operation on flat functors induced by a geometric morphism of toposes. This will be relevant for the characterization of the class of theories classified by a presheaf topos addressed in Chapter 6.

5.2.1 General extensions

Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be essentially small sites and $u : \mathbf{Sh}(\mathcal{D}, K) \rightarrow \mathbf{Sh}(\mathcal{C}, J)$ a geometric morphism. Composition on the left with this morphism induces, via Diaconescu's equivalence, a functor

$$\xi_{\mathcal{E}} : \mathbf{Flat}_K(\mathcal{D}, \mathcal{E}) \rightarrow \mathbf{Flat}_J(\mathcal{C}, \mathcal{E}).$$

This functor can be explicitly described as follows. For any flat K -continuous functor $F : \mathcal{D} \rightarrow \mathcal{E}$ with corresponding geometric morphism $f_F : \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{D}, K)$, the flat functor $\tilde{F} := \xi_{\mathcal{E}}(F) : \mathcal{C} \rightarrow \mathcal{E}$ is given by $f_F^* \circ u^* \circ l_{\mathcal{C}}$, where $l_{\mathcal{C}} : \mathcal{C} \rightarrow \mathbf{Sh}(\mathcal{C}, J)$ is the composite of the Yoneda embedding $y_{\mathcal{C}} : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ with the associated sheaf functor $[\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}, J)$. For any natural transformation $\alpha : F \rightarrow G$ between flat functors in $\mathbf{Flat}_K(\mathcal{D}, \mathcal{E})$, the corresponding natural transformation $f_{\alpha}^* : f_F^* \rightarrow f_G^*$, applied to the functor $u^* \circ l_{\mathcal{C}}$, gives rise to a natural transformation $\tilde{\alpha} : \tilde{F} \rightarrow \tilde{G}$ which is precisely $\xi_{\mathcal{E}}(\alpha)$.

We can express $\tilde{F}$ directly in terms of F by using a colimit construction, as follows.

For any $c \in \mathcal{C}$, the K -sheaf $u^*(l_{\mathcal{C}}(c)) : \mathcal{D}^{\text{op}} \rightarrow \mathbf{Set}$, considered as an object of $[\mathcal{D}^{\text{op}}, \mathbf{Set}]$, can be canonically expressed as a colimit of representables indexed over its category of elements $\mathcal{A}_c$; specifically, $u^*(l_{\mathcal{C}}(c)) \cong \text{colim}(y_{\mathcal{D}} \circ \pi_c)$ in $[\mathcal{D}^{\text{op}}, \mathbf{Set}]$, where $\pi_c : \mathcal{A}_c \rightarrow \mathcal{D}$ is the canonical projection functor and $y_{\mathcal{D}} : \mathcal{D} \rightarrow [\mathcal{D}^{\text{op}}, \mathbf{Set}]$ is the Yoneda embedding. As the associated sheaf functor $a_K : [\mathcal{D}^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{D}, K)$ preserves colimits, the functor $u^*(l_{\mathcal{C}}(c))$ is isomorphic in $\mathbf{Sh}(\mathcal{D}, K)$ to the colimit of the functor $l_{\mathcal{D}} \circ \pi_c$. Therefore $\tilde{F}(c) = (f_F^* \circ u^* \circ l_{\mathcal{C}})(c) = f_F^*(\text{colim}(l_{\mathcal{D}} \circ \pi_c)) \cong \text{colim}(f_F^* \circ l_{\mathcal{D}} \circ \pi_c) = \text{colim}(F \circ \pi_c)$.

For any $(d, z) \in \mathcal{A}_c$, we write $\kappa_{(d,z)}^F : F(d) \rightarrow \tilde{F}(c)$ for the canonical colimit arrow.

Proposition 5.2.1

- (i) For any natural transformation $\alpha : F \rightarrow G$ between functors $F, G : \mathcal{D} \rightarrow \mathcal{E}$ in $\mathbf{Flat}_K(\mathcal{D}, \mathcal{E})$, the natural transformation $\tilde{\alpha} : \tilde{F} \rightarrow \tilde{G}$ is characterized by the following condition: for any object (d, z) of the category $\mathcal{A}_c$, the diagram

$$\begin{array}{ccc} \tilde{F}(c) & \xrightarrow{\tilde{\alpha}(c)} & \tilde{G}(c) \\ \uparrow \kappa_{(d,z)}^F & & \uparrow \kappa_{(d,z)}^G \\ F(d) & \xrightarrow{\alpha(d)} & G(d) \end{array}$$

commutes.

- (ii) The functor $\xi_{\mathcal{E}} : \mathbf{Flat}_K(\mathcal{D}, \mathcal{E}) \rightarrow \mathbf{Flat}_J(\mathcal{C}, \mathcal{E})$ is natural in $\mathcal{E}$; that is, for any geometric morphism $f : \mathcal{F} \rightarrow \mathcal{E}$, the diagram

$$\begin{array}{ccc} \mathbf{Flat}_K(\mathcal{D}, \mathcal{E}) & \xrightarrow{\xi_{\mathcal{E}}} & \mathbf{Flat}_J(\mathcal{C}, \mathcal{E}) \\ \downarrow f^* \circ - & & \downarrow f^* \circ - \\ \mathbf{Flat}_K(\mathcal{D}, \mathcal{F}) & \xrightarrow{\xi_{\mathcal{F}}} & \mathbf{Flat}_J(\mathcal{C}, \mathcal{F}) \end{array}$$

commutes.

- (iii) For any arrow $f : c \rightarrow c'$ in $\mathcal{C}$ and any object (d, z) in $\mathcal{A}_c$, the diagram

$$\begin{array}{ccc} \tilde{F}(c) & \xrightarrow{\tilde{F}(f)} & \tilde{F}(c') \\ \uparrow \kappa_{(d,z)}^F & \nearrow \kappa_{(d, u^*(l_{\mathcal{C}}(f))(z))}^F & \\ F(d) & & \end{array}$$

commutes.

Proof

- (i) The given diagram commutes as it is the naturality square for the natural transformation $f_{\alpha}^* : f_F^* \rightarrow f_G^*$ with respect to the canonical colimit arrow $y_{\mathcal{D}}(d) \rightarrow u^*(l_{\mathcal{C}}(c))$ in $[\mathcal{D}^{\text{op}}, \mathbf{Set}]$ corresponding to the object (d, z) of $\mathcal{A}_c$.
- (ii) This follows as an immediate consequence of the fact that Diaconescu's equivalence is natural in $\mathcal{E}$.
- (iii) Given a natural transformation $\beta : P \rightarrow P'$ of presheaves P and P' in $[\mathcal{D}^{\text{op}}, \mathbf{Set}]$, denoting by $\kappa^P : \int P \rightarrow \mathcal{D}$ and $\kappa^{P'} : \int P' \rightarrow \mathcal{D}$ the canonical projection functors and by $\kappa_{(d,z)}^P : y_{\mathcal{D}}(d) \rightarrow P$, $\kappa_{(d',z')}^{P'} : y_{\mathcal{D}}(d') \rightarrow P'$ the canonical colimit arrows in $[\mathcal{D}^{\text{op}}, \mathbf{Set}]$ (for $(d, z) \in \int P$ and $(d', z') \in \int P'$), we clearly have that $\beta \circ \kappa_{(d,z)}^P = \kappa_{(d, \beta(d)(z))}^{P'}$. To obtain our thesis it suffices to apply this result to the natural transformation $\beta : u^*(l_{\mathcal{C}}(f)) : u^*(l_{\mathcal{C}}(c)) \rightarrow u^*(l_{\mathcal{C}}(c'))$; indeed, by taking the image

under the functor f_F^* of the resulting equality we obtain precisely the identity expressing the commutativity of the diagram in the statement of point (iii) of the proposition. $\square$

5.2.2 Extensions along embeddings of categories

Let $\mathcal{D}$ be a subcategory of a small category $\mathcal{C}$. Then the inclusion functor $i : \mathcal{D} \hookrightarrow \mathcal{C}$ induces a geometric morphism $E(i) : [\mathcal{D}, \mathbf{Set}] \rightarrow [\mathcal{C}, \mathbf{Set}]$ and hence, by Diaconescu's equivalence, a functor

$$\xi_{\mathcal{E}} : \mathbf{Flat}(\mathcal{D}^{\text{op}}, \mathcal{E}) \rightarrow \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E}).$$

For $F \in \mathbf{Flat}(\mathcal{D}^{\text{op}}, \mathcal{E})$, we can describe $\tilde{F} = \xi_{\mathcal{E}}(F)$ directly in terms of F as follows. For any $c \in \mathcal{C}$, the functor $E(i)^*(y_{\mathcal{C}}(c)) : \mathcal{D} \rightarrow \mathbf{Set}$ coincides with the functor $\text{Hom}_{\mathcal{C}}(c, i(-)) : \mathcal{D} \rightarrow \mathbf{Set}$. Its category of elements $\mathcal{A}_c$ has as objects the pairs (d, h) where d is an object of $\mathcal{D}$ and h is an arrow $c \rightarrow d$ in $\mathcal{C}$ and as arrows $(d, h) \rightarrow (d', h')$ the arrows $k : d' \rightarrow d$ in $\mathcal{D}$ such that $k \circ h' = h$ in $\mathcal{C}$. As f_F^* preserves all small colimits and $\text{Hom}_{\mathcal{C}}(c, i(-))$ is the colimit in $[\mathcal{D}, \mathbf{Set}]$ of the composition of the canonical projection $\pi_c : \mathcal{A}_c \rightarrow \mathcal{D}^{\text{op}}$ with the Yoneda embedding $\mathcal{D}^{\text{op}} \rightarrow [\mathcal{D}, \mathbf{Set}]$, the object $\tilde{F}(c)$ can be identified with the colimit of the composite functor $F \circ \pi_c : \mathcal{A}_c \rightarrow \mathcal{E}$. For any object d of $\mathcal{D}$, the pair $(d, 1_d)$ yields an object of the category $\mathcal{A}_d$ and hence a canonical colimit arrow $\chi_d : F(d) \rightarrow \tilde{F}(d)$, also denoted by χ_d^F . For any object (d, h) of $\mathcal{A}_c$, the colimit arrow $\kappa_{(d, h)} : F(d) \rightarrow \tilde{F}(c)$ is equal to the composite $\tilde{F}(h) \circ \chi_d$.

Remarks 5.2.2 (a) The functor $\xi_{\mathcal{E}} : \mathbf{Flat}(\mathcal{D}^{\text{op}}, \mathcal{E}) \rightarrow \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$ coincides with the (restriction to the subcategories of flat functors of the) left Kan extension functor

$$[\mathcal{D}^{\text{op}}, \mathcal{E}] \rightarrow [\mathcal{C}^{\text{op}}, \mathcal{E}]$$

along the inclusion $\mathcal{D}^{\text{op}} \hookrightarrow \mathcal{C}^{\text{op}}$ (cf. Remark 5.2.13(a)).

(b) For any object $d \in \mathcal{D}$ and any natural transformation $\alpha : F \rightarrow G$ between functors in $\mathbf{Flat}(\mathcal{D}^{\text{op}}, \mathcal{E})$, the diagram

$$\begin{array}{ccc} \tilde{F}(d) & \xrightarrow{\tilde{\alpha}(d)} & \tilde{G}(d) \\ \chi_d^F \uparrow & & \uparrow \chi_d^G \\ F(d) & \xrightarrow{\alpha(d)} & G(d) \end{array}$$

commutes (cf. Proposition 5.2.1(a)).

(c) For any object $d \in \mathcal{D}$, the arrow $\chi_d : F(d) \rightarrow \tilde{F}(d)$ is monic. Indeed, it can be identified with the image of the canonical subfunctor inclusion $\text{Hom}_{\mathcal{D}}(d, -) \hookrightarrow \text{Hom}_{\mathcal{C}}(d, i(-))$ under the inverse image f_F^* of the geometric morphism f_F . From this, in view of Remark 5.2.2(b), it follows at once that the functor $\xi_{\mathcal{E}} : \mathbf{Flat}(\mathcal{D}^{\text{op}}, \mathcal{E}) \rightarrow \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$ is faithful.

- (d) For any flat functors $F, G : \mathcal{D}^{\text{op}} \rightarrow \mathcal{E}$ and any natural transformation $\beta : \tilde{F} \rightarrow \tilde{G}$, $\beta = \tilde{\alpha}$ for some natural transformation $\alpha : F \rightarrow G$ if and only if for every $d \in \mathcal{D}$ the diagram

$$\begin{array}{ccc} \tilde{F}(d) & \xrightarrow{\beta(d)} & \tilde{G}(d) \\ \uparrow \chi_d^F & & \uparrow \chi_d^G \\ F(d) & \xrightarrow{\alpha(d)} & G(d) \end{array}$$

commutes. One direction follows from Remark 5.2.2(b). To prove the other one, we observe that $\beta(c) = \tilde{\alpha}(c)$ for all $c \in \mathcal{C}$ if and only if for every object $(a, z) \in \mathcal{A}_c$, $\beta(c) \circ \kappa_{(a,z)} = \tilde{\alpha}(c) \circ \kappa_{(a,z)}$. But $\beta(c) \circ \kappa_{(a,z)} = \beta(c) \circ \tilde{F}(z) \circ \chi_a^F = \tilde{G}(z) \circ \beta(a) \circ \chi_a^F = \tilde{G}(z) \circ \chi_a^G \circ \alpha(a) = \kappa_{(a,z)}^G \circ \alpha(a) = \tilde{\alpha}(c) \circ \kappa_{(a,z)}^F$ (where the third equality follows from the naturality of β and the last from Proposition 5.2.1(a)), as required.

- (e) Let d be an object of $\mathcal{D}$ and $y_{\mathcal{D}}(d) = \text{Hom}_{\mathcal{D}}(-, d) : \mathcal{D}^{\text{op}} \rightarrow \mathbf{Set}$ the representable functor on $\mathcal{D}$ associated with d . Then $\xi_{\mathcal{E}}(y_{\mathcal{D}}(d)) = y_{\mathcal{C}}(d)$, where $y_{\mathcal{C}}(d) = \text{Hom}_{\mathcal{C}}(-, d)$ is the representable functor on $\mathcal{C}$ associated with d (considered here as an object of $\mathcal{C}$). Indeed, the flat functor $y_{\mathcal{D}}(d)$ corresponds to the geometric morphism whose inverse image is the evaluation functor $\text{ev}_d^{\mathcal{D}} : [\mathcal{D}, \mathbf{Set}] \rightarrow \mathbf{Set}$ at the object d , and the composite $\text{ev}_d^{\mathcal{D}} \circ E(i)^*$ coincides with the evaluation functor $\text{ev}_d^{\mathcal{C}} : [\mathcal{C}, \mathbf{Set}] \rightarrow \mathbf{Set}$ at the object d , which corresponds to the flat functor $y_{\mathcal{C}}(d)$.
- (f) Let $\mathcal{E}$ be a Grothendieck topos and $\gamma_{\mathcal{E}} : \mathcal{E} \rightarrow \mathbf{Set}$ the unique (up to isomorphism) geometric morphism from $\mathcal{E}$ to $\mathbf{Set}$. Then, by Proposition 5.2.1(ii) and Remark 5.2.2(e), for any object d of $\mathcal{D}$ the functor $\xi_{\mathcal{E}} : \mathbf{Flat}(\mathcal{D}^{\text{op}}, \mathcal{E}) \rightarrow \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$ sends the flat functor $\gamma_{\mathcal{E}}^* \circ y_{\mathcal{D}}(d)$ to the functor $\gamma_{\mathcal{E}}^* \circ y_{\mathcal{C}}(d)$.

Proposition 5.2.3 *Under the natural equivalences*

$$e_{\mathcal{C}} : \text{Ind-}\mathcal{C} \simeq \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathbf{Set})$$

and

$$e_{\mathcal{D}} : \text{Ind-}\mathcal{D} \simeq \mathbf{Flat}(\mathcal{D}^{\text{op}}, \mathbf{Set}),$$

the functor

$$\xi_{\mathbf{Set}} : \mathbf{Flat}(\mathcal{D}^{\text{op}}, \mathbf{Set}) \rightarrow \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathbf{Set})$$

corresponds to the functor

$$\text{Ind-}i : \text{Ind-}\mathcal{D} \rightarrow \text{Ind-}\mathcal{C}.$$

Proof First, let us recall the definition of the functor $\text{Ind-}i : \text{Ind-}\mathcal{D} \rightarrow \text{Ind-}\mathcal{C}$. For any flat functor $F : \mathcal{D}^{\text{op}} \rightarrow \mathbf{Set}$, $(\text{Ind-}i)(F)$ is the functor given by the colimit in $\text{Ind-}\mathcal{C}$ of the composite functor $i \circ \pi_F$, where $\pi_F : \int F \rightarrow \mathcal{D}$ is the canonical

projection from the category of elements $\int F$ of F to $\mathcal{D}$. For any $c \in \mathcal{C}$, we thus have $\text{Ind-}i(F)(c) = \text{colim}(\text{ev}_c \circ i \circ \pi_F)$, where $\text{ev}_c : \text{Ind-}\mathcal{C} \rightarrow \mathbf{Set}$ is the evaluation functor at c .

Now, the functor $\text{ev}_c \circ i : \mathcal{D} \rightarrow \mathbf{Set}$ is equal to the functor $\text{Hom}_{\mathcal{C}}(c, i(-)) : \mathcal{D} \rightarrow \mathbf{Set}$ considered above, and $\tilde{F}(c) = \text{colim}(F \circ \pi_c) : \mathcal{A}_c \rightarrow \mathbf{Set}$, where $\mathcal{A}_c$ is the category of elements of $\text{ev}_c \circ i$ and $\pi_c : \mathcal{A}_c \rightarrow \mathcal{D}^{\text{op}}$ the associated canonical projection. The commutativity of the tensor product between a presheaf and a covariant set-valued functor (cf. chapter VII of [63] or Theorem 5.1.5 above) thus yields a natural isomorphism between the two sets, as required. $\square$

Corollary 5.2.4 *Let $\mathcal{D} \hookrightarrow \mathcal{C}$ be an embedding of small categories and $F : \mathcal{D}^{\text{op}} \rightarrow \mathcal{E}$ a flat functor. With the above notation, for any object $c \in \mathcal{C}$ we have*

$$\tilde{F}(c) \cong \left(\coprod_{(a,z) \in \mathcal{A}_c} F(a) \right) / R_c,$$

where R_c is the equivalence relation in $\mathcal{E}$ defined by saying that, for any objects $(a, z), (a', z')$ of the category $\mathcal{A}_c$, the geometric bisquent

$$\left(\ulcorner R_c \urcorner (\ulcorner \xi_{(a,z)} \urcorner (x), \ulcorner \xi_{(a',z')} \urcorner (x')) \dashv\vdash_{x^{F(a)}, x'^{F(a')}} \bigvee_{\substack{a \xrightarrow{f} b \xrightarrow{g} a' \\ f \circ z = g \circ z'}} (\exists y^{F(b)}) (\ulcorner F(f) \urcorner (y) = x \wedge \ulcorner F(g) \urcorner (y) = x') \right)$$

is valid in the tautological $\Sigma_{\mathcal{E}}$ -structure $\mathcal{S}_{\mathcal{E}}$, where, for any object (a, z) of the category $\mathcal{A}_c$, $\xi_{(a,z)} : F(a) \rightarrow \coprod_{(a,z) \in \mathcal{A}_c} F(a)$ is the canonical coproduct arrow.

In particular, for any objects $(a, z), (a, z')$ of the category $\mathcal{A}_c$, the geometric bisquent

$$\left(\ulcorner R_c \urcorner (\ulcorner \xi_{(a,z)} \urcorner (x), \ulcorner \xi_{(a,z')} \urcorner (x)) \dashv\vdash_{x^{F(a)}} \bigvee_{\substack{a \xrightarrow{f} b \\ f \circ z = f \circ z'}} (\exists y^{F(b)}) (\ulcorner F(f) \urcorner (y) = x) \right)$$

is valid in $\mathcal{S}_{\mathcal{E}}$.

Semantically, the relation R_c can be characterized by saying that, for any objects $(a, z), (a', z')$ of the category $\mathcal{A}_c$ and any generalized elements $x : E \rightarrow F(a)$, $x' : E \rightarrow F(a')$, $(\xi_{(a,z)} \circ x, \xi_{(a',z')} \circ x') \in R_c$ if and only if there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object $b_i \in \mathcal{D}$, a generalized element $h_i : E_i \rightarrow F(b_i)$ and two arrows $f_i : a \rightarrow b_i$ and $f'_i : a' \rightarrow b_i$ in $\mathcal{D}$ such that $f'_i \circ z' = f_i \circ z$ and $\langle F(f_i), F(f'_i) \rangle \circ h_i = \langle x, x' \rangle \circ e_i$. In particular, for any objects (a, z) and (a, z') of $\mathcal{A}_c$ and any generalized element $x : E \rightarrow F(a)$, $(\xi_{(a,z)} \circ x, \xi_{(a,z')} \circ x) \in R_c$ if and only if there exist an epimorphic

family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object $b_i \in \mathcal{D}$, a generalized element $h_i : E_i \rightarrow F(b_i)$ and an arrow $f_i : a \rightarrow b_i$ in $\mathcal{D}$ such that $f_i \circ z' = f_i \circ z$ and $F(f_i) \circ h_i = x \circ e_i$.

Proof Our thesis follows by applying Proposition 5.1.33 (and its categorical reformulation provided by Proposition 5.1.34) to the flat functor $F : \mathcal{D}^{\text{op}} \rightarrow \mathcal{E}$ and the functor $P : \mathcal{D} \rightarrow \mathbf{Set}$ given by $\text{ev}_c \circ i = \text{Hom}_{\mathcal{D}}(c, i(-)) : \mathcal{D} \rightarrow \mathbf{Set}$, whose category of elements coincides with $\mathcal{A}_c$ and whose associated fibration π_P coincides with $\pi_c : \mathcal{A}_c \rightarrow \mathcal{D}^{\text{op}}$. $\square$

5.2.3 Extensions from categories of set-based models to syntactic categories

Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathcal{K}$ a small category of set-based $\mathbb{T}$ -models. The family of geometric morphisms $\mathbf{Set} \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ corresponding to the $\mathbb{T}$ -models in $\mathcal{K}$ induces a geometric morphism

$$p_{\mathcal{K}} : [\mathcal{K}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

whose associated $\mathbb{T}$ -model in $[\mathcal{K}, \mathbf{Set}]$ is given, at each sort, by the corresponding forgetful functor. We thus have, for each Grothendieck topos $\mathcal{E}$, an induced functor

$$u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}} : \mathbf{Flat}(\mathcal{K}^{\text{op}}, \mathcal{E}) \rightarrow \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E}),$$

which the following theorem describes explicitly. Before stating this result, we need to introduce some notation. For any object $\{\vec{x}. \phi\}$ of $\mathcal{C}_{\mathbb{T}}$, we write $\mathcal{A}_{\{\vec{x}. \phi\}}^{\mathcal{K}}$ (or simply $\mathcal{A}_{\{\vec{x}. \phi\}}$ when the category $\mathcal{K}$ can be obviously inferred from the context) for the category whose objects are the pairs (M, w) , where $M \in \mathcal{K}$ and $w \in [[\vec{x}. \phi]]_M$, and whose arrows $(M, w) \rightarrow (N, z)$ are the $\mathbb{T}$ -model homomorphisms $g : N \rightarrow M$ in $\mathcal{K}$ such that $g(z) = w$. We denote by $\pi_{\{\vec{x}. \phi\}}^{\mathcal{K}} : \mathcal{A}_{\{\vec{x}. \phi\}}^{\mathcal{K}} \rightarrow \mathcal{K}^{\text{op}}$ the canonical projection functor.

Theorem 5.2.5 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathcal{K}$ a small category of set-based $\mathbb{T}$ -models. Then for any flat functor $F : \mathcal{K}^{\text{op}} \rightarrow \mathcal{E}$ the functor $\tilde{F} := u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}}(F) : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{E}$ sends any object $\{\vec{x}. \phi\}$ of $\mathcal{C}_{\mathbb{T}}$ to the colimit $\text{colim}(F \circ \pi_{\{\vec{x}. \phi\}}^{\mathcal{K}})$ and acts on the arrows in the obvious way. In particular, for any formula $\{\vec{x}. \phi\}$ presenting a $\mathbb{T}$ -model $M_{\{\vec{x}. \phi\}}$ in $\mathcal{K}$ (in the sense of Definition 6.1.11), $u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}}(F)(\{\vec{x}. \phi\}) \cong F(M_{\{\vec{x}. \phi\}})$.*

Proof Let $g_F : \mathcal{E} \rightarrow [\mathcal{K}, \mathbf{Set}]$ be the geometric morphism corresponding, via Diaconescu's equivalence, to the flat functor F . Then the functor $u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}}(F)$ is equal to the composite $g_F^* \circ p_{\mathcal{K}}^* \circ y$, where $y : \mathcal{C}_{\mathbb{T}} \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ is the Yoneda embedding.

Now, for any geometric formula $\{\vec{x}. \phi\}$ over Σ , $p_{\mathcal{K}}^*(y(\{\vec{x}. \phi\}))$ is the functor $F_{\{\vec{x}. \phi\}}$ sending any model $M \in \mathcal{K}$ to the set $[[\vec{x}. \phi]]_M$. This functor can be expressed as the colimit $\text{colim}(y' \circ \pi_{\{\vec{x}. \phi\}}^{\mathcal{K}})$, where $y' : \mathcal{K}^{\text{op}} \rightarrow [\mathcal{K}, \mathbf{Set}]$ is the Yoneda embedding, since the functor $\pi_{\{\vec{x}. \phi\}}^{\mathcal{K}}$ coincides with the canonical projection from the category of elements of the functor $F_{\{\vec{x}. \phi\}}$ to $\mathcal{K}^{\text{op}}$.

Therefore we have $u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}}(F)(\{\vec{x} \cdot \phi\}) = g_F^*(\text{colim}(y' \circ \pi_{\{\vec{x} \cdot \phi\}})) \cong \text{colim}(g_F^* \circ y' \circ \pi_{\{\vec{x} \cdot \phi\}}^{\mathcal{K}}) \cong \text{colim}(F \circ \pi_{\{\vec{x} \cdot \phi\}}^{\mathcal{K}})$, as required. $\square$

Remark 5.2.6 From the proof of Theorem 5.2.5 it is clear that the isomorphism $u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}}(F)(\{\vec{x} \cdot \phi\}) \cong F(M_{\{\vec{x} \cdot \phi\}})$ is natural in $\{\vec{x} \cdot \phi\}$, that is for any geometric formulae $\{\vec{x} \cdot \phi\}$ and $\{\vec{y} \cdot \psi\}$ respectively presenting $\mathbb{T}$ -models $M_{\{\vec{x} \cdot \phi\}}$ and $M_{\{\vec{y} \cdot \psi\}}$ and any $\mathbb{T}$ -provably functional formula $\theta : \{\vec{x} \cdot \phi\} \rightarrow \{\vec{y} \cdot \psi\}$ inducing a $\mathbb{T}$ -model homomorphism $M_{[\theta]} : M_{\{\vec{y} \cdot \psi\}} \rightarrow M_{\{\vec{x} \cdot \phi\}}$ (defined by $M_{[\theta]}(\vec{\xi}_{\psi}) = [[\vec{x}, \vec{y}, \theta]]_{M_{\{\vec{x} \cdot \phi\}}}(\vec{\xi}_{\phi})$, where $\vec{\xi}_{\phi}$ and $\vec{\xi}_{\psi}$ are respectively the tuples of generators of $M_{\{\vec{x} \cdot \phi\}}$ and of $M_{\{\vec{y} \cdot \psi\}}$ and $[[\vec{x}, \vec{y}, \theta]]_{M_{\{\vec{x} \cdot \phi\}}}(\vec{\xi}_{\phi})$ is the unique element $z \in [[\vec{y} \cdot \psi]]_{M_{\{\vec{x} \cdot \phi\}}}$ such that $(\vec{\xi}_{\phi}, z) \in [[\vec{x}, \vec{y}, \theta]]_{M_{\{\vec{x} \cdot \phi\}}}$), the arrow $u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}}(F)(\theta)$ corresponds to the arrow $F(M_{[\theta]})$ via the isomorphisms $u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}}(F)(\{\vec{x} \cdot \phi\}) \cong F(M_{\{\vec{x} \cdot \phi\}})$ and $u_{(\mathcal{K}, \mathcal{E})}^{\mathbb{T}}(F)(\{\vec{y} \cdot \psi\}) \cong F(M_{\{\vec{y} \cdot \psi\}})$.

Corollary 5.2.7 Let $\mathbb{T}$ be a geometric theory over a signature Σ , $\mathcal{K}$ a small category of set-based $\mathbb{T}$ -models, $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ a geometric sequent over Σ and $F : \mathcal{K}^{\text{op}} \rightarrow \mathcal{E}$ a flat functor. If σ is valid in every $\mathbb{T}$ -model in $\mathcal{K}$ then $\tilde{F}(\{\vec{x} \cdot \phi\}) \leq \tilde{F}(\{\vec{x} \cdot \psi\})$ as subobjects of $\tilde{F}(\{\vec{x} \cdot \top\})$ in $\mathcal{E}$.

Proof By Theorem 5.2.5 we have $\tilde{F}(\{\vec{x} \cdot \phi\}) = \text{colim}(F \circ \pi_{\{\vec{x} \cdot \phi\}}^{\mathcal{K}})$, $\tilde{F}(\{\vec{x} \cdot \psi\}) = \text{colim}(F \circ \pi_{\{\vec{x} \cdot \psi\}}^{\mathcal{K}})$ and $\tilde{F}(\{\vec{x} \cdot \top\}) = \text{colim}(F \circ \pi_{\{\vec{x} \cdot \top\}}^{\mathcal{K}})$. Now, $\mathcal{A}_{\{\vec{x} \cdot \phi\}}$ and $\mathcal{A}_{\{\vec{x} \cdot \psi\}}$ canonically embed as subcategories of $\mathcal{A}_{\{\vec{x} \cdot \top\}}$ and if σ is valid in every $\mathbb{T}$ -model in $\mathcal{K}$ then we have a canonical functor $i : \mathcal{A}_{\{\vec{x} \cdot \phi\}} \rightarrow \mathcal{A}_{\{\vec{x} \cdot \psi\}}$ which commutes with these embeddings. It thus follows from the functoriality of colimits that $\tilde{F}(\{\vec{x} \cdot \phi\}) \leq \tilde{F}(\{\vec{x} \cdot \psi\})$ as subobjects of $\tilde{F}(\{\vec{x} \cdot \top\})$ in $\mathcal{E}$, as required. $\square$

Let us now apply Proposition 5.1.34 in the context of extensions $F \rightarrow \tilde{F}$ of flat functors induced by the geometric morphism

$$p_{\mathcal{K}} : [\mathcal{K}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}).$$

Proposition 5.2.8 Let $\mathbb{T}$ be a geometric theory over a signature Σ , $\mathcal{K}$ a small category of set-based $\mathbb{T}$ -models, $\mathcal{E}$ a Grothendieck topos with a separating family of objects $\mathcal{S}$ and $F : \mathcal{K}^{\text{op}} \rightarrow \mathcal{E}$ a flat functor. With the above notation, for any geometric formula-in-context $\phi(\vec{x})$ over Σ , we have

$$\tilde{F}(\{\vec{x} \cdot \phi\}) \cong \left(\coprod_{(M, z) \in \mathcal{A}_{\{\vec{x} \cdot \phi\}}} F(M) \right) / R_{\{\vec{x} \cdot \phi\}},$$

where $R_{\{\vec{x} \cdot \phi\}}$ is the equivalence relation in $\mathcal{E}$ defined by saying that for any objects $(M, z), (N, w)$ of the category $\mathcal{A}_{\{\vec{x} \cdot \phi\}}$ and any generalized elements $x : E \rightarrow F(M)$, $x' : E \rightarrow F(N)$ (where $E \in \mathcal{S}$), we have $(\xi_{(M, z)} \circ x, \xi_{(N, w)} \circ x') \in R_{\{\vec{x} \cdot \phi\}}$ if and only if there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$ -model a_i in $\mathcal{K}$, a generalized element

$h_i : E_i \rightarrow F(a_i)$ and two $\mathbb{T}$ -model homomorphisms $f_i : M \rightarrow a_i$ and $f'_i : N \rightarrow a_i$ in $\mathcal{K}$ such that $f_i(z) = f'_i(w)$ and $\langle F(f_i), F(f'_i) \rangle \circ h_i = \langle x, x' \rangle \circ e_i$ (where for any object (M, z) of the category $\mathcal{A}_{\{\bar{x} \cdot \phi\}}$, $\xi_{(M, z)} : F(M) \rightarrow \tilde{F}(\{\bar{x} \cdot \phi\})$ is the canonical colimit arrow).

In particular, for any objects (M, z) and (M, w) of $\mathcal{A}_{\{\bar{x} \cdot \phi\}}$ and any generalized element $x : E \rightarrow F(M)$ (where $E \in \mathcal{S}$), $(\xi_{(M, z)} \circ x, \xi_{(M, w)} \circ x) \in R_{\{\bar{x} \cdot \phi\}}$ if and only if there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$ -model a_i in $\mathcal{K}$, a generalized element $h_i : E_i \rightarrow F(a_i)$ and a $\mathbb{T}$ -model homomorphism $f_i : M \rightarrow a_i$ in $\mathcal{K}$ such that $f_i(z) = f_i(w)$ and $F(f_i) \circ h_i = x \circ e_i$. $\square$

Let $M_{\mathbb{T}}$ be the universal model of $\mathbb{T}$ in its syntactic category $\mathcal{C}_{\mathbb{T}}$ (as in Definition 1.4.4). As we shall see below, for any small subcategory $\mathcal{K}$ of the category $\mathbb{T}\text{-mod}(\mathbf{Set})$ and any flat functor $F : \mathcal{K}^{\text{op}} \rightarrow \mathcal{E}$, we can represent the model $\tilde{F}(M_{\mathbb{T}})$ as an $\mathcal{E}$ -indexed filtered colimit of ‘constant’ $\mathbb{T}$ -models coming from $\mathcal{K}$. For simplicity, we shall first establish an explicit characterization of this colimit representation in the case $\mathcal{E} = \mathbf{Set}$, and then generalize it to an arbitrary Grothendieck topos; finally, we shall discuss the abstract characterization of $\tilde{F}(M_{\mathbb{T}})$ as an $\mathcal{E}$ -indexed colimit by introducing the $\mathcal{E}$ -indexed category $\underline{\mathbb{T}\text{-mod}}(\mathcal{E})$ of models of the theory $\mathbb{T}$ in slices of $\mathcal{E}$.

So let $F : \mathcal{K}^{\text{op}} \rightarrow \mathbf{Set}$ be a flat functor. By Theorem 5.2.5, for any sort A of Σ , $\tilde{F}(M_{\mathbb{T}})A = \tilde{F}(\{x^A \cdot \top\}) = \text{colim}(F \circ \pi_{\{x^A \cdot \top\}})$, where $\pi_{\{x^A \cdot \top\}}$ is the canonical projection functor $\mathcal{A}_{\{x^A \cdot \top\}} \rightarrow \mathcal{K}^{\text{op}}$ from the category of elements $\mathcal{A}_{\{x^A \cdot \top\}}$ of the functor $P_A : \mathcal{K} \rightarrow \mathbf{Set}$ which assigns any model M in $\mathcal{K}$ to the set MA and acts accordingly on the arrows. Now, it follows from the commutativity of the tensor product between a presheaf and a covariant set-valued functor (cf. Chapter VII of [63] or section 5.1.5 above) that $\text{colim}(F \circ \pi_{\{x^A \cdot \top\}}) \cong \text{colim}(P_A \circ \pi_F)$, where $\pi_F : \int F \rightarrow \mathcal{K}$ is the canonical projection functor from the category of elements of the functor F to $\mathcal{K}$. Since F is flat, the category $\int F$ is filtered. Therefore, as filtered colimits in $\mathbb{T}\text{-mod}(\mathbf{Set})$ are computed sortwise as in $\mathbf{Set}$, $\tilde{F}(M_{\mathbb{T}}) \cong \text{colim}(i \circ \pi_F)$, where $i : \mathcal{K} \hookrightarrow \mathbb{T}\text{-mod}(\mathbf{Set})$ is the canonical inclusion functor. So for any object (c, x) of the category $\int F$ we have a $\mathbb{T}$ -model homomorphism $\xi_{(c, x)} : c \rightarrow \tilde{F}(M_{\mathbb{T}})$, which can be expressed in terms of the colimit arrows $\kappa_{(a, y)} : F(c) \rightarrow \tilde{F}(\{x^A \cdot \top\}) = \tilde{F}(M_{\mathbb{T}})A$ (for $y \in cA$ and A a sort of Σ) as follows: for any sort A of Σ , $\xi_{(c, x)}(y) = \kappa_{(c, y)}(x)$. The explicit description of filtered colimits in the category $\mathbf{Set}$ thus yields, for each sort A of Σ , the following characterizing properties of the colimit $\text{colim}(P_A \circ \pi_F)$ in terms of the arrows $\xi_{(c, x)}$:

- (i) For any element x of $\tilde{F}(M_{\mathbb{T}})A$, there exist an object (c, x) of the category $\int F$ and an element y of cA such that $\xi_{(c, x)}(y) = x$.
- (ii) For any objects (c, x) and (c', x') of the category $\int F$ and elements $y \in cA$ and $y' \in c'A$, we have $\xi_{(c, x)}(y) = \xi_{(c', x')}(y')$ if and only if there exist

an object (c'', x'') of $\int F$ and arrows $f : c \rightarrow c''$ and $g : c' \rightarrow c''$ in $\mathcal{K}$ such that $F(f)(x'') = x$, $F(g)(x'') = x'$ and $f_A(y) = g_A(y')$.

Moreover, the filteredness of the category $\int F$ implies the following ‘joint embedding property’: for any objects (c, x) and (c', x') of the category $\int F$, there exist an object (c'', x'') of $\int F$ and arrows $f : c \rightarrow c''$ and $g : c' \rightarrow c''$ in $\mathcal{K}$ such that $F(f)(x'') = x$, $F(g)(x'') = x'$ and $\xi_{(c'', x'')} \circ f = \xi_{(c, x)}$, $\xi_{(c'', x'')} \circ g = \xi_{(c', x')}$.

Let us now proceed to establish the $\mathcal{E}$ -indexed generalization of these characterizing properties of the model $\tilde{F}(M_{\mathbb{T}})$.

Let $F : \mathcal{K}^{\text{op}} \rightarrow \mathcal{E}$ be a flat functor. By Theorem 5.2.5, for any sort A of Σ we have that $\tilde{F}(M_{\mathbb{T}})A = \tilde{F}(\{x^A \cdot \top\}) = \text{colim}(F \circ \pi_{\{x^A \cdot \top\}})$; in particular, for any object (a, y) of the category $\mathcal{A}_{\{x^A \cdot \top\}}$ we have a colimit arrow $\kappa_{(a, y)} : F(a) \rightarrow \tilde{F}(\{x^A \cdot \top\}) = \tilde{F}(M_{\mathbb{T}})A$ in $\mathcal{E}$. More generally, for any formula-in-context $\phi(\vec{x})$ over Σ , any model c in $\mathcal{K}$ and any element $\vec{y}$ of $[[\vec{x} \cdot \phi]]_c$, we have a colimit arrow $\kappa_{(c, \vec{y})} : F(c) \rightarrow \tilde{F}(\{\vec{x} \cdot \phi\})$.

Proposition 5.2.9 *Let $\mathbb{T}$ be a geometric theory over a signature Σ , $\mathcal{K}$ a small subcategory of the category $\mathbb{T}\text{-mod}(\mathbf{Set})$, $\mathcal{E}$ a Grothendieck topos and $F : \mathcal{K}^{\text{op}} \rightarrow \mathcal{E}$ a flat functor. With the above notation, for any pair (c, x) consisting of an object c of $\mathcal{K}$ and a generalized element $x : E \rightarrow F(c)$, there is a Σ -structure homomorphism $\xi_{(c, x)} : c \rightarrow \text{Hom}_{\mathcal{E}}(E, \tilde{F}(M_{\mathbb{T}}))$ defined as follows: for any sort A over Σ , the function $\xi_{(c, x)}_A : cA \rightarrow \text{Hom}_{\mathcal{E}}(E, \tilde{F}(\{x^A \cdot \top\}))$ sends any element $y \in cA$ to the generalized element $\kappa_{(c, y)} \circ x : E \rightarrow \tilde{F}(\{x^A \cdot \top\})$.*

Proof We have to verify that:

- (1) For any function symbol $f : A_1 \cdots A_n \rightarrow B$ of Σ , the diagram

$$\begin{array}{ccc}
 cA_1 \times \cdots \times cA_n & \xrightarrow{cf} & cB \\
 \downarrow \xi_{(c, x)}_{A_1} \times \cdots \times \xi_{(c, x)}_{A_n} & & \downarrow \xi_{(c, x)}_B \\
 \text{Hom}_{\mathcal{E}}(E, \tilde{F}(\{x_1^{A_1} \cdot \top\}) \times \cdots \times \tilde{F}(\{x_n^{A_n} \cdot \top\})) & & \\
 \downarrow i & & \\
 \text{Hom}_{\mathcal{E}}(E, \tilde{F}(\{x_1^{A_1}, \dots, x_n^{A_n} \cdot \top\})) & \xrightarrow{\text{Hom}_{\mathcal{E}}(E, \tilde{F}([f]))} & \text{Hom}_{\mathcal{E}}(E, \tilde{F}(\{x^B \cdot \top\})),
 \end{array}$$

where i is the canonical isomorphism $\tilde{F}(\{x_1^{A_1} \cdot \top\}) \times \cdots \times \tilde{F}(\{x_n^{A_n} \cdot \top\}) \cong \tilde{F}(\{x_1^{A_1}, \dots, x_n^{A_n} \cdot \top\})$ induced by the preservation of finite products by $\tilde{F}$ and $[f]$ is the arrow $[x^B = f(x_1^{A_1}, \dots, x_n^{A_n})] : \{x_1^{A_1}, \dots, x_n^{A_n} \cdot \top\} \rightarrow \{x^B \cdot \top\}$, commutes;

- (2) For any relation symbol $R \rhd A_1 \cdots A_n$ of Σ , we have a commutative diagram

$y \in aA$ and $y' \in bA$, we have that $\xi_{(a,x)}(y) = \xi_{(b,x')}(y')$ if and only if there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$ -model c_i in $\mathcal{K}$, arrows $f_i : a \rightarrow c_i$, $g_i : b \rightarrow c_i$ in $\mathcal{K}$ and a generalized element $x_i : E_i \rightarrow F(c_i)$ such that $\langle x, x' \rangle \circ e_i = \langle F(f_i), F(g_i) \rangle \circ x_i$ and $f_{iA}(y) = g_{iA}(y')$.

Moreover, the following ‘joint embedding property’ holds: for any pairs (a, x) and (b, x') , where a and b are $\mathbb{T}$ -models in $\mathcal{K}$ and $x : E \rightarrow F(a)$, $x' : E \rightarrow F(b)$ are generalized elements, there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each index $i \in I$ a $\mathbb{T}$ -model c_i in $\mathcal{K}$, arrows $f_i : a \rightarrow c_i$, $g_i : b \rightarrow c_i$ in $\mathcal{K}$ and a generalized element $x_i : E_i \rightarrow F(c_i)$ such that $\langle x, x' \rangle \circ e_i = \langle F(f_i), F(g_i) \rangle \circ x_i$ and (by Lemma 5.2.10) the following diagram commutes:

$$\begin{array}{ccc}
 a & \xrightarrow{\xi_{(a,x)}} & \text{Hom}_{\mathcal{E}}(E, M) \\
 & \searrow f_i & \searrow \text{Hom}_{\mathcal{E}}(e_i, M) \\
 & c_i & \xrightarrow{\xi_{(c_i, x_i)}} \text{Hom}_{\mathcal{E}}(E_i, M) \\
 & \nearrow g_i & \nearrow \text{Hom}_{\mathcal{E}}(e_i, M) \\
 b & \xrightarrow{\xi_{(b,x')}} & \text{Hom}_{\mathcal{E}}(E, M)
 \end{array}$$

In fact, such a family can be taken to be the pullback of the family of arrows $\langle F(f), F(g) \rangle : F(c) \rightarrow F(a) \times F(b)$ (indexed by the spans $(f : a \rightarrow c, g : b \rightarrow c)$ in the category $\mathcal{K}$) along the arrow $\langle x, x' \rangle : E \rightarrow F(a) \times F(b)$.

Proof Recall from Theorem 5.2.5 that for any sort A of Σ , $\tilde{F}(M_{\mathbb{T}})A = \text{colim}(F \circ \pi_{\{x^A, \top\}}^{\mathcal{K}})$. But $\pi_{\{x^A, \top\}}^{\mathcal{K}}$ is precisely the projection $\pi_{P_A}^{\mathcal{K}} : \mathcal{K}^{\text{op}} \rightarrow \mathcal{K}^{\text{op}}$ from the category of elements of the functor $P_A : \mathcal{K} \rightarrow \mathbf{Set}$ sending a model N in $\mathcal{C}$ to the set NA . The proposition thus readily follows from Theorem 5.1.25, applied to the functors $P_A : \mathcal{K} \rightarrow \mathbf{Set}$ (with A varying among the sorts of Σ), in view of Theorems 5.1.5 and 5.1.13. $\square$

In order to express the model $\tilde{F}(M_{\mathbb{T}})$ as an $\mathcal{E}$ -indexed filtered colimit of ‘constant’ $\mathbb{T}$ -models coming from $\mathcal{K}$, we need to define the $\mathcal{E}$ -indexed category $\mathbb{T}\text{-mod}(\mathcal{E})$ of models of the theory $\mathbb{T}$ in slices of $\mathcal{E}$.

For any first-order signature Σ and any Grothendieck topos $\mathcal{E}$, we have an $\mathcal{E}$ -indexed category $\Sigma\text{-str}(\mathcal{E})$ whose fibre at $E \in \mathcal{E}$ is the category $\Sigma\text{-str}(\mathcal{E}/E)$ and whose ‘change of base’ functors are the obvious pullback functors. For any sort A of Σ , we have an $\mathcal{E}$ -indexed forgetful functor $U_A : \Sigma\text{-str}(\mathcal{E}) \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ assigning to any Σ -structure M in the topos $\mathcal{E}/E$ the object MA . It is easy to see, by adapting the classical proof for $\mathcal{E} = \mathbf{Set}$ and applying Theorem 5.1.11, that the $\mathcal{E}$ -indexed functors U_A jointly create colimits of $\mathcal{E}$ -indexed diagrams defined on $\mathcal{E}$ -final, $\mathcal{E}$ -filtered subcategories $\underline{\mathcal{A}}_{\mathcal{E}}$ of a small $\mathcal{E}$ -indexed category. Now, for any Grothendieck topos $\mathcal{E}$, every small category $\mathcal{C}$ can be made into an $\mathcal{E}$ -indexed category $\underline{\mathcal{C}}_{\mathcal{E}}$ by setting $\mathcal{C}_E = \mathcal{C}$ for all $E \in \mathcal{E}$ and $\mathcal{C}_{\alpha} = 1_{\mathcal{C}}$ for all arrows α in $\mathcal{E}$.

To any (set-indexed) diagram $D : \mathcal{C} \rightarrow \mathcal{E}$ corresponds an $\mathcal{E}$ -indexed functor $\underline{D}_{\mathcal{E}} : \underline{\mathcal{C}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ sending any object E of $\mathcal{E}$ to the functor $D_E := !^*_E \circ D$, where $!^*_E : \mathcal{E} \rightarrow \mathcal{E}/E$ is the pullback functor along the unique arrow $E \rightarrow 1_{\mathcal{E}}$ in $\mathcal{E}$. Since pullback functors preserve all small limits and colimits, giving a colimiting cocone (resp. a limiting cone) on the diagram D in the classical sense is equivalent to giving an $\mathcal{E}$ -indexed colimiting cocone (resp. limiting cone) over the $\mathcal{E}$ -indexed diagram $\underline{D}_{\mathcal{E}}$. Since the $\mathcal{E}$ -indexed category $\underline{\mathcal{E}}_{\mathcal{E}}$ is complete, for any $\mathcal{E}$ -indexed category $\underline{\mathcal{A}}_{\mathcal{E}}$, the $\mathcal{E}$ -indexed functor category $[\underline{\mathcal{A}}_{\mathcal{E}}, \underline{\mathcal{E}}_{\mathcal{E}}]$ is also complete (since limits are computed pointwise); in particular, it has limits of diagrams defined on $\mathcal{E}$ -indexed categories of the form $\underline{\mathcal{C}}_{\mathcal{E}}$. If $\underline{\mathcal{A}}_{\mathcal{E}}$ is an $\mathcal{E}$ -final, $\mathcal{E}$ -filtered subcategory of a small $\mathcal{E}$ -indexed category, the cocompleteness of $\underline{\mathcal{E}}_{\mathcal{E}}$ as an $\mathcal{E}$ -indexed category ensures that there exists a well-defined $\mathcal{E}$ -indexed colimit functor $\text{colim}_{\mathcal{E}} : [\underline{\mathcal{A}}_{\mathcal{E}}, \underline{\mathcal{E}}_{\mathcal{E}}] \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$. By mimicking the classical proof of the fact that finite limits commute with filtered colimits, it is easy to see that this functor preserves limits of diagrams defined on $\mathcal{E}$ -indexed categories of the form $\underline{\mathcal{C}}_{\mathcal{E}}$ for a finite category $\mathcal{C}$. Since it also preserves colimits, and the structure used for interpreting geometric formulae over Σ only involves set-indexed finite limits and arbitrary colimits, we can conclude that, for any geometric formula $\phi(\vec{x})$ over Σ , the $\mathcal{E}$ -indexed functor $[[\vec{x} \cdot \phi]]_- : \mathbb{T}\text{-mod}(\mathcal{E}) \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ preserves $\mathcal{E}$ -filtered colimits of $\mathcal{E}$ -final and $\mathcal{E}$ -filtered subcategories of a small $\mathcal{E}$ -indexed category. This implies that, for any geometric theory $\mathbb{T}$ over Σ , the $\mathcal{E}$ -indexed full subcategory $\underline{\mathbb{T}\text{-mod}(\mathcal{E})}$ of $\underline{\Sigma\text{-str}(\mathcal{E})}$ is closed in $\underline{\Sigma\text{-str}(\mathcal{E})}$ under $\mathcal{E}$ -indexed colimits of diagrams defined on $\mathcal{E}$ -final, $\mathcal{E}$ -filtered subcategories of a small $\mathcal{E}$ -indexed category.

Now, as we observed in the proof of Proposition 5.2.11, for any sort A of Σ we have a forgetful functor $P_A : \mathcal{K} \rightarrow \mathbf{Set}$ evaluating models in $\mathcal{K}$ at A . Using the notation in the statement of Theorem 5.1.5, we have an associated internal diagram $(P_A)_{\mathcal{E}}$ in $[\mathbb{K}, \mathcal{E}]$ and hence an $\mathcal{E}$ -indexed functor $\underline{\mathbb{K}}_{\mathcal{E}} \rightarrow \underline{\mathcal{E}}_{\mathcal{E}}$ (where $\mathbb{K}$ is the internalization of $\mathcal{K}$ in $\mathcal{E}$). These functors, with A varying among the sorts of Σ , yield an $\mathcal{E}$ -indexed functor $P : \underline{\mathbb{K}}_{\mathcal{E}} \rightarrow \underline{\mathbb{T}\text{-mod}(\mathcal{E})}$. We can thus conclude (in light of the proof of Proposition 5.2.11) that the model $\tilde{F}(M_{\mathbb{T}})$ is the $\mathcal{E}$ -indexed colimit of the diagram $P \circ \pi_{\overline{F}}^f$ (or, equivalently, of its restriction to the $\mathcal{E}$ -strictly final subcategory $\int \underline{F}_{\mathcal{E}}$ of $\int \underline{\overline{F}}_{\mathcal{E}}^f$ – cf. Theorem 5.1.12).

5.2.4 A general adjunction

Let $\mathcal{C}$ be a small category and $\mathcal{E}$ a Grothendieck topos. Recall from section 5.1.1 that the indexed category $[\underline{\mathcal{C}}, \underline{\mathcal{E}}]_{\mathcal{E}}$ is locally small, that is for any two functors $F, G : \mathcal{C} \rightarrow \mathcal{E}$ there exists an object $\text{Hom}_{[\underline{\mathcal{C}}, \underline{\mathcal{E}}]_{\mathcal{E}}}^{\mathcal{E}}(F, G)$ of $\mathcal{E}$ satisfying the universal property that, for any object E of $\mathcal{E}$, the generalized elements $E \rightarrow \text{Hom}_{[\underline{\mathcal{C}}, \underline{\mathcal{E}}]_{\mathcal{E}}}^{\mathcal{E}}(F, G)$ correspond bijectively, naturally in $E \in \mathcal{C}$, to the natural transformations $!^*_E \circ F \rightarrow !^*_E \circ G$.

The following theorem establishes a general adjunction between categories of $\mathcal{E}$ -valued functors induced by a functor $P : \mathcal{C} \rightarrow [\mathcal{D}^{\text{op}}, \mathbf{Set}]$. Before stating it, we need to introduce some notation. For any object c of $\mathcal{C}$, we denote by $\int P(c)$ the

category of elements of the functor $P(c) : \mathcal{D}^{\text{op}} \rightarrow \mathbf{Set}$ and by $\pi_c : \int P(c) \rightarrow \mathcal{D}$ the canonical projection functor. We denote by $y_{\mathcal{D}} : \mathcal{D}^{\text{op}} \rightarrow [\mathcal{D}, \mathbf{Set}]$ the Yoneda embedding. For any functor $H : \mathcal{D} \rightarrow \mathcal{E}$ and any object (d, z) of the category $\int P(c)$, we denote by $\kappa_{(d,z)}^H : H(d) \rightarrow \text{colim}(H \circ \pi_c)$ the canonical colimit arrow.

Theorem 5.2.12 *Let $\mathcal{C}$ and $\mathcal{D}$ be small categories, $\mathcal{E}$ a Grothendieck topos and $P : \mathcal{C} \rightarrow [\mathcal{D}^{\text{op}}, \mathbf{Set}]$ a functor. Let $\widetilde{(-)} : [\mathcal{D}, \mathcal{E}] \rightarrow [\mathcal{C}, \mathcal{E}]$ be the functor sending any functor $F : \mathcal{D} \rightarrow \mathcal{E}$ to the functor $\tilde{F}$ defined as follows (and acting on the arrows of $[\mathcal{D}, \mathcal{E}]$ accordingly):*

*for any $c \in \mathcal{C}$, $\tilde{F}(c) = \text{colim}(F \circ \pi_c)$, and
for any arrow $f : c \rightarrow c'$ in $\mathcal{C}$,*

$$\tilde{F}(f) : \text{colim}(F \circ \pi_c) \rightarrow \text{colim}(F \circ \pi_{c'})$$

is defined by the conditions $\tilde{F}(f) \circ \kappa_{(d,z)}^F = \kappa_{(d, P(f)(z))}^F$ (for any $(d, z) \in \int P(c)$).

Let $(-)_r : [\mathcal{C}, \mathcal{E}] \rightarrow [\mathcal{D}, \mathcal{E}]$ be the functor assigning to any functor $G : \mathcal{C} \rightarrow \mathcal{E}$ the functor $\text{Hom}_{[\mathcal{C}, \mathcal{E}]}^{\mathcal{E}}(\gamma_{\mathcal{E}}^ \circ y_{\mathcal{D}}(-), G)$ and acting on the arrows in the obvious way.*

Then the functors

$$\widetilde{(-)} : [\mathcal{D}, \mathcal{E}] \rightarrow [\mathcal{C}, \mathcal{E}]$$

and

$$(-)_r : [\mathcal{C}, \mathcal{E}] \rightarrow [\mathcal{D}, \mathcal{E}]$$

are adjoint to each other ($\widetilde{(-)}$ on the left and $(-)_r$ on the right). The unit $\eta^F : F \rightarrow (\tilde{F})_r$ and the counit $\varepsilon^G : (\tilde{G}_r) \rightarrow G$ are defined as follows:

- *For any $d \in \mathcal{D}$, $\eta^F(d) : F(d) \rightarrow (\tilde{F})_r(d)$ is the arrow in $\mathcal{E}$ defined by means of generalized elements by saying that it sends any generalized element $E \rightarrow F(d)$, regarded as a natural transformation $\gamma_{\mathcal{E}/E}^* \circ y_{\mathcal{D}}(d) \rightarrow !_E^* \circ F$, to the image $\gamma_{\mathcal{E}/E}^* \circ \widetilde{y_{\mathcal{D}}(d)} \rightarrow !_E^* \circ \tilde{F}$ of this arrow under the functor $\widetilde{(-)}$, regarded as a generalized element $E \rightarrow \text{Hom}_{[\mathcal{C}, \mathcal{E}]}^{\mathcal{E}}(\gamma_{\mathcal{E}}^* \circ \widetilde{y_{\mathcal{D}}(d)}, \tilde{F}) = (\tilde{F})_r(d)$.*
- *For any $c \in \mathcal{C}$, $\varepsilon^G(c) : (\tilde{G}_r)(c) = \text{colim}(G_r \circ \pi_c) \rightarrow G(c)$ is defined by setting, for each object (d, z) of $\int P(c)$, $\varepsilon^G(c) \circ \kappa_{(d,z)}^{G_r}$ equal to the arrow $G_r(d) \rightarrow G(c)$ defined by means of generalized elements by saying that a generalized element $x : E \rightarrow G_r(d)$, corresponding to a natural transformation $\bar{x} : !_E^* \circ \gamma_{\mathcal{E}}^* \circ \widetilde{y_{\mathcal{D}}(d)} \cong \gamma_{\mathcal{E}/E}^* \circ y_{\mathcal{D}}(d) \rightarrow !_E^* \circ \tilde{G}$, is sent to the arrow $E \rightarrow G(c)$ obtained by composing $\bar{x}(c)$ with the component of $\gamma_{\mathcal{E}/E}^*(\widetilde{y_{\mathcal{D}}(d)}(c))$ which corresponds to the element of $\widetilde{y_{\mathcal{D}}(d)}(c)$ given by the image of the identity on d under the colimit arrow $\kappa_{(d,z)}^{y_{\mathcal{D}}(d)} : (y_{\mathcal{D}}(d))(d) \rightarrow \widetilde{y_{\mathcal{D}}(d)}(c)$.*

Proof For simplicity we shall establish the result only in the case $\mathcal{E} = \mathbf{Set}$, the proof of the general case being entirely analogous (the only care that one has to take is to use generalized elements in place of standard set-theoretic elements).

We shall define a bijective correspondence between the natural transformations $\tilde{F} \rightarrow G$ and the natural transformations $F \rightarrow G_r$ which is natural in $F \in [\mathcal{D}, \mathbf{Set}]$ and $G \in [\mathcal{C}, \mathbf{Set}]$.

Given a natural transformation $\alpha : F \rightarrow G_r$, we define $\tau(\alpha) : \tilde{F} \rightarrow G$ by setting, for each $c \in \mathcal{C}$, $\tau_\alpha(c) : \tilde{F}(c) \rightarrow G(c)$ equal to the arrow defined as follows. As $\tilde{F}(c)$ is the colimit of the cone $\{\kappa_{(d,z)}^F : F(d) \rightarrow \tilde{F}(c) \mid (d, z) \in \int P(c)\}$, it suffices to define an arrow $u_{(d,z)} : F(d) \rightarrow G(c)$ for each pair (d, z) where $d \in \mathcal{D}$ and $z \in P(c)(d)$ in such a way that whenever (d, z) and (d', z') are pairs such that $P(c)(f)(z') = z$ for an arrow $f : d \rightarrow d'$ in $\mathcal{D}$ then $u_{(d',z')} \circ F(f) = u_{(d,z)}$; indeed, by the universal property of the colimit, such a family of arrows will induce a unique arrow $\tau_\alpha(c) : \tilde{F}(c) \rightarrow G(c)$ such that $\tau_\alpha(c) \circ \kappa_{(d,z)}^F = u_{(d,z)}$ for each (d, z) in $\int P(c)$. We set $u_{(d,z)} : F(d) \rightarrow G(c)$ equal to the function sending every element $x \in F(d)$ to the element $\alpha(d)(x)(c)(\kappa_{(d,z)}^{y_{\mathcal{D}}(d)}(1_d))$. Given an arrow $f : d \rightarrow d'$ such that $P(c)(f)(z') = z$, we have to verify that, for every $x \in F(d)$, $\alpha(d')(F(f)(x))(c)(\kappa_{(d',z')}^{y_{\mathcal{D}}(d')}(1_{d'})) = \alpha(d)(x)(c)(\kappa_{(d,z)}^{y_{\mathcal{D}}(d)}(1_d))$. This identity immediately follows from the commutativity of the naturality diagram

$$\begin{array}{ccc} F(d) & \xrightarrow{\alpha(d)} & G_r(d) = \text{Hom}_{[\mathcal{C}, \mathbf{Set}]}(\widetilde{y_{\mathcal{D}}(d)}, G) \\ \downarrow F(f) & & \downarrow G_r(f) = - \circ \widetilde{y_{\mathcal{D}}(f)} \\ F(d') & \xrightarrow{\alpha(d')} & G_r(d') = \text{Hom}_{[\mathcal{C}, \mathbf{Set}]}(\widetilde{y_{\mathcal{D}}(d')}, G) \end{array}$$

for α with respect to the arrow f and from that of the diagram

$$\begin{array}{ccc} \widetilde{y_{\mathcal{D}}(d')}(c) & \xrightarrow{\widetilde{y_{\mathcal{D}}(f)}(c)} & \widetilde{y_{\mathcal{D}}(d)}(c) \\ \uparrow \kappa_{(d',z')}^{y_{\mathcal{D}}(d')} & & \uparrow \kappa_{(d,z)}^{y_{\mathcal{D}}(d)} \\ y_{\mathcal{D}}(d') & \xrightarrow{y_{\mathcal{D}}(f)} & y_{\mathcal{D}}(d), \end{array}$$

which is an instance of Proposition 5.2.1(i). Indeed,

$$\begin{aligned} \alpha(d')(F(f)(x))(c)(\kappa_{(d',z')}^{y_{\mathcal{D}}(d')}(1_{d'})) &= \alpha(d)(x)(c)(\widetilde{y_{\mathcal{D}}(f)}(c)(\kappa_{(d',z')}^{y_{\mathcal{D}}(d')}(1_{d'}))) \\ &= \alpha(d)(x)(c)(\kappa_{(d',z')}^{y_{\mathcal{D}}(d)}(f)) \\ &= \alpha(d)(x)(c)(\kappa_{(d,z)}^{y_{\mathcal{D}}(d)}(1_d)), \end{aligned}$$

where the first equality follows from the commutativity of the first diagram, the second follows from the commutativity of the second diagram and the last follows from the fact that the family $\{\kappa_{(d,z)}^{y_{\mathcal{D}}(d)} \mid (d, z) \in \int P(c)\}$ is a cocone.

To complete the definition of $\tau(\alpha)$, it remains to check that the assignment $c \rightarrow \tau(\alpha)(c)$ defines a natural transformation $\tilde{F} \rightarrow G$. We have to verify that, for any arrow $f : c \rightarrow c'$ in $\mathcal{C}$, the diagram

$$\begin{array}{ccc} \tilde{F}(c) & \xrightarrow{\tau(\alpha)(c)} & G(c) \\ \downarrow \tilde{F}(f) & & \downarrow G(f) \\ \tilde{F}(c') & \xrightarrow{\tau(\alpha)(c')} & G(c') \end{array}$$

commutes.

As, by definition of $\tilde{F}(f)$, the diagrams

$$\begin{array}{ccc} \tilde{F}(c) & \xrightarrow{\tilde{F}(f)} & \tilde{F}(c') \\ \kappa_{(d,z)}^F \uparrow & \nearrow \kappa_{(d,P(f)(d)(z))}^F & \\ F(d) & & \end{array}$$

commute and the arrows $\kappa_{(d,z)}^F$ are jointly epimorphic (as they are colimit arrows), the above diagram commutes if and only if the arrows $G(f) \circ \tau(\alpha)(c) \circ \kappa_{(d,z)}^F$ and $\tau(\alpha)(c') \circ \kappa_{(d,P(f)(d)(z))}^F$ are equal, that is if and only if they take the same values at any element $x \in F(d)$. Let us set $z' = P(f)(d)(z)$. By definition of $\tau(\alpha)$,

$$\tau(\alpha)(c)(\kappa_{(d,z)}^F(x)) = \alpha(d)(x)(c)(\kappa_{(d,z)}^{y_{\mathcal{D}}(d)}(1_d)),$$

while

$$\tau(\alpha)(c')(\kappa_{(d,z')}^F(x)) = \alpha(d)(x)(c')(\kappa_{(d,z')}^{y_{\mathcal{D}}(d)}(1_d)).$$

Now, for any $d \in \mathcal{D}$ and any $x \in F(d)$, $\alpha(d)(x)$ is a natural transformation $\widetilde{y_{\mathcal{D}}(d)} \rightarrow G$; in particular, the diagram

$$\begin{array}{ccc} \widetilde{y_{\mathcal{D}}(d)}(c) & \xrightarrow{\alpha(d)(x)(c)} & G(c) \\ \downarrow \widetilde{y_{\mathcal{D}}(d)}(f) & & \downarrow G(f) \\ \widetilde{y_{\mathcal{D}}(d)}(c') & \xrightarrow{\alpha(d)(x)(c')} & G(c') \end{array}$$

commutes.

The commutativity of this diagram, together with that of the diagram

$$\begin{array}{ccc} \widetilde{y_{\mathcal{D}}(d)}(c) & \xrightarrow{\widetilde{y_{\mathcal{D}}(d)}(f)} & \widetilde{y_{\mathcal{D}}(d)}(c') \\ \kappa_{(d,z)}^{y_{\mathcal{D}}(d)} \uparrow & \nearrow \kappa_{(d,z')}^{y_{\mathcal{D}}(d)} & \\ y_{\mathcal{D}}(d)(d) & & \end{array}$$

(which follows by definition of $\widetilde{y_{\mathcal{D}}(d)(f)}$), now immediately imply our thesis.

Let us now define a function χ assigning to any natural transformation $\beta : \tilde{F} \rightarrow G$ a natural transformation $\chi(\beta) : F \rightarrow G_r$. For any $d \in \mathcal{D}$, we define $\chi(\beta)(d) : F(d) \rightarrow G_r(d)$ by setting, for each $x \in F(d)$, $\chi(\beta)(d)(x)$ equal to the natural transformation $\beta \circ \widetilde{a_x} : \widetilde{y_{\mathcal{D}}(d)} \rightarrow G$, where $a_x : y_{\mathcal{D}}(d) \rightarrow F$ is the natural transformation corresponding, via the Yoneda lemma, to the element $x \in F(d)$.

To verify that $\chi(\beta)$ is well-defined, we have to check that, for every arrow $g : d \rightarrow d'$ in $\mathcal{D}$, the diagram

$$\begin{array}{ccc} F(d) & \xrightarrow{\chi(\beta)(d)} & G_r(d) \\ \downarrow F(g) & & \downarrow G_r(g) = - \circ \widetilde{y_{\mathcal{D}}(g)} \\ F(d') & \xrightarrow{\chi(\beta)(d')} & G_r(d') \end{array}$$

commutes, i.e. that, for every $x \in F(d)$, $G_r(g)(\chi(\beta)(d)(x)) = \chi(\beta)(d')(F(g)(x))$. Now, $G_r(g)(\chi(\beta)(d)(x)) = \beta \circ \widetilde{a_x} \circ y_{\mathcal{D}}(g)$, while $\chi(\beta)(d')(F(g)(x)) = \beta \circ \widetilde{a_{F(g)(x)}}$; but $a_{F(g)(x)} = a_x \circ y_{\mathcal{D}}(g)$, as required.

The proof of the fact that the correspondences τ and β are natural in F and G is straightforward and left to the reader.

To conclude the proof of the theorem, it thus remains to show that τ and χ are inverse to each other. The verification of the fact that the unit and counit of the adjunction coincide with those given in the statement of the theorem is straightforward and left to the reader.

Let us show that for any natural transformation $\alpha : F \rightarrow G_r$, $\chi(\tau(\alpha)) = \alpha$. Let us set $\beta = \tau(\alpha)$. We have to prove that for any $d \in \mathcal{D}$, $x \in F(d)$ and $z \in P(c)(d')$, $\alpha(d)(x)(c) \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d)} = \chi(\beta)(d)(x)(c) \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d)}$ as functions $y_{\mathcal{D}}(d)(d') \rightarrow G(c)$, i.e. that for any element $g \in y_{\mathcal{D}}(d)(d') = \text{Hom}_{\mathcal{D}}(d, d')$,

$$\alpha(d)(x)(c)(\kappa_{(d',z)}^{y_{\mathcal{D}}(d)}(g)) = (\chi(\beta)(d)(x)(c) \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d)})(g).$$

By definition of the functor $\widetilde{(-)}$ and of the correspondence τ , the diagram

$$\begin{array}{ccccc} \widetilde{y_{\mathcal{D}}(d)}(c) & \xrightarrow{\widetilde{a_x}(c)} & \tilde{F}(c) & \xrightarrow{\beta(c)} & G(c) \\ \uparrow \kappa_{(d',z)}^{y_{\mathcal{D}}(d)} & & \uparrow \kappa_{(d',z)}^{F} & \nearrow u_{(d',z)} & \\ y_{\mathcal{D}}(d)(d') & \xrightarrow{a_x(d')} & F(d') & & \end{array}$$

commutes, and since $\chi(\beta)(d)(x) = \beta \circ \widetilde{a_x} : \widetilde{y_{\mathcal{D}}(d)} \rightarrow G$, we have that

$$\begin{aligned} (\chi(\beta)(d)(x)(c) \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d)})(g) &= u_{(d',z)} \circ a_x(d')(g) \\ &= \alpha(d')(a_x(d')(g))(c)(\kappa_{(d',z)}^{y_{\mathcal{D}}(d)}(1_{d'})) \\ &= \alpha(d')(F(g)(x))(c)(\kappa_{(d',z)}^{y_{\mathcal{D}}(d')}(1_{d'})). \end{aligned}$$

Now, the naturality diagram for α with respect to the arrow $g : d \rightarrow d'$ yields the equality $\alpha(d)(x) \circ \widetilde{y_{\mathcal{D}}(g)} = \alpha(d')(F(g)(x))$ and hence the equality $\alpha(d)(x)(c) \circ \widetilde{y_{\mathcal{D}}(g)(c)} \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d')} = \alpha(d')(F(g)(x))(c) \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d')}$.

But the commutativity of the diagram

$$\begin{array}{ccc} \widetilde{y_{\mathcal{D}}(d')(c)} & \xrightarrow{\widetilde{y_{\mathcal{D}}(g)(c)}} & \widetilde{y_{\mathcal{D}}(d)(c)} \\ \uparrow \kappa_{(d',z)}^{y_{\mathcal{D}}(d')} & & \uparrow \kappa_{(d',z)}^{y_{\mathcal{D}}(d)} \\ y_{\mathcal{D}}(d')(d') & \xrightarrow{y_{\mathcal{D}}(g)(d')} & y_{\mathcal{D}}(d)(d') \end{array}$$

(which follows by definition of the functor $\widetilde{(-)}$) implies that $\alpha(d)(x)(c) \circ \widetilde{y_{\mathcal{D}}(g)(c)} \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d')} = \alpha(d)(x)(c) \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d)} \circ y_{\mathcal{D}}(g)(d')$. Therefore

$$\alpha(d)(x)(c) \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d)} \circ y_{\mathcal{D}}(g)(d') = \alpha(d')(F(g)(x))(c) \circ \kappa_{(d',z)}^{y_{\mathcal{D}}(d')},$$

and evaluating this at $1_{d'}$ yields the desired equality.

Finally, let us show that the composite $\tau \circ \chi$ is equal to the identity. Let $\beta : \tilde{F} \rightarrow G$ be a natural transformation. We have to show that $\beta = \tau(\chi(\beta))$. Let us set $\alpha = \chi(\beta)$. To prove that $\beta = \tau(\alpha)$, it is equivalent to verify that for every object $c \in \mathcal{C}$, any pair (d, z) with $d \in \mathcal{D}$ and $z \in P(c)(d)$ and any element $x \in F(d)$, $\beta(c)(\kappa_{(d,z)}^F(x)) = \tau(\alpha)(c)(\kappa_{(d,z)}^F(x))$. But $\tau(\alpha)(c)(\kappa_{(d,z)}^F(x)) = \alpha(d)(x)(c)(\kappa_{(d,z)}^{y_{\mathcal{D}}(d)}(1_d)) = \chi(\beta)(d)(x)(c)(\kappa_{(d,z)}^{y_{\mathcal{D}}(d)}(1_d)) = (\beta \circ \widetilde{a_x})(c)(\kappa_{(d,z)}^{y_{\mathcal{D}}(d)}(1_d)) = \beta(c)(\widetilde{a_x}(c)(\kappa_{(d,z)}^{y_{\mathcal{D}}(d)}(1_d)))$. Thus our thesis follows from the commutativity of the diagram

$$\begin{array}{ccc} \widetilde{y_{\mathcal{D}}(d)(c)} & \xrightarrow{\widetilde{a_x}(c)} & \tilde{F}(c) \\ \uparrow \kappa_{(d,z)}^{y_{\mathcal{D}}(d)} & & \uparrow \kappa_{(d,z)}^F \\ y_{\mathcal{D}}(d)(d) & \xrightarrow{a_x(d)} & F(d), \end{array}$$

which is an immediate consequence of the definition of the functor $\widetilde{(-)}$. $\square$

Remarks 5.2.13 (a) Let $f : \mathcal{D} \rightarrow \mathcal{C}$ be a functor between two small categories, and $P : \mathcal{C} \rightarrow [\mathcal{D}^{\text{op}}, \mathbf{Set}]$ be the functor $y_{\mathcal{C}}(-) \circ f^{\text{op}}$, where $y_{\mathcal{C}} : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is the Yoneda embedding. Notice that P is the flat functor corresponding, via Diaconescu's equivalence, to the essential geometric morphism $[\mathcal{D}^{\text{op}}, \mathbf{Set}] \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ induced by the functor $f^{\text{op}} : \mathcal{D}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$. The functor $\widetilde{(-)} : [\mathcal{D}, \mathcal{E}] \rightarrow [\mathcal{C}, \mathcal{E}]$ coincides with the left Kan extension functor along f , while the right adjoint functor $(-)_r$ coincides with the functor $- \circ f : [\mathcal{C}, \mathcal{E}] \rightarrow [\mathcal{D}, \mathcal{E}]$.

(b) Let

$$u : \mathbf{Sh}(\mathcal{D}, K) \rightarrow \mathbf{Sh}(\mathcal{C}, J)$$

be a geometric morphism. Then, for any Grothendieck topos $\mathcal{E}$, u induces as in section 5.2.1 a functor

$$\xi_{\mathcal{E}} : \mathbf{Flat}_K(\mathcal{D}, \mathcal{E}) \rightarrow \mathbf{Flat}_J(\mathcal{C}, \mathcal{E}).$$

Via Diaconescu's equivalence, u corresponds to a flat functor $\mathcal{C} \rightarrow \mathbf{Sh}(\mathcal{D}, K)$ which, composed with the canonical geometric inclusion

$$\mathbf{Sh}(\mathcal{D}, K) \hookrightarrow [\mathcal{D}^{\text{op}}, \mathbf{Set}],$$

yields a functor $P : \mathcal{C} \rightarrow [\mathcal{D}^{\text{op}}, \mathbf{Set}]$. Then $\xi_{\mathcal{E}}$ coincides with the restriction of the functor $\widetilde{(-)} : [\mathcal{D}, \mathcal{E}] \rightarrow [\mathcal{C}, \mathcal{E}]$ induced by P as in Theorem 5.2.12 to the full subcategories $\mathbf{Flat}_K(\mathcal{D}, \mathcal{E}) \hookrightarrow [\mathcal{D}, \mathcal{E}]$ and $\mathbf{Flat}_J(\mathcal{C}, \mathcal{E}) \hookrightarrow [\mathcal{C}, \mathcal{E}]$. In general, the right adjoint functor $(-)_r : [\mathcal{C}, \mathcal{E}] \rightarrow [\mathcal{D}, \mathcal{E}]$ does not restrict to these subcategories, but if it does, it becomes a right adjoint to the functor $\xi_{\mathcal{E}} : \mathbf{Flat}_K(\mathcal{D}, \mathcal{E}) \rightarrow \mathbf{Flat}_J(\mathcal{C}, \mathcal{E})$.

This notably applies in the case of the geometric morphism

$$p_K : [\mathcal{K}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

considered in section 5.2.3. In this case, $\mathcal{C} = \mathcal{C}_{\mathbb{T}}$, $\mathcal{D} = \mathcal{K}^{\text{op}}$ and P is the functor $\mathcal{C}_{\mathbb{T}} \rightarrow [\mathcal{K}, \mathbf{Set}]$ sending any geometric formula $\phi(\vec{x})$ to the functor $M \rightarrow [[\vec{x} \cdot \phi]]_M$; it is easy to see that $\widetilde{y_K(-)}$ is the functor $U : \mathcal{K} \rightarrow [\mathcal{C}_{\mathbb{T}}, \mathbf{Set}]$ sending any model c in $\mathcal{K}$ to the functor $\{\vec{x} \cdot \phi\} \rightarrow [[\vec{x} \cdot \phi]]_c$, and that for any G in $[\mathcal{C}_{\mathbb{T}}, \mathcal{E}]$, $G_r = \text{Hom}_{[\mathcal{C}_{\mathbb{T}}, \mathcal{E}]}^{\mathcal{E}}(\gamma_{\mathcal{E}}^* \circ U(-), G)$.

(c) By a basic property of adjoint functors, the (left adjoint) functor

$$\widetilde{(-)} : [\mathcal{D}, \mathcal{E}] \rightarrow [\mathcal{C}, \mathcal{E}]$$

is full and faithful if and only if the unit $\eta^F : F \rightarrow (\tilde{F})_r$ is an isomorphism for every F in $[\mathcal{D}, \mathcal{E}]$. It readily follows (by purely formal considerations) that for every full subcategory $\mathcal{H}$ of $[\mathcal{D}, \mathcal{E}]$ with canonical embedding $i : \mathcal{H} \hookrightarrow [\mathcal{D}, \mathcal{E}]$, the composite functor $\widetilde{(-)} \circ i$ is full and faithful if and only if the unit $\eta^F : F \rightarrow (\tilde{F})_r$ is an isomorphism for every F in $\mathcal{H}$.

The following result will be useful in the sequel; see section 5.1.2 for the notation employed in it.

Proposition 5.2.14 *Let $f : \mathcal{D} \rightarrow \mathcal{C}$ be a functor between small categories, $\mathcal{E}$ a Grothendieck topos and F a functor $\mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$. Then the $\mathcal{E}$ -indexed functor $\int (F \circ f^{\text{op}})_{\mathcal{E}} \rightarrow \int F_{\mathcal{E}}$ sending any object (c, x) of $\int (F \circ f^{\text{op}})_{\mathcal{E}}$ to the object $(f(c), x)$ of $(\int F)_{\mathcal{E}}$ is $\mathcal{E}$ -final if and only if the $\mathcal{E}$ -indexed functor $\int \varepsilon^F_{\mathcal{E}} : \int (F \circ f^{\text{op}})_{\mathcal{E}} \rightarrow \int F_{\mathcal{E}}$ induced by the natural transformation $\varepsilon^F : \widetilde{F \circ f^{\text{op}}} \rightarrow F$ of Theorem 5.2.12 and Remark 5.2.13(a) is $\mathcal{E}$ -final.*

Proof From the general analysis of section 5.2.1 we know that for any $c \in \mathcal{C}$, $\widetilde{F \circ f^{\text{op}}}(c) = \text{colim}(F \circ f^{\text{op}} \circ \pi_c)$, where π_c is the canonical projection to $\mathcal{D}^{\text{op}}$ from the category $\mathcal{A}_c$ whose objects are the pairs (d, h) , where d is an object of the category $\mathcal{D}$ and h is an arrow $c \rightarrow f(d)$ in $\mathcal{C}$, and whose arrows are the obvious ones. The natural transformation $\varepsilon^F : \widetilde{F \circ f^{\text{op}}} \rightarrow F$ is defined by the condition that, for any objects c of $\mathcal{C}$ and (d, h) of $\mathcal{A}_c$, $\varepsilon^F(c) \circ \kappa_{(d, h)}^F = F(h)$, where $\kappa_{(d, h)}^F : F(f(d)) \rightarrow \text{colim}(F \circ f^{\text{op}} \circ \pi_c)$ is the canonical colimit arrow.

Now, our thesis follows immediately from the fact that for any object of the category $\int (F \circ f^{\text{op}})_E$ of the form $(c, \kappa_{(d, h)}^F \circ y)$ for a generalized element $y : E \rightarrow F(f(d))$, $(\int \varepsilon^F)_E((c, \kappa_{(d, h)}^F \circ y)) = (c, F(h) \circ y)$, by invoking the fact that the colimit arrows are jointly epimorphic. $\square$

5.3 Yoneda representations of flat functors

In this section, which is based on [17], we introduce a technique, based on ideas from indexed category theory, for representing flat functors into arbitrary Grothendieck toposes; this technique will be applied in Chapters 6 and 8 to obtain results in the theory of classifying toposes.

We shall borrow from section 5.1.1 the notation and terminology concerning indexed categories.

By the Yoneda lemma, for each $F \in [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ there is a natural isomorphism of functors

$$F \cong \text{Hom}_{[\mathcal{C}^{\text{op}}, \mathbf{Set}]}^{\mathbf{Set}}(y_{\mathcal{C}}(-), F),$$

where $\text{Hom}_{[\mathcal{C}^{\text{op}}, \mathbf{Set}]}^{\mathbf{Set}}(y_{\mathcal{C}}(-), F)$ is the functor given by the composite

$$\mathcal{C}^{\text{op}} \xrightarrow{y_{\mathcal{C}}^{\text{op}} \times \Delta F} [\mathcal{C}^{\text{op}}, \mathbf{Set}]^{\text{op}} \times [\mathcal{C}^{\text{op}}, \mathbf{Set}] \xrightarrow{\text{Hom}_{[\mathcal{C}^{\text{op}}, \mathbf{Set}]}^{\mathbf{Set}}} \mathbf{Set},$$

ΔF is the constant functor equal to F and $y_{\mathcal{C}} : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is the Yoneda embedding.

By exploiting the local smallness of $\mathcal{E}$ -indexed diagram categories (cf. section 5.1.1), we can generalize this result to the case of functors with values in an arbitrary Grothendieck topos. More specifically, we have the following result:

Theorem 5.3.1 (cf. Theorem 2.3 [17]). *Let $\mathcal{C}$ be a small category and $\mathcal{E}$ a Grothendieck topos. Then for every functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ there is a natural isomorphism of functors*

$$F \cong \text{Hom}_{[\mathcal{C}^{\text{op}}, \mathcal{E}]}^{\mathcal{E}}(y_{\mathcal{C}}^{\mathcal{E}}(-), F),$$

where $y_{\mathcal{C}}^{\mathcal{E}} : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathcal{E}]$ is the functor given by the composite

$$\mathcal{C} \xrightarrow{y_{\mathcal{C}}} [\mathcal{C}^{\text{op}}, \mathbf{Set}] \xrightarrow{\gamma_{\mathcal{E}}^* \circ -} [\mathcal{C}^{\text{op}}, \mathcal{E}]$$

and $\text{Hom}_{[\underline{\mathcal{C}^{\text{op}}, \mathcal{E}}]}^{\mathcal{E}}(y_{\mathcal{C}}^{\mathcal{E}}(-), F)$ is the functor given by the composite

$$\mathcal{C}^{\text{op}} \xrightarrow{(y_{\mathcal{C}}^{\mathcal{E}})^{\text{op}} \times \Delta F} [\mathcal{C}^{\text{op}}, \mathcal{E}]^{\text{op}} \times [\mathcal{C}^{\text{op}}, \mathcal{E}] \xrightarrow{\text{Hom}_{[\underline{\mathcal{C}^{\text{op}}, \mathcal{E}}]}^{\mathcal{E}}} \mathcal{E}.$$

Moreover, the above isomorphism is natural in F .

In the case of flat functors, Theorem 5.3.1 specializes to the following result:

Corollary 5.3.2 *Let $\mathcal{C}$ be a small category and $\mathcal{E}$ a Grothendieck topos. Then for every flat functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$, there is a natural isomorphism of functors*

$$F \cong \text{Hom}_{\underline{\mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})}}^{\mathcal{E}}(y_{\mathcal{C}}^{\mathcal{E}}(-), F),$$

where $y_{\mathcal{C}}^{\mathcal{E}} : \mathcal{C} \rightarrow \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$ is the functor given by the composite

$$\mathcal{C} \xrightarrow{y_{\mathcal{C}}} \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathbf{Set}) \xrightarrow{\gamma_{\mathcal{E}}^* \circ -} \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})$$

and $\text{Hom}_{\underline{\mathbf{Flat}(\mathcal{C}, \mathcal{E})}}^{\mathcal{E}}(y_{\mathcal{C}}^{\mathcal{E}}(-), F)$ is the functor given by the composite

$$\mathcal{C}^{\text{op}} \xrightarrow{(y_{\mathcal{C}}^{\mathcal{E}})^{\text{op}} \times \Delta F} \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})^{\text{op}} \times \mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E}) \xrightarrow{\text{Hom}_{\underline{\mathbf{Flat}(\mathcal{C}^{\text{op}}, \mathcal{E})}}^{\mathcal{E}}} \mathcal{E}.$$

Moreover, the above isomorphism is natural in F .

From now on we shall refer to this result as the *Yoneda representation of flat functors*.

5.3.1 Cauchy completion of sites

When dealing with theories classified by a presheaf topos $[\mathcal{C}, \mathbf{Set}]$, it is natural to replace the category $\mathcal{C}$ with its Cauchy completion $\hat{\mathcal{C}}$, since $[\mathcal{C}, \mathbf{Set}]$ and $[\hat{\mathcal{C}}, \mathbf{Set}]$ are equivalent (cf. Corollary A1.1.9 [55]). Recall that $\hat{\mathcal{C}}$ can be equivalently characterized as the closure of $\mathcal{C}$ under retracts in $\text{Ind-}\mathcal{C}$.

From the equivalence of toposes

$$[\mathcal{C}^{\text{op}}, \mathbf{Set}] \simeq [\hat{\mathcal{C}}^{\text{op}}, \mathbf{Set}]$$

it follows that for any Grothendieck topology J on $\mathcal{C}$ there exists a unique Grothendieck topology $\hat{J}$ on $\hat{\mathcal{C}}$ such that subtopos

$$\mathbf{Sh}(\hat{\mathcal{C}}, \hat{J}) \hookrightarrow [\hat{\mathcal{C}}^{\text{op}}, \mathbf{Set}]$$

corresponds to the subtopos

$$\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$$

under the equivalence

$$[\hat{\mathcal{C}}^{\text{op}}, \mathbf{Set}] \simeq [\mathcal{C}^{\text{op}}, \mathbf{Set}].$$

We shall describe this topology explicitly in Theorem 5.3.3. For this, let us adopt the following conventions:

- If S is a sieve in $\mathcal{C}$, we denote by $\overline{S}$ the sieve in $\hat{\mathcal{C}}$ generated by the arrows in S .
- If R is a sieve in $\hat{\mathcal{C}}$, we denote by $R \cap \text{Arr}(\mathcal{C})$ the sieve in $\mathcal{C}$ formed by the elements of R which are arrows in $\mathcal{C}$.
- Given an arrow $g : d \rightarrow c$ in $\mathcal{C}$ and sieves S and R on c respectively in $\mathcal{C}$ and $\hat{\mathcal{C}}$, we denote by $g_{\mathcal{C}}^*(S)$ and $g_{\hat{\mathcal{C}}}^*(R)$ the sieves obtained by pulling back S and R along g respectively in the categories $\mathcal{C}$ and $\hat{\mathcal{C}}$.

Theorem 5.3.3 (Theorem 2.15 [17]). *Let $\mathcal{C}$ be a small category and $\hat{\mathcal{C}}$ its Cauchy completion. Given a Grothendieck topology J on $\mathcal{C}$ there exists a unique Grothendieck topology $\hat{J}$ on $\hat{\mathcal{C}}$ that induces J on $\mathcal{C}$, which is defined as follows: for each sieve R on $d \in \hat{\mathcal{C}}$, $R \in \hat{J}(d)$ if and only if there exist a retract $d \xrightarrow{i} a \xrightarrow{r} d$ with $a \in \mathcal{C}$ and a sieve $S \in J(a)$ such that $R = i_{\hat{\mathcal{C}}}^*(\overline{S})$. Furthermore, if $d \in \mathcal{C}$ then $R \in \hat{J}(d)$ if and only if there exists a sieve S in $\mathcal{C}$ on d such that $R = \overline{S}$.*

Proof Since the full embedding $\mathcal{C} \hookrightarrow \hat{\mathcal{C}}$ is (trivially) dense with respect to every Grothendieck topology on $\hat{\mathcal{C}}$, it follows from the Comparison Lemma (Theorem 1.1.8) that there is at most one Grothendieck topology on $\hat{\mathcal{C}}$ that induces J on $\mathcal{C}$. Therefore, to prove the first part of the thesis it will be enough to show that the assignment $\hat{J}$ in the statement of the theorem is a Grothendieck topology that induces J on $\mathcal{C}$. This, as well as the second part of the thesis, can be easily deduced from the following fact (whose proof is left to the reader): given an object $c \in \mathcal{C}$, the assignments $R \rightarrow R \cap \text{Arr}(\mathcal{C})$ and $S \rightarrow \overline{S}$ are inverse to each other and define a bijection between the set of sieves in $\mathcal{C}$ on c and the set of sieves in $\hat{\mathcal{C}}$ on c ; moreover, these bijections are natural with respect to the operations of pullback of sieves in $\mathcal{C}$ and in $\hat{\mathcal{C}}$ along an arrow in $\mathcal{C}$.

By way of example, we provide the details of the proof that $\hat{J}$ satisfies the ‘stability axiom’ for Grothendieck topologies. Given $R \in \hat{J}(d)$ and $g : e \rightarrow d$ in $\hat{\mathcal{C}}$, we want to prove that $g^*(R) \in \hat{J}(e)$. Since $R \in \hat{J}(d)$, there exist a retract $d \xrightarrow{i} a \xrightarrow{r} d$ with $a \in \mathcal{C}$ and a sieve $S \in J(a)$ such that $R = i_{\hat{\mathcal{C}}}^*(\overline{S})$. On the other hand, there exists a retract $e \xrightarrow{j} b \xrightarrow{z} e$ with $b \in \mathcal{C}$. Now, $g^*(R) = g^*(i_{\hat{\mathcal{C}}}^*(\overline{S})) = (i \circ g)^*(\overline{S}) = ((i \circ g \circ z) \circ j)^*(\overline{S}) = j^*((i \circ g \circ z)^*(\overline{S})) = j^*((i \circ g \circ z)_{\hat{\mathcal{C}}}^*(\overline{S})) = j^*(\overline{(i \circ g \circ z)^*(S)})$. Our claim thus follows at once from the stability axiom for J . $\square$